Optimal Transport for Machine Learners

Compact teaching notes

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1 Optimal Matching between Point Clouds

1.1 Monge Problem for Discrete Points

Matching Problem. Given a cost matrix $(\mathrm{C}_{i,j})_{i\in[\![n]\!],j\in[\![m]\!]}$ and assuming $n=m$, the optimal assignment problem aims to find a bijection σ within the set $\mathrm{Perm}(n)$ of permutations of n elements that solves

$$\min_{\sigma\in\mathrm{Perm}(n)} \frac{1}{n}\sum_{i=1}^{n} \mathrm{C}_{i,\sigma(i)}. \tag{1}$$

1-D Case.

Proposition 1 (Monotone matching on the line). *Assume that the points $(x_i)_i$ and $(y_j)_j$ are pairwise distinct. If the cost is of the form $\mathrm{C}_{i,j}=h(x_i-y_j)$, where $h:\mathbb{R}\to\mathbb{R}^+$ is strictly convex (for example, $\mathrm{C}_{i,j}=|x_i-y_j|^p$ for $p>1$), then any optimal σ defines a strictly increasing map $x_i\mapsto y_{\sigma(i)}$ (and thus is unique), i.e., $\forall (i,i'),\quad (x_i-x_{i'})(y_{\sigma(i)}-y_{\sigma(i')})>0$.*

Proof. Indeed, if this property is violated, i.e., there exists (i,i') such that $(x_i-x_{i'})(y_{\sigma(i)}-y_{\sigma(i')})<0$, then one can define a permutation $\tilde\sigma$ by swapping the match, i.e., $\tilde\sigma(i)=\sigma(i')$ and $\tilde\sigma(i')=\sigma(i)$, yielding a strictly better cost, as proved in the following fact.
Let $h:\mathbb{R}\to\mathbb{R}$ be strictly convex and let $x<x'$ and $y<y'$. Then $h(x-y)+h(x'-y')<h(x-y')+h(x'-y)$. We set the gap $d:=y'-y>0$ and define for every $s\in\mathbb{R}$ $D(s):=\frac{h(s)-h(s-d)}{d}$ and $\Delta:=h(x-y')+h(x'-y)-h(x-y)-h(x'-y')=d(D(x'-y)-D(x-y))$. Because h is strictly convex, D is strictly increasing. Since $x-y<x'-y$, monotonicity yields $D(x-y)<D(x'-y)$, that is $\Delta>0$. □

$x_{\sigma_X(1)}\leqslant x_{\sigma_X(2)}\leqslant\dots$ and $y_{\sigma_Y(1)}\leqslant y_{\sigma_Y(2)}\leqslant\dots$ Note that if h is strictly convex, then all optimal assignments are increasing, and if the points are all distinct, this increasing map is unique. If h is not strictly convex, for instance $c(x,y)=|x-y|$, non-increasing optimal assignments can also exist.

Proposition 2 (Non-crossing optimal matchings). *In dimension 2, for $c(x,y)=\|x-y\|$, if σ is an optimal assignment, then segments $[x_i,y_{\sigma(i)}]$ cannot cross.*

Proof. If two segments $[x_i,y_{\sigma(i)}]$ and $[x_j,y_{\sigma(j)}]$ cross at an interior point z, then the triangle inequality gives $\|x_i-y_{\sigma(j)}\|+\|x_j-y_{\sigma(i)}\|<\|x_i-y_{\sigma(i)}\|+\|x_j-y_{\sigma(j)}\|$. The assignment obtained by swapping $(\sigma(i),\sigma(j))$ therefore has a strictly smaller cost, which contradicts optimality. □

$C_n=\frac{1}{n+1}\binom{2n}{n}\sim\frac{4^n}{\sqrt{\pi}n^{3/2}}$.

1.2 Matching Algorithms

Hungarian primal-dual method. $u_i+v_j\leqslant C_{i,j}\qquad\forall i,j.$

$$\delta=\min_{i\in S,\ j\notin T}\big(C_{i,j}-u_i-v_j\big),\qquad u_i\leftarrow u_i+\delta\ (i\in S),\qquad v_j\leftarrow v_j-\delta\ (j\in T).$$

Proposition 1 (Correctness and complexity of the Hungarian primal-dual method). *Assume the Hungarian method terminates with a perfect matching σ contained in the equality graph $E(u,v)=\{(i,j)\ ;\ u_i+v_j=C_{i,j}\}$, where (u,v) is dual feasible, i.e. $u_i+v_j\leqslant C_{i,j}$ for all (i,j). Then σ is an optimal assignment. Moreover, the usual Hungarian updates terminate after finitely many augmentations. With maintained slacks, the method uses $O(n^3)$ arithmetic operations.*

Proof. For any permutation τ, dual feasibility gives $\sum_i C_{i,\tau(i)}\geqslant\sum_i(u_i+v_{\tau(i)})=\sum_i u_i+\sum_j v_j$. This is the weak duality lower bound. If σ is contained in the equality graph, then $\sum_i C_{i,\sigma(i)}=\sum_i u_i+\sum_j v_j$, so the primal cost of σ reaches the dual lower bound and is optimal.
At each successful augmentation, the matching cardinality increases by one, so there are at most n augmentations. During one augmentation phase, the algorithm grows an alternating tree in the equality graph. If no augmenting path is available, the dual update uses the smallest slack of an edge leaving the current tree. For edges inside the tree, adding δ to source labels and subtracting δ from target labels preserves tightness; for edges from S to T^c, the definition of δ preserves feasibility and makes at least one new edge tight; all other inequalities are unchanged or become looser. Thus the reachable sets strictly grow between two failed augmentation attempts, and they can grow at most n times within one phase. If the current slacks $\min_{i\in S}(C_{i,j}-u_i-v_j)$ are updated when a source enters S, each tree expansion costs $O(n)$. A phase therefore costs $O(n^2)$, and the n phases give $O(n^3)$ operations. Hence the method reaches a perfect optimal matching. □

2 Monge Problem between Measures

2.1 Measures

Histograms. $\Sigma_n:=\left\{\mathrm{a}\in\mathbb{R}^n_+\ ;\ \sum_{i=1}^n\mathrm{a}_i=1\right\}.$

Discrete measure, empirical measure. A discrete measure with weights a and locations $x_1, \ldots, x_n \in \mathcal{X}$ reads

$$\alpha = \sum_{i=1}^{n} \mathrm{a}_i \delta_{x_i} \tag{2}$$

where δ_x is the Dirac mass at position x, intuitively a unit of mass infinitely concentrated at location x. Such a measure is a probability measure if, additionally, $\mathrm{a} \in \Sigma_n$, and more generally a positive measure if all weights a_i are nonnegative.

General measures. A Dirac measure δ_x is then defined as $\delta_x(A) = 1$ if $x \in A$ and 0 otherwise, and this extends by linearity for discrete measures of the form (2) as
$\alpha(A) = \sum_{x_i \in A} \mathrm{a}_i$ We denote $\mathcal{M}_+(\mathcal{X})$ the subset of all positive measures on $\mathcal{X}$, i.e. $\alpha(A) \geqslant 0$ (and $\alpha(\mathcal{X}) < +\infty$ for the measure to be finite). The set of probability measures is denoted $\mathcal{M}_+^1(\mathcal{X})$, which means that any $\alpha \in \mathcal{M}_+^1(\mathcal{X})$ is positive, and that $\alpha(\mathcal{X}) = 1$.

Radon measures. Using Lebesgue integration, a Borel measure can be used to compute the integral of measurable functions (i.e. such that level sets $\{x \,;\, f(x) < t\}$ are Borel sets), and we denote this pairing as
$\langle f, \alpha \rangle \coloneqq \int f(x) \mathrm{d}\alpha(x)$. $\int_{\mathcal{X}} f(x) \mathrm{d}\alpha(x) = \sum_{i=1}^n \mathrm{a}_i f(x_i)$.

$$\|\alpha\|_{\mathrm{TV}} = \|\ell\|_{\mathcal{C}(\mathcal{X}) \to \mathbb{R}} = \sup_{f \in \mathcal{C}(\mathcal{X})} \{ \langle f, \alpha \rangle \,;\, \|f\|_\infty \leqslant 1 \}.$$

$|\alpha|(A) = \sup\limits_{A = \cup_i B_i} \sum_i |\alpha(B_i)|$

Proposition 2 (Dual and measure definitions of total variation)**.** *For a finite signed Radon measure α on a compact space, $\|\alpha\|_{\mathrm{TV}} = |\alpha|(\mathcal{X})$.*

Proof. The inequality $\|\alpha\|_{\mathrm{TV}} \leqslant |\alpha|(\mathcal{X})$ follows from $\left|\int f \,\mathrm{d}\alpha\right| \leqslant \int |f| \,\mathrm{d}|\alpha| \leqslant \|f\|_\infty |\alpha|(\mathcal{X})$. For the reverse inequality, write the Jordan decomposition $\alpha = \alpha^+ - \alpha^-$, so that $|\alpha| = \alpha^+ + \alpha^-$. The measurable function $s = \frac{\mathrm{d}\alpha}{\mathrm{d}|\alpha|}$ takes values in $\{-1, 1\}$ and satisfies $\mathrm{d}\alpha = s \,\mathrm{d}|\alpha|$. By regularity of Radon measures on compact spaces, s can be approximated in $L^1(|\alpha|)$ by continuous functions f with $\|f\|_\infty \leqslant 1$. Hence $|\alpha|(\mathcal{X}) = \int s \,\mathrm{d}\alpha$ is the supremum of $\int f \,\mathrm{d}\alpha$ over continuous f with $\|f\|_\infty \leqslant 1$. □

Relative densities. A measure α which is a weighting of another reference one $\mathrm{d}x$ is said to have a density, which is denoted $\mathrm{d}\alpha(x) = \rho_\alpha(x)\mathrm{d}x$ (on $\mathbb{R}^d$ $\mathrm{d}x$ is often the Lebesgue measure), often also denoted $\rho_\alpha = \frac{\mathrm{d}\alpha}{\mathrm{d}x}$, which means that
$\forall\, h \in \mathcal{C}(\mathbb{R}^d), \quad \int_{\mathbb{R}^d} h(x) \mathrm{d}\alpha(x) = \int_{\mathbb{R}^d} h(x) \rho_\alpha(x) \mathrm{d}x.$

Probabilistic interpretation.

2.2 Push Forward

For some continuous map $T : \mathcal{X} \to \mathcal{Y}$, we define the pushforward operator $T_\sharp : \mathcal{M}(\mathcal{X}) \to \mathcal{M}(\mathcal{Y})$.
For a Dirac mass, one has $T_\sharp \delta_x = \delta_{T(x)}$, and this formula is extended to arbitrary measures by linearity. In some sense, moving from T to $T_\sharp$ is a way to linearize any map at the price of moving from a (possibly) finite-dimensional space $\mathcal{X}$ to the infinite-dimensional space $\mathcal{M}(\mathcal{X})$, and this idea is central to many convex relaxation methods, most notably Lasserre's relaxation.
For discrete measures (2), the pushforward operation consists simply in moving the positions of all the points in the support of the measure
$T_\sharp \alpha \coloneqq \sum_i \mathrm{a}_i \delta_{T(x_i)}$.

Definition 1 (Push-forward)**.** *For $T : \mathcal{X} \to \mathcal{Y}$, the push forward measure $\beta = T_\sharp \alpha \in \mathcal{M}(\mathcal{Y})$ of some $\alpha \in \mathcal{M}(\mathcal{X})$ satisfies*

$$\forall\, h \in \mathcal{C}(\mathcal{Y}), \quad \int_{\mathcal{Y}} h(y) \mathrm{d}\beta(y) = \int_{\mathcal{X}} h(T(x)) \mathrm{d}\alpha(x). \tag{3}$$

Equivalently, for any measurable set $B \subset \mathcal{Y}$, one has

$$\beta(B) = \alpha(\{x \in \mathcal{X} \,;\, T(x) \in B\}). \tag{4}$$

Note that $T_\sharp$ preserves positivity and total mass, so that if $\alpha \in \mathcal{M}_+^1(\mathcal{X})$ then $T_\sharp \alpha \in \mathcal{M}_+^1(\mathcal{Y})$.

Proposition 3 (Push-forward formula for densities)**.** *Let α and β have densities ρ_α and ρ_β on open subsets of $\mathbb{R}^d$. Assume that T is a C^1 diffeomorphism and that $\beta = T_\sharp \alpha$. Then, for all x,*

$$\rho_\alpha(x) = |\det(T'(x))| \rho_\beta(T(x)). \tag{5}$$

Equivalently, for $y = T(x)$, $\rho_\beta(y) = \rho_\alpha(T^{-1}y) \frac{1}{|\det T'(T^{-1}y)|}$.

Proof. Explicitly doing the change of variable $y = T(x)$, so that $\mathrm{d}y = |\det(T'(x))| \mathrm{d}x$ in formula (3) for measures with densities $(\rho_\alpha, \rho_\beta)$ on $\mathbb{R}^d$ (assuming T is smooth and a bijection), one has for all $h \in \mathcal{C}(\mathcal{Y})$

$$\begin{aligned} \int_{\mathcal{Y}} h(y) \rho_\beta(y) \mathrm{d}y &= \int_{\mathcal{Y}} h(y) \mathrm{d}\beta(y) = \int_{\mathcal{X}} h(T(x)) \mathrm{d}\alpha(x) = \int_{\mathcal{X}} h(T(x)) \rho_\alpha(x) \mathrm{d}x \\ &= \int_{\mathcal{Y}} h(y) \rho_\alpha(T^{-1}y) \frac{\mathrm{d}y}{|\det(T'(T^{-1}y))|}, \end{aligned}$$

which shows that $\rho_\beta(y) = \rho_\alpha(T^{-1}y)\frac{1}{|\det(T'(T^{-1}y))|}$. Since T is a diffeomorphism, one obtains equivalently $\rho_\alpha(x) = |\det(T'(x))|\rho_\beta(T(x))$ where $T'(x) \in \mathbb{R}^{d\times d}$ is the Jacobian matrix of T (the matrix formed by taking the gradient of each coordinate of T).

This implies, denoting $y = T(x)$ $|\det(T'(x))| = \frac{\rho_\alpha(x)}{\rho_\beta(y)}$. □

Remark 1 (Probabilistic interpretation). The law (i.e. probability distribution) of a random variable X, equivalently, is the push-forward of $\mathbb{P}$ by X, $\alpha = X_\sharp\mathbb{P}$.

Applying another push-forward $\beta = T_\sharp\alpha$ for $T : \mathcal{X} \to \mathcal{Y}$, following (3), is equivalent to defining another random variable $Y = T(X)$, namely $\omega \in \Omega \mapsto T(X(\omega)) \in \mathcal{Y}$. Thus β is the distribution of Y.

Drawing a random sample y from Y is thus simply achieved by computing $y = T(x)$ where x is drawn from X.

2.3 Monge's Formulation

Monge problem.

$$\inf_T \left\{ \int_{\mathcal{X}} c(x, T(x))\mathrm{d}\alpha(x) \; ; \; T_\sharp\alpha = \beta \right\}. \tag{6}$$

Proposition 4 (Empirical Monge maps are matchings)**.** *Assume that $x_1, \dots, x_n$ are distinct, that $y_1, \dots, y_n$ are distinct, and that $\alpha = \frac{1}{n}\sum_{i=1}^n \delta_{x_i}$ and $\beta = \frac{1}{n}\sum_{j=1}^n \delta_{y_j}$. If $T_\sharp\alpha = \beta$, then there exists a permutation $\sigma \in \mathrm{Perm}(n)$ such that $T(x_i) = y_{\sigma(i)}$ for all i. Conversely, every such permutation defines a feasible Monge map on the support of α, and $\int_{\mathcal{X}} c(x, T(x))\mathrm{d}\alpha(x) = \frac{1}{n}\sum_i c(x_i, y_{\sigma(i)})$.*

Proof. Since $T_\sharp\alpha = \beta$, all the images $T(x_i)$ must belong to $\{y_1, \dots, y_n\}$; otherwise the push-forward would give positive mass to a point outside the support of β. Moreover, each y_j must receive exactly one preimage among the x_i, because $\beta(\{y_j\}) = 1/n$ and each atom x_i carries mass $1/n$. This defines a bijection σ between source and target indices. The converse and the cost identity follow by direct substitution in the definition of the push-forward and of the integral with respect to α. □

Remark 2 (Existence of Monge maps). Even if the constraint set is not empty, the infimum might not be reached. The most celebrated example is the case where α is distributed uniformly on a single segment and β is distributed on two segments on either side.

Example 1 (Semi-discrete Monge maps). $C_j \coloneqq T^{-1}(y_j) \quad \text{where} \quad \alpha(C_j) = b_j$. If one exchanges the roles of α and β so that α is discrete, then no valid T exists in general: it is not possible to push forward a discrete measure to a measure with density.

Monge distance. In the special case $c(x, y) = d^p(x, y)$ where d is a distance, we denote

$$\tilde{\mathcal{W}}_p^p(\alpha, \beta) \coloneqq \inf_T \left\{ \mathcal{E}_\alpha(T) \coloneqq \int_{\mathcal{X}} d(x, T(x))^p \mathrm{d}\alpha(x) \; ; \; T_\sharp\alpha = \beta \right\}. \tag{7}$$

If the constraint set is empty, then we set $\tilde{\mathcal{W}}_p^p(\alpha, \beta) = +\infty$.

Proposition 5 (Directed Monge distance)**.** *Assume $\mathcal{X} = \mathcal{Y}$ is a compact metric space. The quantity $\tilde{\mathcal{W}}_p$ is nonnegative, vanishes only on the diagonal, and satisfies the triangle inequality. It is therefore a directed extended distance: it need not be symmetric and may take the value $+\infty$.*

Proof. Nonnegativity is immediate. If $\tilde{\mathcal{W}}_p(\alpha, \beta) = 0$, choose feasible maps T_k with $\int d(x, T_k(x))^p \mathrm{d}\alpha(x) \to 0$. For every continuous function h, compactness gives uniform continuity, hence

$$\left| \int h \mathrm{d}\beta - \int h \mathrm{d}\alpha \right| = \left| \int h(T_k(x)) - h(x) \mathrm{d}\alpha(x) \right| \longrightarrow 0.$$

Thus $\alpha = \beta$.

If $\tilde{\mathcal{W}}_p(\alpha, \beta) = +\infty$, then either $\tilde{\mathcal{W}}_p(\alpha, \gamma) = +\infty$ or $\tilde{\mathcal{W}}_p(\gamma, \beta) = +\infty$, because otherwise there are maps (S, T) such that $S_\sharp\alpha = \gamma$ and $T_\sharp\gamma = \beta$ and then $(T \circ S)_\sharp\alpha = \beta$ so that $\tilde{\mathcal{W}}_p^p(\alpha, \beta) \leqslant \mathcal{E}_\alpha(T \circ S) < +\infty$.

We can therefore restrict our attention to the case where $\tilde{\mathcal{W}}_p^p(\alpha, \gamma) < +\infty$ and $\tilde{\mathcal{W}}_p^p(\gamma, \beta) < +\infty$ because otherwise the inequality is trivial.

For any $\varepsilon > 0$, consider ε-minimizers $S_\sharp\alpha = \gamma$ and $T_\sharp\gamma = \beta$ such that $\mathcal{E}_\alpha(S)^{\frac{1}{p}} \leqslant \tilde{\mathcal{W}}_p(\alpha, \gamma) + \varepsilon \quad \text{and} \quad \mathcal{E}_\gamma(T)^{\frac{1}{p}} \leqslant \tilde{\mathcal{W}}_p(\gamma, \beta) + \varepsilon$.

Using $(T \circ S)_\sharp\alpha = \beta$, the feasibility of this composed map and the triangle inequality give

$$\begin{aligned} \tilde{\mathcal{W}}_p(\alpha, \beta) &\leqslant \left(\int d(x, T(S(x)))^p \mathrm{d}\alpha(x) \right)^{1/p} \\ &\leqslant \left(\int (d(x, S(x)) + d(S(x), T(S(x))))^p \mathrm{d}\alpha(x) \right)^{1/p}. \end{aligned}$$

Then, using Minkowski inequality for the L^p spaces with measure α $\|f+g\|_{L^p(\alpha)} \leqslant \|f\|_{L^p(\alpha)} + \|g\|_{L^p(\alpha)}$ and with $f(x) \triangleq d(x, S(x))$ and $g(x) \triangleq d(S(x), T(S(x)))$ one has

$$\begin{aligned} \tilde{\mathcal{W}}_p(\alpha, \beta) &\leqslant \left(\int d(x, S(x))^p \mathrm{d}\alpha(x) \right)^{1/p} + \left(\int d(S(x), T(S(x)))^p \mathrm{d}\alpha(x) \right)^{1/p} \\ &= \left(\int d(x, S(x))^p \mathrm{d}\alpha(x) \right)^{1/p} + \left(\int d(y, T(y))^p \mathrm{d}\gamma(y) \right)^{1/p} \\ &\leqslant \tilde{\mathcal{W}}_p(\alpha, \gamma) + \tilde{\mathcal{W}}_p(\gamma, \beta) + 2\varepsilon. \end{aligned}$$

Letting $\varepsilon \to 0$ gives the result. $\square$

2.4 Existence and Uniqueness of the Monge Map

Brenier's theorem.
Theorem 1 (Brenier)**.** *In the case $\mathcal{X} = \mathcal{Y} = \mathbb{R}^d$ and $c(x,y) = \|x-y\|^2$, if α has a density with respect to the Lebesgue measure, then there exists a unique optimal Monge map T. This map is characterized by being the unique gradient of a convex function $T = \nabla\varphi$ such that $(\nabla\varphi)_\sharp\alpha = \beta$.*

Proof. Kantorovich duality provides optimal potentials (f,g) with equality $f(x)+g(y) = \|x-y\|^2$ on the support of any optimal plan. After subtracting the harmless quadratic terms, this equality is equivalent to the Fenchel equality $\varphi(x) + \varphi^*(y) = \langle x,\, y\rangle$ for a convex function φ. Hence the support of every optimal plan lies in the graph of the subdifferential $\partial\varphi$. Since α has a density and convex functions are differentiable Lebesgue-almost everywhere, $\partial\varphi(x)$ is a singleton for α-almost every x. The plan is therefore concentrated on the graph of $T = \nabla\varphi$ and defines a Monge map. Any two optimal plans are concentrated on the same single-valued graph α-almost everywhere, which gives uniqueness of the map. $\square$

Remark 3 (Beyond the quadratic Euclidean cost)*.* $T(x) = \exp_x(-\nabla\varphi(x))$, where φ is c-convex. The main additional issues are the cut locus and regularity of geodesics; this is why the Euclidean statement is usually presented first.
$\forall\,(x,x') \in \mathbb{R}^d \times \mathbb{R}^d, \quad \langle \nabla\varphi(x) - \nabla\varphi(x'),\, x-x'\rangle \geqslant 0.$
Remark 4 (Monotone fields need not be gradients)*.*

$$R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}.$$

$\langle R_\theta x - R_\theta x',\, x-x'\rangle = \langle R_\theta(x-x'),\, x-x'\rangle = \cos(\theta)\|x-x'\|^2 \geqslant 0$. However, for $\theta \neq 0$, R_θ is not symmetric and therefore cannot be the gradient of a scalar potential. Indeed, if a linear field Ax equals $\nabla\varphi(x)$, then its Jacobian A must be symmetric; equivalently, for a quadratic potential $\varphi(x) = \langle Bx,\, x\rangle/2$, one has $\nabla\varphi(x) = ((B+B^\top)/2)x$.
One can also consider costs of the form $c(x,y) = h(x-y)$ where h is a strictly convex smooth function, for instance, $c(x,y) = \|x-y\|^p$ with $1 < p < +\infty$.
Remark 5 (Caffarelli regularity)*.* Let $\Omega, \Lambda \subset \mathbb{R}^d$ be bounded domains, with Λ convex, and let $\alpha = \rho_\alpha \mathrm{d}x$ and $\beta = \rho_\beta \mathrm{d}y$ have densities bounded above and below by positive constants. If $T = \nabla\varphi$ is the Brenier map from α to β, then φ is strictly convex and T is locally Hölder continuous.
Indeed, φ is in general non-smooth, but it is convex and therefore locally Lipschitz on the interior of its domain, so $\nabla\varphi$ is well defined α-almost everywhere.
If α does not have a density, then a Monge map might fail to exist. The correct relaxed object is an optimal Kantorovich plan concentrated on the graph of the set-valued map $\partial\varphi$.

Monge-Ampère equation. For measures with densities, using (5), one obtains that φ is the unique (up to the addition of a constant) convex function which solves the following Monge-Ampère-type equation

$$\det(\partial^2\varphi(x))\rho_\beta(\nabla\varphi(x)) = \rho_\alpha(x) \tag{8}$$

where $\partial^2\varphi(x) \in \mathbb{R}^{d\times d}$ is the Hessian of φ.
$\varphi(x) = \frac{\|x\|^2}{2} + \varepsilon\psi(x), \qquad T(x) = \nabla\varphi(x) = x + \varepsilon\nabla\psi(x).$ $\rho_\alpha(x) = \det(I + \varepsilon\nabla^2\psi(x))\,\rho_\beta(x + \varepsilon\nabla\psi(x)).$

$$\rho_\alpha = \rho_\beta + \varepsilon\big(\rho_\beta\Delta\psi + \langle\nabla\rho_\beta,\, \nabla\psi\rangle\big) + o(\varepsilon) = \rho_\beta + \varepsilon\,\mathrm{div}(\rho_\beta\nabla\psi) + o(\varepsilon).$$

$-\mathrm{div}(\rho\nabla\psi) = \eta$. For the uniform reference density $\rho = 1$, this reduces to $-\Delta\psi = \eta$, which is the precise sense in which Monge–Ampère linearizes to a Laplacian.

OT in 1-D. For a measure α on $\mathbb{R}$, we introduce the cumulative function

$$\forall\, x \in \mathbb{R}, \quad \mathcal{C}_\alpha(x) := \int_{-\infty}^x \mathrm{d}\alpha, \tag{9}$$

$\forall\, r \in [0,1], \quad \mathcal{C}_\alpha^{-1}(r) = \min_x \,\{x \in \mathbb{R}\cup\{-\infty\}\,;\, \mathcal{C}_\alpha(x) \geqslant r\}\,.$
Proposition 6 (Quantile push-forward)**.** *One has $(\mathcal{C}_\alpha)_\sharp^{-1}\mathcal{U} = \alpha$, where $\mathcal{U}$ is the uniform distribution in $[0,1]$. If α has a density, then $(\mathcal{C}_\alpha)_\sharp\alpha = \mathcal{U}$.*

Proof. For simplicity, we assume α has a strictly positive density, so that $\mathcal{C}_\alpha$ is a strictly increasing continuous function. Denoting $\gamma := (\mathcal{C}_\alpha)_\sharp^{-1}\mathcal{U}$ we aim at proving $\gamma = \alpha$, which is equivalent to $\mathcal{C}_\gamma = \mathcal{C}_\alpha$. One has

$$\mathcal{C}_\gamma(x) = \int_{-\infty}^x \mathrm{d}\gamma = \int_{\mathbb{R}} 1_{]-\infty,x]}\mathrm{d}((\mathcal{C}_\alpha^{-1})_\sharp\mathcal{U}) = \int_0^1 1_{]-\infty,x]}(\mathcal{C}_\alpha^{-1}(z))\mathrm{d}z = \int_0^1 1_{[0,\mathcal{C}_\alpha(x)]}(z)\mathrm{d}z = \mathcal{C}_\alpha(x)$$

where we use the fact that $-\infty \leqslant \mathcal{C}_\alpha^{-1}(z) \leqslant x \iff 0 \leqslant z \leqslant \mathcal{C}_\alpha(x)$. $\square$

If α has a density, this shows that the map

$$T = \mathcal{C}_\beta^{-1} \circ \mathcal{C}_\alpha \tag{10}$$

satisfies $T_\sharp \alpha = \beta$.
For the cost $c(x, y) = |x - y|^2$, since this T is increasing (hence the gradient of a convex function since we are in 1-D), by Brenier's theorem, T is the solution to Monge problem (at least if we impose that α has a density, otherwise it might lead to a solution of the Kantorovich problem by properly defining the pseudo-inverse).
For discrete measures, one cannot apply directly this reasoning (because α does not have a density), but if the measures are uniform on the same number of Dirac masses, then this approach is equivalent to the sorting formula.
Proposition 7 (One-dimensional Wasserstein formulas)**.** *Let $\alpha, \beta \in \mathcal{M}_+^1(\mathbb{R})$ have finite p-th moments. For every $p \geqslant 1$,*

$$\mathcal{W}_p(\alpha, \beta)^p = \int_0^1 |\mathcal{C}_\alpha^{-1}(r) - \mathcal{C}_\beta^{-1}(r)|^p \mathrm{d}r = \|\mathcal{C}_\alpha^{-1} - \mathcal{C}_\beta^{-1}\|_{L^p([0,1])}^p. \tag{11}$$

For $p = 1$, this is equivalently

$$\mathcal{W}_1(\alpha, \beta) = \|\mathcal{C}_\alpha - \mathcal{C}_\beta\|_{L^1(\mathbb{R})} = \int_{\mathbb{R}} |\mathcal{C}_\alpha(x) - \mathcal{C}_\beta(x)| \mathrm{d}x = \int_{\mathbb{R}} \left| \int_{-\infty}^x \mathrm{d}(\alpha - \beta) \right| \mathrm{d}x. \tag{12}$$

Proof. When α has a density, the map $T = \mathcal{C}_\beta^{-1} \circ \mathcal{C}_\alpha$ pushes α onto β and is nondecreasing. Hence it is optimal for every convex increasing cost $h(|x-y|)$, in particular for $|x-y|^p$. Since $r = \mathcal{C}_\alpha(x)$ is uniformly distributed on $[0, 1]$ under $x \sim \alpha$, the change of variables $x = \mathcal{C}_\alpha^{-1}(r)$ gives the first formula. General measures follow by approximating them narrowly with smooth measures while preserving the p-th moments, or equivalently by using the monotone rearrangement theorem for generalized quantiles.
For $p = 1$, use the layer-cake identity. If q_α and q_β are the two quantile functions, then $\int_0^1 |q_\alpha(r) - q_\beta(r)| \mathrm{d}r = \int_{\mathbb{R}} \lambda\big(\{r : q_\alpha(r) \leqslant x < q_\beta(r)\} \cup \{r : q_\beta(r) \leqslant x < q_\alpha(r)\}\big) \mathrm{d}x$. The measure of the displayed set is exactly $|\mathcal{C}_\alpha(x) - \mathcal{C}_\beta(x)|$ for almost every x. This proves (12). □

For $p = 1$, formula (12) shows that $\mathcal{W}_1$ is a norm on signed measures with zero total mass once they are identified with their cumulative primitives.

OT on 1-D Gaussians. We first consider the case where $\alpha = \mathcal{N}(m_\alpha, s_\alpha^2)$ and $\beta = \mathcal{N}(m_\beta, s_\beta^2)$ are two Gaussians in $\mathbb{R}$.
$T(x) = \frac{s_\beta}{s_\alpha}(x - m_\alpha) + m_\beta$ satisfies $T_\sharp \alpha = \beta$. Furthermore, it is the derivative of the convex function
$\varphi(x) = \frac{s_\beta}{2 s_\alpha}(x - m_\alpha)^2 + m_\beta x$, $\tilde{\mathcal{W}}_2^2(\alpha, \beta) = \int_{\mathbb{R}} \left(\frac{s_\beta}{s_\alpha}(x - m_\alpha) + m_\beta - x \right)^2 \mathrm{d}\alpha(x) = (m_\alpha - m_\beta)^2 + (s_\alpha - s_\beta)^2$. This should be contrasted with the geometry of KL, where singular Gaussians (for which $s = 0$) are infinitely distant.

OT on Gaussians. If $\alpha = \mathcal{N}(\mathbf{m}_\alpha, \mathbf{\Sigma}_\alpha)$ and $\beta = \mathcal{N}(\mathbf{m}_\beta, \mathbf{\Sigma}_\beta)$ are two Gaussians in $\mathbb{R}^d$, we now look for an affine map

$$T : x \mapsto \mathbf{m}_\beta + A(x - \mathbf{m}_\alpha). \tag{13}$$

Proposition 8 (Affine push-forward of Gaussians)**.** *One has $T_\sharp \alpha = \beta$ if and only if*

$$A \mathbf{\Sigma}_\alpha A^\top = \mathbf{\Sigma}_\beta. \tag{14}$$

Proof. An affine function maps a Gaussian to a Gaussian so that it remains to show that the mean and covariance of $T_\sharp \alpha$ are those of β.

$$\mathbb{E}(Y) = \mathbb{E}(\mathbf{m}_\beta + A(X - \mathbf{m}_\alpha)) = \mathbf{m}_\beta + A\mathbb{E}(X - \mathbf{m}_\alpha) = \mathbf{m}_\beta$$

while its covariance is

$$\mathbb{E}((Y - \mathbf{m}_\beta)(Y - \mathbf{m}_\beta)^\top) = \mathbb{E}([A(X - \mathbf{m}_\alpha)][A(X - \mathbf{m}_\alpha)]^\top) = A\mathbb{E}((X - \mathbf{m}_\alpha)(X - \mathbf{m}_\alpha)^\top)A^\top = A \mathbf{\Sigma}_\alpha A^\top.$$

□

Proposition 9 (Gaussian $\mathcal{W}_2$ formula and Bures covariance term)**.** *Assume that $\mathbf{\Sigma}_\alpha$ and $\mathbf{\Sigma}_\beta$ are positive definite. The unique symmetric positive-definite solution of $A \mathbf{\Sigma}_\alpha A = \mathbf{\Sigma}_\beta$ is*

$$A = \mathbf{\Sigma}_\alpha^{-\frac{1}{2}} \left(\mathbf{\Sigma}_\alpha^{\frac{1}{2}} \mathbf{\Sigma}_\beta \mathbf{\Sigma}_\alpha^{\frac{1}{2}} \right)^{\frac{1}{2}} \mathbf{\Sigma}_\alpha^{-\frac{1}{2}}. \tag{15}$$

The affine map $T(x) = \mathbf{m}_\beta + A(x - \mathbf{m}_\alpha)$ is the optimal quadratic-cost transport from $\mathcal{N}(\mathbf{m}_\alpha, \mathbf{\Sigma}_\alpha)$ to $\mathcal{N}(\mathbf{m}_\beta, \mathbf{\Sigma}_\beta)$, and

$$\mathcal{W}_2^2(\alpha, \beta) = \|\mathbf{m}_\alpha - \mathbf{m}_\beta\|^2 + \mathcal{B}(\mathbf{\Sigma}_\alpha, \mathbf{\Sigma}_\beta)^2, \tag{16}$$

where

$$\mathcal{B}(\mathbf{\Sigma}_\alpha, \mathbf{\Sigma}_\beta)^2 \coloneqq \mathrm{tr} \left(\mathbf{\Sigma}_\alpha + \mathbf{\Sigma}_\beta - 2(\mathbf{\Sigma}_\alpha^{1/2} \mathbf{\Sigma}_\beta \mathbf{\Sigma}_\alpha^{1/2})^{1/2} \right). \tag{17}$$

Proof. Multiplying $A\mathbf{\Sigma}_\alpha A = \mathbf{\Sigma}_\beta$ on the left and right by $\mathbf{\Sigma}_\alpha^{1/2}$ gives $(\mathbf{\Sigma}_\alpha^{1/2} A \mathbf{\Sigma}_\alpha^{1/2})^2 = \mathbf{\Sigma}_\alpha^{1/2}\mathbf{\Sigma}_\beta\mathbf{\Sigma}_\alpha^{1/2}$. Since A is symmetric positive, $\mathbf{\Sigma}_\alpha^{1/2} A \mathbf{\Sigma}_\alpha^{1/2}$ is symmetric positive and is therefore the unique positive square root of the right-hand side. Conversely, the matrix in (15) is symmetric positive and satisfies the covariance equation.
By Proposition 3 or by the covariance calculation above, this affine map pushes α to β. It is the gradient of the convex quadratic potential $\varphi(x) = \langle \mathbf{m}_\beta, x\rangle + \frac{1}{2}\langle A(x-\mathbf{m}_\alpha), x-\mathbf{m}_\alpha\rangle$, so Brenier's theorem implies optimality. If $X \sim \alpha$, then

$$\begin{aligned}
\mathbb{E}\|X - T(X)\|^2 &= \|\mathbf{m}_\alpha - \mathbf{m}_\beta\|^2 + \mathbb{E}\|(I-A)(X-\mathbf{m}_\alpha)\|^2 \\
&= \|\mathbf{m}_\alpha - \mathbf{m}_\beta\|^2 + \operatorname{tr}\left((I-A)\mathbf{\Sigma}_\alpha(I-A)^\top\right) \\
&= \|\mathbf{m}_\alpha - \mathbf{m}_\beta\|^2 + \operatorname{tr}(\mathbf{\Sigma}_\alpha) + \operatorname{tr}(A\mathbf{\Sigma}_\alpha A) - 2\operatorname{tr}(A\mathbf{\Sigma}_\alpha) \\
&= \|\mathbf{m}_\alpha - \mathbf{m}_\beta\|^2 + \operatorname{tr}(\mathbf{\Sigma}_\alpha) + \operatorname{tr}(\mathbf{\Sigma}_\beta) - 2\operatorname{tr}\left((\mathbf{\Sigma}_\alpha^{1/2}\mathbf{\Sigma}_\beta\mathbf{\Sigma}_\alpha^{1/2})^{1/2}\right),
\end{aligned}$$

which is the desired expression. The formula for singular covariance matrices follows by adding ηI and letting $\eta \downarrow 0$. ☐

Proposition 10 (Metric and convexity properties of the Bures term)**.** *The function $\mathcal{B}$ is a distance on positive semidefinite covariance matrices. Moreover, $\mathcal{B}^2$ is jointly convex: for $t \in [0,1]$, $\mathcal{B}^2((1-t)\Sigma_0 + t\Sigma_1, (1-t)\Lambda_0 + t\Lambda_1) \leqslant (1-t)\mathcal{B}^2(\Sigma_0,\Lambda_0) + t\mathcal{B}^2(\Sigma_1,\Lambda_1)$.*

Proof. The key identity is the Procrustes representation $\mathcal{B}^2(\Sigma,\Lambda) = \min_{Q^\top Q = I}\|\Sigma^{1/2} - \Lambda^{1/2}Q\|_F^2$. Indeed, expanding the square gives $\operatorname{tr}\Sigma + \operatorname{tr}\Lambda - 2\max_{Q^\top Q=I}\operatorname{tr}(\Sigma^{1/2}Q^\top\Lambda^{1/2})$, and the orthogonal Procrustes formula identifies the maximum with $\operatorname{tr}((\Sigma^{1/2}\Lambda\Sigma^{1/2})^{1/2})$. Symmetry, positivity and separation follow immediately from this representation. The triangle inequality follows by choosing two almost optimal orthogonal matrices Q_1, Q_2 and writing $\|\Sigma^{1/2} - \Gamma^{1/2}Q_2Q_1\|_F \leqslant \|\Sigma^{1/2} - \Lambda^{1/2}Q_1\|_F + \|\Lambda^{1/2} - \Gamma^{1/2}Q_2\|_F$. Letting the two choices become optimal proves the metric property.
For convexity, use the equivalent factor formulation $\mathcal{B}^2(\Sigma,\Lambda) = \min_{UU^\top=\Sigma,\, VV^\top=\Lambda}\|U-V\|_F^2$. Choose nearly optimal factors (U_0,V_0) and (U_1,V_1) for the two pairs. Define block factors $U_t = [\sqrt{1-t}\,U_0, \sqrt{t}\,U_1]$, $\quad V_t = [\sqrt{1-t}\,V_0, \sqrt{t}\,V_1]$. Then $U_tU_t^\top = (1-t)\Sigma_0 + t\Sigma_1$ and $V_tV_t^\top = (1-t)\Lambda_0 + t\Lambda_1$, while $\|U_t - V_t\|_F^2 = (1-t)\|U_0 - V_0\|_F^2 + t\|U_1 - V_1\|_F^2$. Taking the infimum over the initial factors proves joint convexity. ☐

Remark 6 (Diagonal covariances and Hellinger geometry). In the case where $\mathbf{\Sigma}_\alpha = \operatorname{diag}(r_i)_i$ and $\mathbf{\Sigma}_\beta = \operatorname{diag}(s_i)_i$ are diagonal, the Bures metric reduces to the Euclidean distance between square roots,
$\mathcal{B}(\mathbf{\Sigma}_\alpha, \mathbf{\Sigma}_\beta) = \|\sqrt{r} - \sqrt{s}\|_2$.

3 Kantorovich Relaxation

3.1 Discrete Relaxation

$$\mathrm{U}(\mathrm{a},\mathrm{b}) := \left\{\mathrm{P} \in \mathbb{R}_+^{n\times m}\ ;\ \mathrm{P}\mathbb{1}_m = \mathrm{a} \quad \text{and} \quad \mathrm{P}^\top\mathbb{1}_n = \mathrm{b}\right\}, \tag{18}$$

$$\mathrm{P}\mathbb{1}_m = \left(\sum_j \mathrm{P}_{i,j}\right)_i \in \mathbb{R}^n \quad \text{and} \quad \mathrm{P}^\top\mathbb{1}_n = \left(\sum_i \mathrm{P}_{i,j}\right)_j \in \mathbb{R}^m.$$

The set of matrices $\mathrm{U}(\mathrm{a},\mathrm{b})$ is bounded, defined by $n+m$ equality constraints, and therefore a convex polytope (the convex hull of a finite set of matrices).

$$\mathrm{L}_\mathrm{C}(\mathrm{a},\mathrm{b}) := \min_{\mathrm{P}\in\mathrm{U}(\mathrm{a},\mathrm{b})} \langle \mathrm{C}, \mathrm{P}\rangle := \sum_{i,j}\mathrm{C}_{i,j}\mathrm{P}_{i,j}. \tag{19}$$

Proposition 11 (Sparse optimal plans)**.** *Problem (19) admits an optimal coupling $\mathrm{P}^\star$ with at most $n+m-1$ nonzero entries.*

Proof. The feasible set $\mathrm{U}(\mathrm{a},\mathrm{b})$ is a non-empty compact polytope, hence the linear objective admits an extreme-point minimizer by Proposition 13. The affine marginal constraints have rank $n+m-1$: there are $n+m$ row and column equations, and the only linear redundancy is that both sets imply the same total mass identity $\sum_{i,j} P_{i,j} = 1$.
At an extreme point of a polytope defined by equality constraints and non-negativity constraints, the columns corresponding to positive variables must be linearly independent in the constraint matrix. Otherwise, one could move a small amount of mass along a non-zero signed perturbation supported only on the positive entries while preserving all marginals, contradicting extremality. Since the rank is $n+m-1$, the support of an extreme point has cardinality at most $n+m-1$. ☐

Linear programming algorithms.

1-D cases.

Permutation Matrices as Couplings. For a permutation $\sigma \in \mathrm{Perm}(n)$, we write P_σ for the corresponding permutation matrix,

$$\forall\,(i,j) \in [\![n]\!]^2, \quad (\mathrm{P}_\sigma)_{i,j} = \begin{cases} 1 & \text{if} \quad j = \sigma_i \\ 0 & \text{otherwise.}\end{cases} \tag{20}$$

$\mathrm{Perm}_n := \{\mathrm{P}_\sigma \,;\, \sigma \in \mathrm{Perm}(n)\}$, $\langle \mathrm{C}, \mathrm{P}_\sigma\rangle = \sum_{i=1}^n \mathrm{C}_{i,\sigma_i}$ $\min_{\mathrm{P}\in\mathrm{Perm}_n} \langle \mathrm{C}, \mathrm{P}\rangle$. $\mathcal{B}_n := \left\{\mathrm{P} \in \mathbb{R}_+^{n\times n} \,;\, \mathrm{P}\mathbb{1}_n = \mathrm{P}^\top\mathbb{1}_n = \mathbb{1}_n\right\}$ $\min_{\mathrm{P}\in\mathcal{B}_n} \langle \mathrm{C}, \mathrm{P}\rangle$. $\mathrm{Perm}_n = \mathcal{B}_n \cap \{0,1\}^{n\times n}$. $\min_{\mathrm{P}\in\mathcal{B}_n} \langle \mathrm{C}, \mathrm{P}\rangle \leqslant \min_{\mathrm{P}\in\mathrm{Perm}_n} \langle \mathrm{C}, \mathrm{P}\rangle$. $\mathrm{Extr}(\mathcal{C}) := \left\{\mathrm{P} \,;\, \forall (Q,R) \in \mathcal{C}^2, \mathrm{P} = \frac{Q+R}{2} \Rightarrow Q = R\right\}$.

Proposition 12 (Existence of extreme points)**.** *If $\mathcal{C}$ is a non-empty compact convex subset of a finite-dimensional vector space, then* $\mathrm{Extr}(\mathcal{C}) \neq \emptyset$.

Proof. Among all non-empty faces of $\mathcal{C}$, choose one of minimal affine dimension (this is possible in finite dimension). If this face contained two distinct points, it would have a non-trivial exposed subface obtained by maximizing a linear functional that is not constant on the face, contradicting minimality. Hence the minimal face is a singleton, and its unique point is an extreme point of $\mathcal{C}$. □

Proposition 13 (Linear programs have extreme minimizers)**.** *If $\mathcal{C}$ is compact, then $\mathrm{Extr}(\mathcal{C}) \cap \left(\underset{\mathrm{P}\in\mathcal{C}}{\operatorname{argmin}}\, \langle \mathrm{C}, \mathrm{P}\rangle\right) \neq \emptyset$.*

Proof. Let $\mathcal{S} := \underset{\mathrm{P}\in\mathcal{C}}{\operatorname{argmin}}\, \langle \mathrm{C}, \mathrm{P}\rangle$.
We first note that $\mathcal{S}$ is convex (as always for an argmin) and compact because $\mathcal{C}$ is compact and the objective function is continuous so that $\mathrm{Extr}(\mathcal{S}) \neq \emptyset$.
Indeed, let $\mathrm{P} \in \mathrm{Extr}(\mathcal{S})$ and assume that $\mathrm{P} = (Q+R)/2$ with $Q, R \in \mathcal{C}$. By linearity of the objective and optimality of P, both Q and R must also minimize $\langle \mathrm{C}, \cdot\rangle$ over $\mathcal{C}$, hence $Q, R \in \mathcal{S}$. Since P is extremal in $\mathcal{S}$, this implies $Q = R = \mathrm{P}$, so $\mathrm{P} \in \mathrm{Extr}(\mathcal{C})$. □

Theorem 2 (Birkhoff and von Neumann)**.** *One has $\mathrm{Extr}(\mathcal{B}_n) = \mathrm{Perm}_n$.*

Proof. We first show the simplest inclusion $\mathrm{Perm}_n \subset \mathrm{Extr}(\mathcal{B}_n)$. Indeed it follows from the fact that $\mathrm{Extr}([0,1]) = \{0,1\}$. Take $\mathrm{P} \in \mathrm{Perm}_n$, if $\mathrm{P} = (Q+R)/2$ with $Q_{i,j}, R_{i,j} \in [0,1]$, since $\mathrm{P}_{i,j} \in \{0,1\}$ then necessarily $Q_{i,j} = R_{i,j} \in \{0,1\}$.
Pick $\mathrm{P} \in \mathcal{B}_n \backslash \mathrm{Perm}_n$; we need to split $\mathrm{P} = (Q+R)/2$ where Q, R are distinct bistochastic matrices.
P defines a bipartite graph linking two sets of n vertices.
This graph is composed of isolated edges when $\mathrm{P}_{i,j} = 1$ and connected edges corresponding to $0 < \mathrm{P}_{i,j} < 1$. If i is such a connected vertex on the left (similarly for j on the right), because $\sum_j \mathrm{P}_{i,j} = 1$, there are necessarily at least two edges (i, j_1) and (i, j_2) emanating from it (similarly on the right there are at least two converging edges (i_1, j) and (i_2, j)). This means that by following these connections, one necessarily can extract a cycle (if not, one could always extend it by the previous remarks) of the form $(i_1, j_1, i_2, j_2, \ldots, i_p, j_p)$, i.e. $i_{p+1} = i_1$. We assume this cycle is the shortest one among all this (finite) ensemble of cycles. Along this cycle, the left-right and right-left edges satisfy $0 < \mathrm{P}_{i_s,j_s}, \mathrm{P}_{i_{s+1},j_s} < 1$. The $(i_s)_s$ and $(j_s)_s$ are also all distinct because the cycle is the shortest. $\varepsilon := \min_{1\leqslant s\leqslant p} \{\mathrm{P}_{i_s,j_s}, \mathrm{P}_{i_{s+1},j_s}, 1-\mathrm{P}_{i_s,j_s}, 1-\mathrm{P}_{i_{s+1},j_s}\}$ so that $0 < \varepsilon < 1$. We split the graph into two sets of edges, left-right and right-left $\mathcal{A} := \{(i_s, j_s)\}_{s=1}^p$ and $\mathcal{B} := \{(i_{s+1}, j_s)\}_{s=1}^p$. We define the two matrices as

$$Q_{i,j} := \begin{cases} \mathrm{P}_{i,j} & \text{if} \quad (i,j) \notin \mathcal{A}\cup\mathcal{B} \\ \mathrm{P}_{i,j} + \varepsilon/2 & \text{if} \quad (i,j) \in \mathcal{A} \\ \mathrm{P}_{i,j} - \varepsilon/2 & \text{if} \quad (i,j) \in \mathcal{B} \end{cases} \quad \text{and} \quad R_{i,j} := \begin{cases} \mathrm{P}_{i,j} & \text{if} \quad (i,j) \notin \mathcal{A}\cup\mathcal{B} \\ \mathrm{P}_{i,j} - \varepsilon/2 & \text{if} \quad (i,j) \in \mathcal{A} \\ \mathrm{P}_{i,j} + \varepsilon/2 & \text{if} \quad (i,j) \in \mathcal{B} \end{cases}.$$

Because of the choice of ε, one has $0 \leqslant Q_{i,j}, R_{i,j} \leqslant 1$. Because each left-right edge in $\mathcal{A}$ is associated with a right-left edge in $\mathcal{B}$, (and the other way) the sum constraint on the row (and on the column) is maintained, so that $Q, R \in \mathcal{B}_n$. Finally, note that $\mathrm{P} = (Q+R)/2$. □

Corollary 1 (Kantorovich for matching)**.** *If $m = n$ and $\mathrm{a} = \mathrm{b} = \mathbb{1}_n$, then there exists an optimal solution for Problem* (19) *$\mathrm{P}_{\sigma^\star}$, which is a permutation matrix associated to an optimal permutation $\sigma^\star \in \mathrm{Perm}(n)$ for Problem* (1).

Proof. The feasible set of (19) is the compact convex polytope $\mathcal{B}_n$. Proposition 13 ensures that the linear objective has an optimal extreme point of $\mathcal{B}_n$. By Theorem 2, this extreme point is a permutation matrix $\mathrm{P}_{\sigma^\star}$. Since the cost of $\mathrm{P}_{\sigma^\star}$ is exactly $\sum_i C_{i,\sigma^\star(i)}$, the associated permutation solves the assignment problem. □

Remark 7 (General case).

3.2 Relaxation for Arbitrary Measures

Continuous couplings.

Definition 2 (Marginals of a joint measure)**.** *Let $\pi \in \mathcal{M}_+^1(\mathcal{X}\times\mathcal{Y})$ and let $P_1(x,y) = x$, $P_2(x,y) = y$ be the coordinate projections. The marginals of π are $\pi_1 := P_{1\sharp}\pi \in \mathcal{M}_+^1(\mathcal{X})$, $\pi_2 := P_{2\sharp}\pi \in \mathcal{M}_+^1(\mathcal{Y})$. Equivalently, for all bounded continuous test functions (f,g), $\int_{\mathcal{X}\times\mathcal{Y}} f(x)\mathrm{d}\pi(x,y) = \int_{\mathcal{X}} f\mathrm{d}\pi_1$, $\int_{\mathcal{X}\times\mathcal{Y}} g(y)\mathrm{d}\pi(x,y) = \int_{\mathcal{Y}} g\mathrm{d}\pi_2$.*

A useful mnemonic for the marginal constraint $\pi_1 = \alpha$ and $\pi_2 = \beta$, mimicking the discrete case where one sums along rows and columns, is
$\int_{\mathcal{Y}} \mathrm{d}\pi(x,y) = \mathrm{d}\alpha(x)$ and $\int_{\mathcal{X}} \mathrm{d}\pi(x,y) = \mathrm{d}\beta(y)$,

$$\forall\, (f,g) \in \mathcal{C}_b(\mathcal{X}) \times \mathcal{C}_b(\mathcal{Y}), \quad \int_{\mathcal{X}\times\mathcal{Y}} f(x)\mathrm{d}\pi(x,y) = \int_{\mathcal{X}} f\mathrm{d}\alpha \quad \text{and} \quad \int_{\mathcal{X}\times\mathcal{Y}} g(y)\mathrm{d}\pi(x,y) = \int_{\mathcal{Y}} g\mathrm{d}\beta.$$

$\forall\, A \subset \mathcal{X},\ B \subset \mathcal{Y}, \quad \pi(A\times\mathcal{Y}) = \alpha(A) \quad \text{and} \quad \pi(\mathcal{X}\times B) = \beta(B).$

Definition 3 (Couplings)**.** *Given $\alpha \in \mathcal{M}_+^1(\mathcal{X})$ and $\beta \in \mathcal{M}_+^1(\mathcal{Y})$, the set of couplings between α and β is*

$$\mathcal{U}(\alpha,\beta) \coloneqq \left\{\pi \in \mathcal{M}_+^1(\mathcal{X}\times\mathcal{Y})\ ;\ \pi_1 = \alpha \quad \text{and} \quad \pi_2 = \beta\right\}. \tag{21}$$

It is the continuous analogue of the transportation polytope (18).
Remark 8 (Probabilistic interpretation of couplings).
The set $\mathcal{U}(\alpha,\beta)$ is always non-empty because it contains at least the tensor product coupling $\alpha\otimes\beta$ defined by $\mathrm{d}(\alpha\otimes\beta)(x,y) = \mathrm{d}\alpha(x)\mathrm{d}\beta(y)$, i.e.

$$\forall\, h \in \mathcal{C}_b(\mathcal{X}\times\mathcal{Y}),\quad \begin{aligned} \int_{\mathcal{X}\times\mathcal{Y}} h(x,y)\mathrm{d}(\alpha\otimes\beta)(x,y) &= \int_{\mathcal{X}}\left(\int_{\mathcal{Y}} h(x,y)\mathrm{d}\beta(y)\right)\mathrm{d}\alpha(x) \\ &= \int_{\mathcal{Y}}\left(\int_{\mathcal{X}} h(x,y)\mathrm{d}\alpha(x)\right)\mathrm{d}\beta(y). \end{aligned}$$

$$\forall\, f \in \mathcal{C}_b(\mathcal{X}),\quad \int_{\mathcal{X}} f(x)\mathrm{d}(\alpha\otimes\beta)_1(x) = \int_{\mathcal{X}\times\mathcal{Y}} f(x)\mathrm{d}\alpha(x)\mathrm{d}\beta(y) = \int_{\mathcal{X}} f(x)\mathrm{d}\alpha(x)\int_{\mathcal{Y}}\mathrm{d}\beta = \int_{\mathcal{X}} f(x)\mathrm{d}\alpha(x)$$

A very different (concentrated) type of coupling is defined when there exists a map $T:\mathcal{X}\to\mathcal{Y}$ such that $T_\sharp\alpha = \beta$ (i.e. the constraint set of Monge's problem (6) is non-empty). This coupling is defined through the integrated definition of push-forward as
$\forall\, h \in \mathcal{C}_b(\mathcal{X}\times\mathcal{Y}),\quad \int_{\mathcal{X}\times\mathcal{Y}} h(x,y)\mathrm{d}\pi(x,y) = \int_{\mathcal{X}} h(x,T(x))\mathrm{d}\alpha$. A last important class of examples are semi-discrete problems (for instance to perform quantization or to fit statistical models), where α has a density with respect to Lebesgue and β is discrete.

Continuous Kantorovich problem.

$$\mathcal{L}_c(\alpha,\beta) \coloneqq \min_{\pi\in\mathcal{U}(\alpha,\beta)} \int_{\mathcal{X}\times\mathcal{Y}} c(x,y)\mathrm{d}\pi(x,y). \tag{22}$$

Remark 9 (Probabilistic interpretation of Kantorovich's problem).

$$\mathcal{L}_c(\alpha,\beta) = \min_{X\sim\alpha,\, Y\sim\beta} \mathbb{E}(c(X,Y)). \tag{23}$$

Proposition 14 (Existence on compact spaces)**.** *Assume that $\mathcal{X}$ and $\mathcal{Y}$ are compact metric spaces and that $c \in \mathcal{C}(\mathcal{X}\times\mathcal{Y})$. Then the Kantorovich problem (22) admits at least one minimizer.*

Proof. The constraint set is non-empty because it contains the product coupling $\alpha\otimes\beta$. It is also closed for the weak topology of measures because the marginal constraints are preserved under weak convergence. Since $\mathcal{X}\times\mathcal{Y}$ is compact, the set of probability measures on it is compact for the weak topology, and therefore $\mathcal{U}(\alpha,\beta)$ is compact. Finally, the functional $\pi \mapsto \int c\mathrm{d}\pi$ is weakly continuous because c is continuous and bounded. The minimum is thus attained. ☐

$\mathcal{P}_p(\mathcal{X}) = \left\{\mu \in \mathcal{M}_+^1(\mathcal{X})\ ;\ \int d(x,x_0)^p\mathrm{d}\mu(x) < +\infty\right\}$

Monge-Kantorovich equivalence.
Corollary 2 (Monge–Kantorovich equivalence under Brenier)**.** *We assume α has a density with respect to Lebesgue measure, and that $c(x,y) = \|x-y\|^2$. We denote by T the optimal transport solving Monge's formulation. Then $\pi = (\mathrm{Id},T)_\sharp\alpha$ is the unique optimal coupling solving the Kantorovich formulation. Monge and Kantorovich costs are the same.*

Proof. The proof of Brenier's theorem shows that the support of an optimal π is contained in the subdifferential $\partial\varphi$ of a convex function φ, which in general is a set-valued mapping.
When α does not have a density, then φ may be non-smooth, and non-smooth points where $\alpha(\{x\}) > 0$ lead to mass splitting. If α has a density, then this φ is differentiable α-almost everywhere. Denoting by $T = \nabla\varphi$ the unique optimal transport (which is a valid definition almost everywhere, and can be assigned arbitrary values at points of non-differentiability), the coupling

$$\pi = (\mathrm{Id},T)_\sharp\alpha \quad \text{i.e.} \quad \forall\, h \in \mathcal{C}(\mathcal{X}\times\mathcal{Y}),\quad \int_{\mathcal{X}\times\mathcal{Y}} h\mathrm{d}\pi = \int_{\mathcal{X}} h(x,T(x))\mathrm{d}\alpha(x)$$

is optimal. Since any optimal coupling is supported on this graph and T is unique α-almost everywhere, the optimal Kantorovich coupling is unique and the two costs coincide. ☐

In terms of random vectors, denoting (X,Y) a random vector with law π, it means that any such optimal random vector satisfies $Y = T(X)$ where $X\sim\alpha$ (and of course $T(X)\sim\beta$ by the marginal constraint).

3.3 c-cyclical monotonicity

Support and c-cyclical monotonicity.
Definition 4 (Support)**.** *For a Radon measure π on $\mathcal{X}\times\mathcal{Y}$* $\operatorname{supp}(\pi) \coloneqq \left\{(x,y)\ :\ \pi(U\times V) > 0 \text{ for every open } U \ni x,\ V \ni y\right\}$.
Definition 5 (c-cyclical monotonicity)**.** *A set $\Gamma \subset \mathcal{X}\times\mathcal{Y}$ is* c-cyclically monotone *if, for every $k \geq 2$, every finite family $\{(x_i,y_i)\}_{i=1}^k \subset \Gamma$, and every permutation $\sigma \in \mathfrak{S}_k$, $\sum_{i=1}^k c(x_i,y_i) \leq \sum_{i=1}^k c\big(x_i,y_{\sigma(i)}\big)$. Any permutation is a product of cycles, so it suffices to verify the inequality for cycles, $\sum_{i=1}^k c(x_i,y_i) \leq \sum_{i=1}^k c\big(x_i,y_{i+1}\big)$.*

Optimal matching to optimal transport. Let the marginals be uniform on n points: $\alpha = \frac{1}{n}\sum_{i=1}^n \delta_{x_i}$, $\beta = \frac{1}{n}\sum_{i=1}^n \delta_{y_i}$. Such a permutation defines, by optimality, a coupling with support $\Gamma := (x_i, y_{\sigma(i)})_i$ that is c-cyclically monotone.

Theorem 3 (Rockafellar)**.** *For any optimal plan π solving Kantorovich's problem,* $\mathrm{supp}(\pi)$ *is c-cyclically monotone.*

Proof. Suppose that $\mathrm{supp}(\pi)$ is not c-cyclically monotone. Then there exist points $(x_i, y_i)_{i=1}^k$ in the support and a permutation σ such that $\sum_i c(x_i, y_i) > \sum_i c(x_i, y_{\sigma(i)})$. By continuity of c, this strict inequality persists on small neighborhoods $U_i \times V_i$ of the points (x_i, y_i), chosen disjoint enough so that each has positive π-mass. Let $\eta > 0$ be smaller than all these masses. Removing mass η from each $U_i \times V_i$ and re-inserting the same mass from U_i to $V_{\sigma(i)}$ preserves both marginals, because the operation only permutes target neighborhoods among the same source neighborhoods. The cost strictly decreases by the displayed gap up to the continuity error. This contradicts optimality of π. □

Monotonicity. Assume the plan is induced by a measurable map $T : \mathcal{X} \to \mathcal{Y}$, i.e. $\pi = (\mathrm{Id}, T)_\# \alpha$. For any k points $x_1, \ldots, x_k$ in the domain, cyclical monotonicity reads
$\sum_{i=1}^k c\big(x_i, T(x_i)\big) \ \le\ \sum_{i=1}^k c\big(x_i, T(x_{i+1})\big), \qquad x_{k+1} = x_1$. For $c(x,y) = \frac{1}{2}|x-y|^2$, taking $k = 2$, shows that for any x, y,
$\langle T(x) - T(y),\, x - y\rangle \ \ge\ 0,$

One dimension. $|x - T(x)|^p + |y - T(y)|^p \ \le\ |x - T(y)|^p + |y - T(x)|^p$, which is equivalent to $T(x) \le T(y)$ whenever $x < y$. Thus T must be non-decreasing, recovering the classical monotone rearrangement.

3.4 Metric Properties: Wasserstein Distances

OT defines a distance.

Lemma 1 (Discrete gluing lemma)**.** *Given* $(\mathrm{a}, \mathrm{b}, \mathrm{c}) \in \Sigma_n \times \Sigma_p \times \Sigma_m$*, let* $\mathrm{P} \in \mathrm{U}(\mathrm{a}, \mathrm{b})$ *and* $\mathrm{Q} \in \mathrm{U}(\mathrm{b}, \mathrm{c})$*. Then there exists a 3-D tensor coupling* $\mathrm{S} \in \mathbb{R}_+^{n\times p\times m}$ *such that the 2-D marginals satisfy* $\sum_k \mathrm{S}_{i,j,k} = \mathrm{P}_{i,j}$ *and* $\sum_i \mathrm{S}_{i,j,k} = \mathrm{Q}_{j,k}$*. Note that this implies that the three 1-D marginals of* S *are* $(\mathrm{a}, \mathrm{b}, \mathrm{c})$*.*

Proof. One verifies that

$$\mathrm{S}_{i,j,k} = \begin{cases} \frac{\mathrm{P}_{i,j}\mathrm{Q}_{j,k}}{\mathrm{b}_j} & \text{if} \quad \mathrm{b}_j \neq 0 \\ 0 \text{ otherwise} \end{cases} \tag{24}$$

is acceptable. Indeed, if $\mathrm{b}_j \neq 0$ $\sum_k \mathrm{S}_{i,j,k} = \sum_k \frac{\mathrm{P}_{i,j}\mathrm{Q}_{j,k}}{\mathrm{b}_j} = \frac{\mathrm{P}_{i,j}}{\mathrm{b}_j}(\mathrm{Q}\mathbb{1}_m)_j = \frac{\mathrm{P}_{i,j}}{\mathrm{b}_j}\mathrm{b}_j$. If $\mathrm{b}_j = 0$, then necessarily $\mathrm{P}_{i,j} = 0$ and $\sum_k \mathrm{S}_{i,j,k} = 0 = \mathrm{P}_{i,j}$. □

Proposition 15 (Discrete Wasserstein distance)**.** *We suppose $n = m$, and that for some $p \geqslant 1$,* $\mathrm{C} = \mathrm{D}^p = (\mathrm{D}_{i,j}^p)_{i,j} \in \mathbb{R}^{n\times n}$ *where* $\mathrm{D} \in \mathbb{R}_+^{n\times n}$ *is a distance on* $[\![n]\!]$, i.e.

1. $\mathrm{D} \in \mathbb{R}_+^{n\times n}$ *is symmetric;*
2. $\mathrm{D}_{i,j} = 0$ *if and only if* $i = j$*;*
3. $\forall\, (i,j,k) \in [\![n]\!]^3, \mathrm{D}_{i,k} \leqslant \mathrm{D}_{i,j} + \mathrm{D}_{j,k}$.

Then

$$\mathrm{W}_p(\mathrm{a}, \mathrm{b}) := \mathrm{L}_{\mathrm{D}^p}(\mathrm{a}, \mathrm{b})^{1/p} \tag{25}$$

(note that W_p *depends on* D*) defines the p-Wasserstein distance on* Σ_n*, i.e.* W_p *is symmetric, positive,* $\mathrm{W}_p(\mathrm{a}, \mathrm{b}) = 0$ *if and only if* $\mathrm{a} = \mathrm{b}$*, and it satisfies the triangle inequality* $\forall\, \mathrm{a}, \mathrm{b}, \mathrm{c} \in \Sigma_n, \quad \mathrm{W}_p(\mathrm{a}, \mathrm{c}) \leqslant \mathrm{W}_p(\mathrm{a}, \mathrm{b}) + \mathrm{W}_p(\mathrm{b}, \mathrm{c})$.

Proof. For symmetry, since D^p is symmetric, we use the fact that if $\mathrm{P} \in \mathrm{U}(\mathrm{a}, \mathrm{b})$ is optimal for $\mathrm{W}_p(\mathrm{a}, \mathrm{b})$, then $\mathrm{P}^\top \in \mathrm{U}(\mathrm{b}, \mathrm{a})$ is optimal for $\mathrm{W}_p(\mathrm{b}, \mathrm{a})$. For definiteness, since $\mathrm{C} = \mathrm{D}^p$ has a null diagonal, $\mathrm{W}_p(\mathrm{a}, \mathrm{b}) = 0$ is achieved by the diagonal coupling $\mathrm{P}^\star = \mathrm{diag}(\mathrm{a}) = \mathrm{diag}(\mathrm{b})$ when $\mathrm{a} = \mathrm{b}$; by positivity of all off-diagonal elements of D^p, $\mathrm{W}_p(\mathrm{a}, \mathrm{b}) > 0$ whenever $\mathrm{a} \neq \mathrm{b}$ because any admissible coupling then has a nonzero element outside the diagonal.

To prove the triangle inequality of Wasserstein distances for arbitrary measures, we consider $\mathrm{a}, \mathrm{b}, \mathrm{c} \in \Sigma_n$, and let P and Q be two optimal solutions of the transport problems between a and b, and b and c respectively.

We use the gluing Lemma 1 which defines $\mathrm{S} \in \mathbb{R}_+^{n^3}$ with marginals $\sum_k \mathrm{S}_{\cdot,\cdot,k} = \mathrm{P}$ and $\sum_i \mathrm{S}_{i,\cdot,\cdot} = \mathrm{Q}$. We define $\mathrm{R} = \sum_j \mathrm{S}_{\cdot,j,\cdot}$, which is an element of $\mathrm{U}(\mathrm{a}, \mathrm{c})$.

$$\begin{tikzcd} a \arrow[r, "P"] \arrow[rr, bend right, "R"'] & b \arrow[r, "Q"] & c \end{tikzcd}$$

Note that if one assumes $\mathrm{b} > 0$ then $\mathrm{R} = \mathrm{P}\,\mathrm{diag}(1/\mathrm{b})\mathrm{Q}$.

The triangle inequality follows from

$$\begin{aligned}
\mathrm{W}_p(\mathrm{a},\mathrm{c}) &= \Big(\min_{\tilde{\mathrm{R}}\in\mathrm{U}(\mathrm{a},\mathrm{c})} \langle \tilde{\mathrm{R}},\, \mathrm{D}^p\rangle\Big)^{1/p} \leqslant \langle \mathrm{R},\, \mathrm{D}^p\rangle^{1/p} \\
&= \Big(\sum_{i,k} \mathrm{D}^p_{ik} \sum_j \mathrm{S}_{i,j,k}\Big)^{1/p} \leqslant \Big(\sum_{i,j,k} \Big(\mathrm{D}_{ij} + \mathrm{D}_{j,k}\Big)^p \mathrm{S}_{i,j,k}\Big)^{1/p} \\
&\leqslant \Big(\sum_{i,j,k} \mathrm{D}^p_{ij}\mathrm{S}_{i,j,k}\Big)^{1/p} + \Big(\sum_{i,j,k} \mathrm{D}^p_{j,k}\mathrm{S}_{i,j,k}\Big)^{1/p} \\
&= \Big(\sum_{i,j} \mathrm{D}^p_{i,j} \sum_k \mathrm{S}_{i,j,k}\Big)^{1/p} + \Big(\sum_{j,k} \mathrm{D}^p_{j,k} \sum_i \mathrm{S}_{i,j,k}\Big)^{1/p} \\
&= \Big(\sum_{i,j} \mathrm{D}^p_{i,j}\mathrm{P}_{i,j}\Big)^{1/p} + \Big(\sum_{j,k} \mathrm{D}^p_{j,k}\mathrm{Q}_{j,k}\Big)^{1/p} = \mathrm{W}_p(\mathrm{a},\mathrm{b}) + \mathrm{W}_p(\mathrm{b},\mathrm{c}).
\end{aligned}$$

The first inequality follows from the feasibility of R, the second is the usual triangle inequality for elements in D, and the third comes from Minkowski's inequality. □

Lemma 2 (Gluing lemma)**.** *Let $(\alpha,\beta,\gamma) \in \mathcal{M}_+^1(\mathcal{X}) \times \mathcal{M}_+^1(\mathcal{Y}) \times \mathcal{M}_+^1(\mathcal{Z})$ where $(\mathcal{X},\mathcal{Y},\mathcal{Z})$ are Polish spaces, i.e. separable topological spaces that can be metrized by complete distances. Given $\pi \in \mathcal{U}(\alpha,\beta)$ and $\xi \in \mathcal{U}(\beta,\gamma)$, then there exists a tensor coupling measure $\sigma \in \mathcal{M}_+(\mathcal{X}\times\mathcal{Y}\times\mathcal{Z})$ such that $(P_{\mathcal{X},\mathcal{Y}})_\sharp\sigma = \pi \quad$ and $\quad (P_{\mathcal{Y},\mathcal{Z}})_\sharp\sigma = \xi$ where we denoted the projector $P_{\mathcal{X},\mathcal{Y}}(x,y,z) = (x,y)$ and $P_{\mathcal{Y},\mathcal{Z}}(x,y,z) = (y,z)$.*

Proof. The proof of this fundamental result is involved since it requires using the disintegration of measure (which corresponds to conditional probabilities).
The disintegration of measures is applicable because the spaces are Polish.
We disintegrate π and ξ against β to obtain two families $(\pi_y)_{y\in\mathcal{Y}}$ and $(\xi_y)_{y\in\mathcal{Y}}$ of probability distributions on $\mathcal{X}$ and $\mathcal{Z}$. These families are defined by the fact that $\forall\, h \in \mathcal{C}(\mathcal{X}\times\mathcal{Y})$, $\quad \int_{\mathcal{Y}} \Big(\int_{\mathcal{X}} h(x,y)\mathrm{d}\pi_y(x)\Big)\mathrm{d}\beta(y) = \int h(x,y)\mathrm{d}\pi(x,y)$. and similarly for ξ.
When $\beta = \sum_j \mathrm{b}_j\delta_{y_j}$ and $\pi = \sum_{i,j} \mathrm{P}_{i,j}\delta_{(x_i,y_j)}$, then this conditional distribution is defined on the support of β as $\pi_{y_j} = \sum_i \frac{\mathrm{P}_{i,j}}{\mathrm{b}_j}\delta_{x_i}$ (and similarly for ξ).
The glued measure is then defined by the conditional-product formula $\forall\, g \in \mathcal{C}(\mathcal{X}\times\mathcal{Y}\times\mathcal{Z})$, $\int g(x,y,z)\mathrm{d}\sigma(x,y,z) = \int g(x,y,z)\mathrm{d}\pi_y(x)\mathrm{d}\xi_y(z)\mathrm{d}\beta(y)$. For discrete measures, this matches the definition (24), since $\sigma = \sum_{i,j,k} \mathrm{S}_{i,j,k}\delta_{x_i,y_j,z_k}$ where $\mathrm{S}_{i,j,k} = \frac{\mathrm{P}_{i,j}}{\mathrm{b}_j}\frac{\mathrm{Q}_{j,k}}{\mathrm{b}_j}\mathrm{b}_j$. □

Proposition 16 (Wasserstein distance on measures)**.** *We assume $\mathcal{X} = \mathcal{Y}$, and that for some $p \geqslant 1$, $c(x,y) = d(x,y)^p$ where d is a distance on $\mathcal{X}$,* i.e.
(i) $d(x,y) = d(y,x) \geqslant 0$;
(ii) $d(x,y) = 0$ if and only if $x = y$;
(iii) $\forall\,(x,y,z) \in \mathcal{X}^3, d(x,z) \leqslant d(x,y) + d(y,z)$.
Then

$$\mathcal{W}_p(\alpha,\beta) \coloneqq \mathcal{L}_{d^p}(\alpha,\beta)^{1/p} \tag{26}$$

(note that $\mathcal{W}_p$ depends on d) defines the p-Wasserstein distance on $\mathcal{X}$, i.e. *$\mathcal{W}_p$ is symmetric, positive, $\mathcal{W}_p(\alpha,\beta) = 0$ if and only if $\alpha = \beta$, and it satisfies the triangle inequality $\forall\,(\alpha,\beta,\gamma) \in \mathcal{M}_+^1(\mathcal{X})^3, \quad \mathcal{W}_p(\alpha,\gamma) \leqslant \mathcal{W}_p(\alpha,\beta) + \mathcal{W}_p(\beta,\gamma)$.*

Proof. The symmetry follows from the fact that since d is symmetric, if $\pi(x,y)$ is optimal for $\mathcal{L}_{d^p}(\alpha,\beta)$, then $\pi(y,x) \in \mathcal{U}(\beta,\alpha)$ is optimal for $\mathcal{L}_{d^p}(\beta,\alpha)$.
If $\mathcal{L}_{d^p}(\alpha,\beta) = 0$, then necessarily an optimal coupling π is supported on the diagonal $\Delta \coloneqq \{(x,x)\}_x \subset \mathcal{X}^2$.
We denote $\lambda(x)$ the corresponding measure on the diagonal, i.e. such that $\int h(x,y)\mathrm{d}\pi(x,y) = \int h(x,x)\mathrm{d}\lambda(x)$.
Then since $\pi \in \mathcal{U}(\alpha,\beta)$ necessarily $\lambda = \alpha$ and $\lambda = \beta$ so that $\alpha = \beta$.
For the triangle inequality, we consider optimal couplings $\pi \in \mathcal{U}(\alpha,\beta)$ and $\xi \in \mathcal{U}(\beta,\gamma)$ and we glue them according to the Lemma 2.
We define the composition of the two couplings (π,ξ) as $\rho \coloneqq (P_{\mathcal{X},\mathcal{Z}})_\sharp\sigma$.
Note that if π and ξ are couplings induced by two Monge maps $T_{\mathcal{X}}(x)$ and $T_{\mathcal{Y}}(y)$, then ρ is itself induced by the Monge map $T_{\mathcal{Y}} \circ T_{\mathcal{X}}$, so that this notion of composition of coupling generalizes the composition of maps.
The triangular inequality follows from

$$\begin{aligned}
\mathcal{W}_p(\alpha,\gamma) &\leqslant \Big(\int_{\mathcal{X}\times\mathcal{Z}} d(x,z)^p\mathrm{d}\rho(x,z)\Big)^{1/p} = \Big(\int_{\mathcal{X}\times\mathcal{Y}\times\mathcal{Z}} d(x,z)^p\mathrm{d}\sigma(x,y,z)\Big)^{1/p} \\
&\leqslant \Big(\int_{\mathcal{X}\times\mathcal{Y}\times\mathcal{Z}} (d(x,y) + d(y,z))^p\mathrm{d}\sigma(x,y,z)\Big)^{1/p} \\
&\leqslant \Big(\int_{\mathcal{X}\times\mathcal{Y}\times\mathcal{Z}} d(x,y)^p\mathrm{d}\sigma(x,y,z)\Big)^{1/p} + \Big(\int_{\mathcal{X}\times\mathcal{Y}\times\mathcal{Z}} d(y,z)^p\mathrm{d}\sigma(x,y,z)\Big)^{1/p} \\
&= \Big(\int_{\mathcal{X}\times\mathcal{Y}} d(x,y)^p\mathrm{d}\pi(x,y)\Big)^{1/p} + \Big(\int_{\mathcal{Y}\times\mathcal{Z}} d(y,z)^p\mathrm{d}\xi(y,z)\Big)^{1/p} = \mathcal{W}_p(\alpha,\beta) + \mathcal{W}_p(\beta,\gamma).
\end{aligned}$$

□

3.5 Metric Properties: Topology and Applications

Convergence in law topology. On a bounded metric space, all $\mathcal{W}_p$ distances define the same topology, although they are not equivalent as distances.

Proposition 17 (Equivalence of Wasserstein distances on compact spaces)**.** *One has for* $p \leqslant q$ $\mathcal{W}_p(\alpha,\beta) \leqslant \mathcal{W}_q(\alpha,\beta) \leqslant \mathrm{diam}(\mathcal{X})^{\frac{q-p}{q}} \mathcal{W}_p(\alpha,\beta)^{\frac{p}{q}}$ *where* $\mathrm{diam}(\mathcal{X}) \triangleq \sup_{x,y} d(x,y)$.

Proof. The left inequality follows from Jensen inequality, $\varphi(\int c(x,y)\mathrm{d}\pi(x,y)) \leqslant \int \varphi(c(x,y))\mathrm{d}\pi(x,y)$, applied to any probability distribution π and to the convex function $\varphi(r) = r^{q/p}$ with $c(x,y) = d(x,y)^p$, so that one gets $\left(\int d(x,y)^p \mathrm{d}\pi(x,y)\right)^{\frac{q}{p}} \leqslant \int d(x,y)^q \mathrm{d}\pi(x,y)$. The right inequality follows from $d(x,y)^q \leqslant \mathrm{diam}(\mathcal{X})^{q-p} d(x,y)^p$. □

Definition 6 (Weak* topology)**.** *$(\alpha_k)_k$ converges weakly* to α in $\mathcal{M}_+^1(\mathcal{X})$ (denoted $\alpha_k \rightharpoonup \alpha$) if and only if for any bounded continuous function $f \in \mathcal{C}_b(\mathcal{X})$, $\int_{\mathcal{X}} f \mathrm{d}\alpha_k \to \int_{\mathcal{X}} f \mathrm{d}\alpha$. On compact spaces, $\mathcal{C}_b(\mathcal{X}) = \mathcal{C}(\mathcal{X})$, which is why the boundedness is often left implicit there.*

Remark 10 (A Riemann-sum weak limit)*.*

$$\frac{1}{n}\sum_{k=1}^{n} \delta_{k/n} \rightharpoonup \mathcal{U}_{[0,1]}.$$

$\frac{1}{n}\sum_{k=1}^{n} f(k/n) \longrightarrow \int_0^1 f(x)\mathrm{d}x$,

Remark 11 (Weak convergence for discrete measures)*.*

In terms of random vectors, if $X_n \sim \alpha_n$ and $X \sim \alpha$ (not necessarily defined on the same probability space), weak convergence corresponds to convergence in law of X_n toward X.

Remark 12 (Central limit theorem)*.*

Definition 7 (Strong topology)**.** *The simplest distance on Radon measures is the total variation norm, which is the dual norm of the L^∞ norm on $\mathcal{C}(\mathcal{X})$ and whose topology is often called the "strong" topology $\|\alpha-\beta\|_{\mathrm{TV}} := \sup_{\|f\|_\infty \leqslant 1} \int f \mathrm{d}(\alpha-\beta) = |\alpha-\beta|(\mathcal{X})$ where $|\alpha-\beta|(\mathcal{X})$ is the mass of the absolute value of the difference measure. When $\alpha-\beta = \rho \mathrm{d}x$ has a density, then $\|\alpha-\beta\|_{\mathrm{TV}} = \int |\rho(x)|\mathrm{d}x = \|\rho\|_{L^1(\mathrm{d}x)}$ is the L^1 norm associated to $\mathrm{d}x$. When $\alpha-\beta = \sum_i u_i \delta_{z_i}$ is discrete, then $\|\alpha-\beta\|_{\mathrm{TV}} = \sum_i |u_i| = \|u\|_{\ell^1}$ is the discrete ℓ^1 norm.*

Proposition 18 (Total variation as Wasserstein for the discrete metric)**.** *Denoting d the 0/1 distance such that $d(x,x) = 0$ and $d(x,y) = 1$ if $x \neq y$, then $\mathcal{W}_p(\alpha,\beta)^p = \frac{1}{2}\|\alpha-\beta\|_{\mathrm{TV}}$.*

Proof. For the sake of simplicity, we do the proof for discrete measures with weights (a,b) and without loss of generality assume they have the same support $(x_i)_i$ and we denote $\mathrm{D} \triangleq (d(x_i,x_j))_{i,j}$ which is 0 on the diagonal and one outside.

Also since $d^p = d$ we consider $p = 1$. We denote $\mathrm{c}_i = \min(\mathrm{a}_i, \mathrm{b}_i)$. By conservation of mass, for every $\mathrm{P} \in \mathrm{U}(\mathrm{a},\mathrm{b})$, $\mathrm{P}_{i,i} \leqslant \mathrm{c}_i$, thus $\langle \mathrm{P}, \mathrm{D}\rangle = \sum_{i\neq j} \mathrm{P}_{i,j} = 1 - \sum_i \mathrm{P}_{i,i} \geqslant 1 - \sum_i \mathrm{c}_i$. We need to show that this bound is tight, namely to construct $\hat{P} \in \mathrm{U}(\mathrm{a},\mathrm{b})$ such that $\mathrm{diag}(\hat{P}) = \mathrm{c}$. Let $\bar{\mathrm{a}} \triangleq \mathrm{a} - \mathrm{c} = (\mathrm{a}-\mathrm{b})_+ \geqslant 0$ and $\bar{\mathrm{b}} \triangleq \mathrm{b} - \mathrm{c} = (\mathrm{b}-\mathrm{a})_+ \geqslant 0$ If $\bar{\mathrm{a}} = \bar{\mathrm{b}} = 0$, then $\mathrm{a} = \mathrm{b}$ and the diagonal coupling is optimal. Otherwise, one has $\frac{\bar{\mathrm{a}}\otimes\bar{\mathrm{b}}}{\langle \bar{\mathrm{a}}, \mathbb{1}\rangle} \in \mathrm{U}(\bar{\mathrm{a}},\bar{\mathrm{b}})$ and we remark that $\langle \bar{\mathrm{a}}, \mathbb{1}\rangle = \langle \bar{\mathrm{b}}, \mathbb{1}\rangle = 1 - \langle \mathrm{c}, \mathbb{1}\rangle$.

Thus denoting $\hat{\mathrm{P}} \triangleq \mathrm{diag}(\mathrm{c}) + \frac{\bar{\mathrm{a}}\otimes\bar{\mathrm{b}}}{\langle \bar{\mathrm{a}}, \mathbb{1}\rangle} \geqslant 0$ satisfies $\hat{\mathrm{P}}\mathbb{1} = \mathrm{c} + \bar{\mathrm{a}} = \mathrm{a}$ and $\hat{\mathrm{P}}^\top \mathbb{1} = \mathrm{c} + \bar{\mathrm{b}} = \mathrm{b}$ so that $\hat{P} \in \mathrm{U}(\mathrm{a},\mathrm{b})$ is a coupling so that $\mathrm{diag}(\hat{P}) = \mathrm{diag}(\mathrm{c})$ since $\mathrm{diag}(\bar{\mathrm{a}}\otimes\bar{\mathrm{b}}) = 0$. We thus conclude that

$$\mathrm{W}_1(\mathrm{a},\mathrm{b}) = \langle \mathrm{D}, \hat{\mathrm{P}}\rangle = \sum_{i,j} \frac{\bar{\mathrm{a}}_i \bar{\mathrm{b}}_j}{\langle \bar{\mathrm{a}}, \mathbb{1}\rangle} = \sum_i \bar{\mathrm{a}}_i = \sum_i \bar{\mathrm{b}}_i = \frac{1}{2}\sum_i (\bar{\mathrm{a}}_i + \bar{\mathrm{b}}_i) = \frac{1}{2}\|\mathrm{a}-\mathrm{b}\|_{\mathrm{TV}}.$$

□

$\|\delta_{x_n} - \delta_x\|_{\mathrm{TV}} = 2$ and $\mathcal{W}_p(\delta_{x_n}, \delta_x) = d(x_n, x)$.

Proposition 19 (Wasserstein metrizes weak convergence on compact spaces)**.** *If $\mathcal{X}$ is compact, $\alpha_k \rightharpoonup \alpha$ if and only if $\mathcal{W}_p(\alpha_k,\alpha) \to 0$.*

Proof. For $p = 1$, this is exactly Corollary 3, which follows from the Kantorovich–Rubinstein dual representation and compactness of the unit Lipschitz ball modulo constants. For a compact space, Proposition 17 shows that all Wasserstein distances $\mathcal{W}_p$ induce the same convergent sequences. Hence weak convergence is equivalent to convergence in $\mathcal{W}_p$ for every $p \geqslant 1$. □

$\int d(x,x_0)^p \mathrm{d}\alpha_k(x) \longrightarrow \int d(x,x_0)^p \mathrm{d}\alpha(x)$.

Proposition 20 (Comparison with total variation on discrete spaces)**.** *One has*

$$\frac{d_{\min}}{2}\|\alpha-\beta\|_{\mathrm{TV}} \leqslant \mathcal{W}_1(\alpha,\beta) \leqslant \frac{d_{\max}}{2}\|\alpha-\beta\|_{\mathrm{TV}} \quad \textit{where} \quad \begin{cases} d_{\min} := \inf_{x\neq y} d(x,y) \\ d_{\max} := \sup_{x,y} d(x,y) \end{cases}$$

Proof. We denote $d_0(x,y)$ the distance such that $d_0(x,x) = 0$ and $d_0(x,y) = 1$ for $x \neq y$. One has $d_{\min} d_0(x,y) \leqslant d(x,y) \leqslant d_{\max} d_0(x,y)$ so that integrating this against any $\pi \in \mathcal{U}(\alpha,\beta)$ and taking the minimum among those π gives the result using Proposition 18. □

This bound is sharp, as this can be observed by taking $\alpha = \delta_x$ and $\beta = \delta_y$, in which case the bound simply reads if $x \neq y$ $d_{\min} \leqslant d(x,y) \leqslant d_{\max}$.

Distributional robustness and $\mathcal{W}_\infty$. Wasserstein distances are also used to define ambiguity sets around an empirical law. A distributionally robust optimization (DRO) problem replaces the empirical risk $\int \ell_\theta \mathrm{d}\hat{\alpha}$ by $\sup_{\beta:\, \mathcal{W}_p(\beta,\hat{\alpha})\leqslant\rho} \int \ell_\theta \mathrm{d}\beta$, or, equivalently in penalized form, by $\sup_\beta \int \ell_\theta \mathrm{d}\beta - \lambda\, \mathcal{W}_p(\beta,\hat{\alpha})^p$. This asks for performance against all distributions that can be reached by transporting the empirical mass within a budget.

$$\mathcal{W}_\infty(\alpha,\beta) \coloneqq \inf_{\pi\in\mathcal{U}(\alpha,\beta)} \operatorname*{ess\,sup}_{(x,y)\sim\pi} d(x,y) \tag{27}$$

is the limit of $\mathcal{W}_p(\alpha,\beta)$ as $p\to\infty$ on bounded spaces, not the limit of the convex programs defining $\mathcal{W}_p^p$. It minimizes the worst displacement rather than an average displacement, and the resulting optimization is no longer a linear convex program because of the essential-supremum objective.

Theoretical application: quantitative central limit theorems.

Proposition 21 (Berry–Esseen bound in $\mathcal{W}_1$)**.** *Let $(X_i)_{i=1}^n$ be i.i.d. real random variables such that $\mathbb{E}X_i = 0$, $\mathbb{E}X_i^2 = 1$ and $\mathbb{E}|X_i|^3 < +\infty$. If α_n is the law of $n^{-1/2}\sum_i X_i$ and γ is the standard Gaussian law, then $\mathcal{W}_1(\alpha_n,\gamma) \leqslant \frac{C\,\mathbb{E}|X_1|^3}{\sqrt{n}}$, where C is a universal constant.*

Proof. By Kantorovich–Rubinstein duality,

$$\mathcal{W}_1(\alpha_n,\gamma) = \sup_{\mathrm{Lip}(h)\leqslant 1} |\mathbb{E}h(S_n) - \mathbb{E}h(G)|, \qquad S_n = n^{-1/2}\sum_i X_i,\ G\sim\gamma.$$

For each such h, solve Stein's equation $f_h'(x) - xf_h(x) = h(x) - \mathbb{E}h(G)$. The solution satisfies uniform derivative bounds depending only on the Lipschitz constant of h. Hence $\mathbb{E}h(S_n) - \mathbb{E}h(G) = \mathbb{E}[f_h'(S_n) - S_n f_h(S_n)]$. Expanding $f_h'(S_n)$ and $f_h(S_n)$ by replacing the summands one at a time, the first- and second-order terms cancel because $\mathbb{E}X_i = 0$ and $\mathbb{E}X_i^2 = 1$. The Taylor remainder is bounded by $C\sum_i \mathbb{E}|X_i/\sqrt{n}|^3$, which gives the displayed $n^{-1/2}$ rate. This is the Wasserstein form of the Berry–Esseen theorem; sharper constants and higher-order transport-distance refinements are studied in. □

Applications and implications.

4 Sinkhorn

4.1 Entropic Regularization for Discrete Measures

Entropic Regularization for Discrete Measures. $\mathrm{H}(\mathrm{P}) \triangleq -\sum_{i,j} \mathrm{P}_{i,j}\log(\mathrm{P}_{i,j})$,

$$\mathrm{L}_{\mathrm{C}}^{\varepsilon}(\mathrm{a},\mathrm{b}) \coloneqq \min_{\mathrm{P}\in\mathrm{U}(\mathrm{a},\mathrm{b})} \langle \mathrm{P},\,\mathrm{C}\rangle - \varepsilon\mathrm{H}(\mathrm{P}). \tag{28}$$

Proposition 22 (Existence and uniqueness of entropic OT)**.** *Assume that* a,b *are probability histograms and that* C *is finite. For every $\varepsilon > 0$, problem* (28) *admits a unique minimizer. If all entries of* a *and* b *are positive, then this minimizer is positive on every entry.*

Proof. The transport polytope $\mathrm{U}(\mathrm{a},\mathrm{b})$ is non-empty and compact, and the objective is continuous on it with the convention $0\log 0 = 0$, so a minimizer exists. On the relative interior of the polytope, $-\partial^2\mathrm{H}(\mathrm{P}) = \mathrm{diag}(1/\mathrm{P}_{i,j})$ is positive definite on every non-zero feasible direction. Hence $-\mathrm{H}$ is strictly convex on the polytope and $\langle \mathrm{P},\,\mathrm{C}\rangle - \varepsilon\mathrm{H}(\mathrm{P})$ is strictly convex, which implies uniqueness.
If $\mathrm{a}_i,\mathrm{b}_j > 0$ and a minimizer had $\mathrm{P}_{i,j} = 0$, then for small $t > 0$ the perturbation $\mathrm{P}_t = (1-t)\mathrm{P} + t\,\mathrm{a}\otimes\mathrm{b}$ remains feasible. The directional derivative of the entropic part at $t = 0$ is $-\infty$ because the derivative of $r\log r$ at 0 is $-\infty$ along a positive direction. Thus the objective decreases for small t, contradicting optimality. Therefore the minimizer is strictly positive. □

Smoothing effect.

4.2 Sinkhorn's Algorithm

Proposition 23 (Scaling form of entropic OT)**.** P *is the unique solution to* (28) *if and only if there exists* $(\mathrm{u},\mathrm{v}) \in \mathbb{R}_+^n \times \mathbb{R}_+^m$ *such that*

$$\forall\,(i,j)\in[\![n]\!]\times[\![m]\!], \quad \mathrm{P}_{i,j} = \mathrm{u}_i\mathrm{K}_{i,j}\mathrm{v}_j \quad \textit{where} \quad \mathrm{K}_{i,j} \coloneqq e^{-\frac{\mathrm{C}_{i,j}}{\varepsilon}}, \tag{29}$$

and $\mathrm{P}\in\mathcal{U}(\mathrm{a},\mathrm{b})$.

Proof. Without loss of generality, we assume $\mathrm{a}_i,\mathrm{b}_j > 0$ (otherwise, rows or columns with zero mass are fixed to zero and can be removed). By Proposition 22, the minimizer is strictly positive on the remaining support.
We can thus ignore the positivity constraint when introducing two dual variables $\mathrm{f}\in\mathbb{R}^n, \mathrm{g}\in\mathbb{R}^m$ for each marginal constraint so that the Lagrangian of (28) reads

$$\mathcal{E}(\mathrm{P},\mathrm{f},\mathrm{g}) = \langle \mathrm{P},\,\mathrm{C}\rangle + \varepsilon\sum_{i,j}\mathrm{P}_{i,j}\log(\mathrm{P}_{i,j}) + \langle \mathrm{f},\,\mathrm{a} - \mathrm{P}\mathbb{1}_m\rangle + \langle \mathrm{g},\,\mathrm{b} - \mathrm{P}^\top\mathbb{1}_n\rangle.$$

Considering first-order conditions (where we ignore the positivity constraint as explained above), we have

$$\frac{\partial \mathcal{E}(\mathrm{P}, \mathrm{f}, \mathrm{g})}{\partial \mathrm{P}_{i,j}} = \mathrm{C}_{i,j} + \varepsilon(\log(\mathrm{P}_{i,j}) + 1) - \mathrm{f}_i - \mathrm{g}_j = 0.$$

which results, in an optimal P coupling of the regularized problem, in the expression $\mathrm{P}_{i,j} = e^{\frac{\mathrm{f}_i + \mathrm{g}_j - \mathrm{C}_{i,j}}{\varepsilon} - 1}$ which can be rewritten in the form provided in the proposition using non-negative vectors $\mathrm{u}_i := e^{\mathrm{f}_i/\varepsilon - 1}$ and $\mathrm{v}_j := e^{\mathrm{g}_j/\varepsilon}$. □

$$\mathrm{diag}(\mathrm{u})\mathrm{K}\,\mathrm{diag}(\mathrm{v})\mathbb{1}_m = \mathrm{a}, \quad \text{and} \quad \mathrm{diag}(\mathrm{v})\mathrm{K}^\top \mathrm{diag}(\mathrm{u})\mathbb{1}_n = \mathrm{b}, \tag{30}$$

$$\mathrm{u} \odot (\mathrm{K}\mathrm{v}) = \mathrm{a} \quad \text{and} \quad \mathrm{v} \odot (\mathrm{K}^\top \mathrm{u}) = \mathrm{b} \tag{31}$$

where $\odot$ corresponds to the entry-wise multiplication of vectors.
An intuitive way to try to solve these equations is to solve them iteratively, by modifying first u so that it satisfies the left-hand side of Equation (31) and then v to satisfy its right-hand side. These two updates define Sinkhorn's algorithm

$$\mathrm{u}^{(\ell+1)} := \frac{\mathrm{a}}{\mathrm{K}\mathrm{v}^{(\ell)}} \quad \text{and} \quad \mathrm{v}^{(\ell+1)} := \frac{\mathrm{b}}{\mathrm{K}^\top \mathrm{u}^{(\ell+1)}}, \tag{32}$$

initialized with an arbitrary positive vector, for instance $\mathrm{v}^{(0)} = \mathbb{1}_m$. The division operator used above between two vectors is to be understood entry-wise.
A chief computational advantage of Sinkhorn's algorithm, besides its simplicity, is that the only expensive step is multiplication by the Gibbs kernel. Its complexity therefore scales like Cnm, where C is the number of Sinkhorn iterations.

4.3 Reformulation using relative entropy

$$\mathrm{KL}(\mathrm{P}|\mathrm{Q}) := \sum_{i,j} \mathrm{P}_{i,j} \log\left(\frac{\mathrm{P}_{i,j}}{\mathrm{Q}_{i,j}}\right) - \mathrm{P}_{i,j} + \mathrm{Q}_{i,j}. \tag{33}$$

$\mathrm{KL}(\mathrm{P}|\mathrm{Q}) = \sum_{i,j} \mathrm{P}_{i,j} \log\left(\frac{\mathrm{P}_{i,j}}{\mathrm{Q}_{i,j}}\right)$. $\mathrm{H}(\mathrm{P}) = -\mathrm{KL}(\mathrm{P}|\mathbb{1}_{n\times m})$. KL is a particular instance of both a φ-divergence (as defined in Section 7.2) and a Bregman divergence; up to standard affine rescalings, it is the canonical overlap between these two families. This special property is at the heart of the fact that this regularization leads to elegant algorithms and a tractable mathematical analysis.

Proposition 24 (Relative entropy is distance-like)**.** *Let $\mathrm{P}, \mathrm{Q} \in \mathbb{R}_+^{n\times m}$ have the same total mass and assume $\mathrm{Q}_{i,j} > 0$ on the support of* P*. Then $\mathrm{KL}(\mathrm{P}|\mathrm{Q}) \geqslant 0$, with equality if and only if $\mathrm{P} = \mathrm{Q}$.*

Proof. Write $\varphi(s) = s\log s - s + 1$. Convexity gives $\varphi(s) \geqslant \varphi(1) + \varphi'(1)(s-1) = 0$, and strict convexity gives equality only at $s = 1$. Hence $\mathrm{KL}(\mathrm{P}|\mathrm{Q}) = \sum_{i,j} \mathrm{Q}_{i,j}\varphi(\mathrm{P}_{i,j}/\mathrm{Q}_{i,j}) \geqslant 0$, with equality only when $\mathrm{P}_{i,j}/\mathrm{Q}_{i,j} = 1$ for all entries with $\mathrm{Q}_{i,j} > 0$. The support convention rules out positive P where $\mathrm{Q} = 0$, so equality is equivalent to $\mathrm{P} = \mathrm{Q}$. □

Equivalently, when P and Q have the same total mass, it reads
$\mathrm{KL}(\mathrm{P}|\mathrm{Q}) = \sum_{i,j} \varphi(\mathrm{P}_{i,j}/\mathrm{Q}_{i,j})\mathrm{Q}_{i,j}$. where $\varphi(s) = s\log(s)$. For any convex φ such that $\varphi(1) = 0$, one has indeed by Jensen $\sum_{i,j} \varphi(\mathrm{P}_{i,j}/\mathrm{Q}_{i,j})\mathrm{Q}_{i,j} \geqslant \varphi(\sum_{i,j} \mathrm{P}_{i,j}/\mathrm{Q}_{i,j}\mathrm{Q}_{i,j}) = \varphi(\sum_{i,j} \mathrm{P}_{i,j}) = \varphi(1) = 0$.

$$\min_{\mathrm{P}\in\mathrm{U}(\mathrm{a},\mathrm{b})} \langle \mathrm{P},\, \mathrm{C}\rangle + \varepsilon \mathrm{KL}(\mathrm{P}|\mathrm{a}\otimes\mathrm{b}). \tag{34}$$

Proposition 25 (Reference measure shift for KL)**.** *For $\mathrm{P} \in \mathrm{U}(\mathrm{a}, \mathrm{b})$, one has $\mathrm{KL}(\mathrm{P}|\mathrm{a}\otimes\mathrm{b}) = \mathrm{KL}(\mathrm{P}|\mathrm{a}'\otimes\mathrm{b}') - \mathrm{KL}(\mathrm{a}|\mathrm{a}') - \mathrm{KL}(\mathrm{b}|\mathrm{b}')$. In particular,* (34) *and* (28) *have the same unique minimizer.*

Proof. This follows from

$$\sum_{i,j} \mathrm{P}_{i,j} \log \frac{\mathrm{P}_{i,j}}{\mathrm{a}_i \mathrm{b}_j} = \sum_{i,j} \mathrm{P}_{i,j} \log \frac{\mathrm{P}_{i,j}}{\mathrm{a}'_i \mathrm{b}'_j} + \sum_{i,j} \mathrm{P}_{i,j} \left(\log \frac{\mathrm{a}'_i}{\mathrm{a}_i} + \log \frac{\mathrm{b}'_j}{\mathrm{b}_j}\right).$$

□

Proposition 26 (Convergence with ε)**.** *Assume, after removing zero-mass rows and columns, that* a *and* b *are positive and that* C *is finite. The unique solution P_ε of* (28) *converges to the optimal solution with maximal entropy within the set of all optimal solutions of the Kantorovich problem, namely*

$$\mathrm{P}_\varepsilon \overset{\varepsilon\to 0}{\longrightarrow} \underset{\mathrm{P}}{\mathrm{argmin}} \; \{-\mathrm{H}(\mathrm{P})\;;\; \mathrm{P} \in \mathrm{U}(\mathrm{a}, \mathrm{b}), \langle \mathrm{P},\, \mathrm{C}\rangle = \mathrm{L}_\mathrm{C}(\mathrm{a}, \mathrm{b})\} \tag{35}$$

so that in particular

$$\mathrm{L}^\varepsilon_\mathrm{C}(\mathrm{a}, \mathrm{b}) \overset{\varepsilon\to 0}{\longrightarrow} \mathrm{L}_\mathrm{C}(\mathrm{a}, \mathrm{b}).$$

One has

$$\mathrm{P}_\varepsilon \overset{\varepsilon\to\infty}{\longrightarrow} \mathrm{a}\otimes\mathrm{b}. \tag{36}$$

Proof. **Case $\varepsilon \to 0$.** We denote P_ℓ the solution of (28) for $\varepsilon = \varepsilon_\ell$.
Since $\mathrm{U}(\mathrm{a}, \mathrm{b})$ is bounded, we can extract a sequence (that we do not relabel for the sake of simplicity) such that $\mathrm{P}_\ell \to \mathrm{P}^\star$. Since $\mathrm{U}(\mathrm{a}, \mathrm{b})$ is closed, $\mathrm{P}^\star \in \mathrm{U}(\mathrm{a}, \mathrm{b})$. Using the equivalent KL-normalized formulation (34), optimality of P and P_ℓ for their respective optimization problems (for $\varepsilon = 0$ and $\varepsilon = \varepsilon_\ell$) gives

$$0 \leqslant \langle \mathrm{C},\, \mathrm{P}_\ell \rangle - \langle \mathrm{C},\, \mathrm{P} \rangle \leqslant \varepsilon_\ell (\mathrm{KL}(\mathrm{P}|\mathrm{a} \otimes \mathrm{b}) - \mathrm{KL}(\mathrm{P}_\ell|\mathrm{a} \otimes \mathrm{b})). \tag{37}$$

Since KL is continuous, taking the limit $\ell \to +\infty$ in this expression shows that $\langle \mathrm{C},\, \mathrm{P}^\star \rangle = \langle \mathrm{C},\, \mathrm{P} \rangle$ so that $\mathrm{P}^\star$ is a feasible point of (35). Furthermore, dividing by ε_ℓ in (37) and taking the limit shows that $\mathrm{KL}(\mathrm{P}^\star|\mathrm{a} \otimes \mathrm{b}) \leqslant \mathrm{KL}(\mathrm{P}|\mathrm{a} \otimes \mathrm{b})$, which shows that $\mathrm{P}^\star$ is a solution of (35). Since the solution $\mathrm{P}_0^\star$ to this program is unique by strict convexity of $\mathrm{KL}(\cdot|\mathrm{a} \otimes \mathrm{b})$ on the optimal face, one has $\mathrm{P}^\star = \mathrm{P}_0^\star$, and the whole sequence is converging.
Case $\varepsilon \to +\infty$. Evaluating at $\mathrm{a} \otimes \mathrm{b}$ (which belongs to the constraint set $\mathrm{U}(\mathrm{a}, \mathrm{b})$) the energy, one has $\langle \mathrm{C},\, \mathrm{P}_\varepsilon \rangle + \varepsilon \mathrm{KL}(\mathrm{P}_\varepsilon|\mathrm{a} \otimes \mathrm{b}) \leqslant \langle \mathrm{C},\, \mathrm{a} \otimes \mathrm{b} \rangle + \varepsilon \times 0$ and since $\langle \mathrm{C},\, \mathrm{P}_\varepsilon \rangle \geqslant 0$, this leads to $\mathrm{KL}(\mathrm{P}_\varepsilon|\mathrm{a} \otimes \mathrm{b}) \leqslant \varepsilon^{-1} \langle \mathrm{C},\, \mathrm{a} \otimes \mathrm{b} \rangle \leqslant \frac{\|\mathrm{C}\|_\infty}{\varepsilon}$ so that $\mathrm{KL}(\mathrm{P}_\varepsilon|\mathrm{a} \otimes \mathrm{b}) \to 0$ and thus $\mathrm{P}_\varepsilon \to \mathrm{a} \otimes \mathrm{b}$ since KL is a valid divergence. □

4.4 General Formulation

$$\mathcal{L}_c^\varepsilon(\alpha, \beta) \coloneqq \min_{\pi \in \mathcal{U}(\alpha, \beta)} \int_{\mathcal{X} \times \mathcal{Y}} c(x, y) \mathrm{d}\pi(x, y) + \varepsilon \, \mathrm{KL}(\pi|\alpha \otimes \beta) \tag{38}$$

where the relative entropy is a generalization of the discrete Kullback-Leibler divergence (33)

$$\mathrm{KL}(\pi|\xi) \coloneqq \int_{\mathcal{X} \times \mathcal{Y}} \log\Big(\frac{\mathrm{d}\pi}{\mathrm{d}\xi}(x, y)\Big) \mathrm{d}\pi(x, y) + \int_{\mathcal{X} \times \mathcal{Y}} (\mathrm{d}\xi(x, y) - \mathrm{d}\pi(x, y)), \tag{39}$$

Remark 13 (Probabilistic interpretation). If $(X, Y) \sim \pi$ have marginals $X \sim \alpha$ and $Y \sim \beta$, then $\mathrm{KL}(\pi|\alpha \otimes \beta) = \mathcal{I}(X, Y)$ is the mutual information of the couple, which is 0 if and only if X and Y are independent. The entropic problem (38) is thus equivalent to
$\min_{(X,Y), X \sim \alpha, Y \sim \beta} \mathbb{E}(c(X, Y)) + \varepsilon \mathcal{I}(X, Y).$

4.5 Convergence of Sinkhorn

Alternating KL projections.
Definition 8 (Bregman divergence)**.** *Let Φ be a differentiable strictly convex function on a convex domain Ω. The Bregman divergence generated by Φ is $D_\Phi(P|Q) \coloneqq \Phi(P) - \Phi(Q) - \langle \nabla\Phi(Q),\, P - Q \rangle$. For the negative entropy $\Phi(P) = \sum_{i,j} P_{i,j} \log P_{i,j}$ on the positive orthant, one obtains $D_\Phi(P|Q) = \mathrm{KL}(P|Q)$ up to the harmless convention at the boundary.*
Proposition 27 (Gibbs-KL reformulation)**.** *One has $\langle \mathrm{P},\, \mathrm{C} \rangle + \varepsilon \mathrm{KL}(\mathrm{P}|\mathrm{a} \otimes \mathrm{b}) = \varepsilon \mathrm{KL}(\mathrm{P}|\mathrm{K}) + cst$,*

Proof. The objective is indeed equal to

$$\varepsilon \sum_{i,j} \mathrm{P}_{i,j} \frac{\mathrm{C}_{i,j}}{\varepsilon} + \log\left(\frac{\mathrm{P}_{i,j}}{\mathrm{a}_i \mathrm{b}_j}\right) \mathrm{P}_{i,j} = \varepsilon \sum_{i,j} \mathrm{P}_{i,j} \log\left(\frac{\mathrm{P}_{i,j}}{\mathrm{a}_i \mathrm{b}_j e^{-\mathrm{C}_{i,j}/\varepsilon}}\right) = \varepsilon \sum_{i,j} \mathrm{P}_{i,j} \log\left(\frac{\mathrm{P}_{i,j}}{\mathrm{K}_{i,j}}\right) + \mathrm{cst}.$$

□

$$\mathrm{P}_\varepsilon = \mathrm{Proj}^{\mathrm{KL}}_{\mathrm{U}(\mathrm{a},\mathrm{b})}(\mathrm{K}) \coloneqq \underset{\mathrm{P} \in \mathrm{U}(\mathrm{a},\mathrm{b})}{\mathrm{argmin}} \; \mathrm{KL}(\mathrm{P}|\mathrm{K}). \tag{40}$$

Proposition 28 (Cyclic KL projections on affine constraints)**.** *Let $\mathcal{C}_1, \mathcal{C}_2$ be two affine subspaces whose intersection with the positive orthant is non-empty. Define $P_{k+1} = \mathrm{Proj}^{\mathrm{KL}}_{\mathcal{C}_2} \mathrm{Proj}^{\mathrm{KL}}_{\mathcal{C}_1}(P_k)$, starting from a positive P_0. If the KL projections are well-defined, then P_k converges to the KL projection of P_0 onto $\mathcal{C}_1 \cap \mathcal{C}_2$.*

Proof. For an affine set $\mathcal{C}$, the KL projection $P^+ = \mathrm{Proj}^{\mathrm{KL}}_{\mathcal{C}}(P)$ satisfies the Pythagorean identity $\mathrm{KL}(Q|P) = \mathrm{KL}(Q|P^+) + \mathrm{KL}(P^+|P) \quad \forall Q \in \mathcal{C}$. This follows by writing the first-order optimality condition: $\nabla\Phi(P^+) - \nabla\Phi(P)$ is orthogonal to the tangent space of $\mathcal{C}$, so the usual three-point identity for Bregman divergences has no cross term. Applying this identity successively with any $Q \in \mathcal{C}_1 \cap \mathcal{C}_2$ shows that $\mathrm{KL}(Q|P_k)$ decreases and that the projection drops $\mathrm{KL}(P_{k+1}|P_k)$ are summable. Hence the sequence is bounded in KL geometry and every cluster point lies in both affine sets. The Pythagorean identity then identifies any cluster point as the unique KL projection onto $\mathcal{C}_1 \cap \mathcal{C}_2$, by strict convexity. Thus the whole sequence converges. □

$\mathcal{C}^1_{\mathrm{a}} \coloneqq \{\mathrm{P} \,;\, \mathrm{P}\mathbb{1}_m = \mathrm{a}\}$ and $\mathcal{C}^2_{\mathrm{b}} \coloneqq \{\mathrm{P} \,;\, \mathrm{P}^\top \mathbb{1}_n = \mathrm{b}\}$

$$\mathrm{P}^{(\ell+1)} \coloneqq \mathrm{Proj}^{\mathrm{KL}}_{\mathcal{C}^1_{\mathrm{a}}}(\mathrm{P}^{(\ell)}) \quad \text{and} \quad \mathrm{P}^{(\ell+2)} \coloneqq \mathrm{Proj}^{\mathrm{KL}}_{\mathcal{C}^2_{\mathrm{b}}}(\mathrm{P}^{(\ell+1)}). \tag{41}$$

Proposition 29 (KL projections are scalings)**.** *One has $\mathrm{Proj}^{\mathrm{KL}}_{\mathcal{C}^1_{\mathrm{a}}}(\mathrm{P}) = \mathrm{diag}\left(\frac{\mathrm{a}}{\mathrm{P}\mathbb{1}_m}\right)\mathrm{P}$ and $\mathrm{Proj}^{\mathrm{KL}}_{\mathcal{C}^2_{\mathrm{b}}}(\mathrm{P}) = \mathrm{P}\,\mathrm{diag}\left(\frac{\mathrm{b}}{\mathrm{P}^\top \mathbb{1}_n}\right)$.*

Proof. Consider the problem along each row or column vector to impose a fixed sum $s \in \mathbb{R}_+$ $\min_p \{\mathrm{KL}(p|q) \,;\, \langle p,\, \mathbb{1} \rangle = s\}$. The Lagrange multiplier equation for this problem reads $\log(p/q) + \lambda\mathbb{1} = 0 \implies p = uq$ where $u = e^{-\lambda} > 0$. The constraint $\langle p,\, \mathbb{1} \rangle = s$ is equivalent to $\langle uq,\, \mathbb{1} \rangle = s$, i.e. $u = s/\sum_i q_i$, which gives the desired scaling formula $p = sq/\sum_i q_i$. □

$$\mathrm{P}^{(2\ell)} \coloneqq \mathrm{diag}(\mathrm{u}^{(\ell)})\mathrm{K}\,\mathrm{diag}(\mathrm{v}^{(\ell)}),$$

$$\mathrm{P}^{(2\ell+1)} \coloneqq \mathrm{diag}(\mathrm{u}^{(\ell+1)})\mathrm{K}\,\mathrm{diag}(\mathrm{v}^{(\ell)})$$
$$\text{and}\quad \mathrm{P}^{(2\ell+2)} \coloneqq \mathrm{diag}(\mathrm{u}^{(\ell+1)})\mathrm{K}\,\mathrm{diag}(\mathrm{v}^{(\ell+1)})$$

Convergence for the Hilbert metric.

$$\forall\,(\mathrm{u},\mathrm{u}') \in (\mathbb{R}^n_{+,*})^2, \quad d_{\mathcal{H}}(\mathrm{u},\mathrm{u}') \coloneqq \|\log(\mathrm{u}) - \log(\mathrm{u}')\|_V \tag{42}$$

$\|z\|_V = \max(z) - \min(z)$.

Proposition 30 (Hilbert metric on the projective cone). *The function $d_{\mathcal{H}}$ defines a complete distance on the projective cone $\mathbb{R}^n_{+,*}/\sim$, where $\mathrm{u} \sim \mathrm{u}'$ means that $\mathrm{u} = s\mathrm{u}'$ for some $s > 0$.*

Proof. The map $\mathrm{u} \mapsto \log \mathrm{u}$ identifies $\mathbb{R}^n_{+,*}/\sim$ with the quotient vector space $\mathbb{R}^n/\,\mathrm{Span}(\mathbb{1}_n)$, because multiplying u by $s > 0$ adds the constant vector $\log(s)\mathbb{1}_n$. The variation semi-norm $\|z\|_V = \max_i z_i - \min_i z_i$ vanishes exactly on constant vectors, so it induces a norm on this quotient. Symmetry, the triangle inequality and separation therefore follow from the corresponding norm properties. Completeness follows because $\mathbb{R}^n/\,\mathrm{Span}(\mathbb{1}_n)$ is finite-dimensional and all finite-dimensional normed spaces are complete. □

Theorem 4 (Birkhoff contraction theorem). *Let $\mathrm{K} \in \mathbb{R}^{n\times m}_{+,*}$, then for $(\mathrm{v},\mathrm{v}') \in (\mathbb{R}^m_{+,*})^2$*

$$d_{\mathcal{H}}(\mathrm{Kv},\mathrm{Kv}') \leqslant \lambda(\mathrm{K}) d_{\mathcal{H}}(\mathrm{v},\mathrm{v}') \text{ where } \begin{cases} \lambda(\mathrm{K}) \coloneqq \frac{\sqrt{\eta(\mathrm{K})}-1}{\sqrt{\eta(\mathrm{K})}+1} < 1 \\ \eta(\mathrm{K}) \coloneqq \max\limits_{i,j,k,\ell} \frac{\mathrm{K}_{i,k}\mathrm{K}_{j,\ell}}{\mathrm{K}_{j,k}\mathrm{K}_{i,\ell}}. \end{cases}$$

Proof. For a positive linear map A on a cone, define its projective diameter $\Delta(A) \coloneqq \sup_{u,v>0} d_{\mathcal{H}}(Au, Av)$. Then $d_{\mathcal{H}}(Au, Av) \leqslant \tanh(\Delta(A)/4) d_{\mathcal{H}}(u, v)$. Indeed, after quotienting by positive scalings, write $r_k = u_k/v_k$ and normalize so that $e^{-h/2} \leqslant r_k \leqslant e^{h/2}$, where $h = d_{\mathcal{H}}(u, v)$. The ratio between two coordinates of Au/Av is a quotient of two weighted averages of the numbers r_k. A two-point extremal argument shows that the largest possible contraction is obtained when the mass of the two weights is placed on the two endpoints $e^{-h/2}$ and $e^{h/2}$; the cross-ratio bound defining $\Delta(A)$ then gives

$$d_{\mathcal{H}}(Au, Av) \leqslant 2\log \frac{e^{\Delta(A)/4}e^{h/2} + e^{-\Delta(A)/4}e^{-h/2}}{e^{\Delta(A)/4}e^{-h/2} + e^{-\Delta(A)/4}e^{h/2}} \leqslant \tanh(\Delta(A)/4)h.$$

For the matrix K, its projective diameter is $\Delta(\mathrm{K}) = \log \eta(\mathrm{K})$, $\eta(\mathrm{K}) = \max_{i,j,k,\ell} \frac{\mathrm{K}_{i,k}\mathrm{K}_{j,\ell}}{\mathrm{K}_{j,k}\mathrm{K}_{i,\ell}}$. Therefore $\tanh(\Delta(\mathrm{K})/4) = (\sqrt{\eta(\mathrm{K})} - 1)/(\sqrt{\eta(\mathrm{K})} + 1)$, which is the claimed contraction factor. □

Theorem 5 (Linear convergence of Sinkhorn). *One has $(\mathrm{u}^{(\ell)}, \mathrm{v}^{(\ell)}) \to (\mathrm{u}^\star, \mathrm{v}^\star)$ and*

$$d_{\mathcal{H}}(\mathrm{u}^{(\ell)}, \mathrm{u}^\star) = O(\lambda(\mathrm{K})^{2\ell}), \quad d_{\mathcal{H}}(\mathrm{v}^{(\ell)}, \mathrm{v}^\star) = O(\lambda(\mathrm{K})^{2\ell}). \tag{43}$$

One also has

$$d_{\mathcal{H}}(\mathrm{u}^{(\ell)}, \mathrm{u}^\star) \leqslant \frac{d_{\mathcal{H}}(\mathrm{P}^{(\ell)}\mathbb{1}_m, \mathrm{a})}{1-\lambda(\mathrm{K})} \quad \text{and} \quad d_{\mathcal{H}}(\mathrm{v}^{(\ell)}, \mathrm{v}^\star) \leqslant \frac{d_{\mathcal{H}}(\mathrm{P}^{(\ell),\top}\mathbb{1}_n, \mathrm{b})}{1-\lambda(\mathrm{K})}, \tag{44}$$

where we denoted $\mathrm{P}^{(\ell)} \coloneqq \mathrm{diag}(\mathrm{u}^{(\ell)})\mathrm{K}\,\mathrm{diag}(\mathrm{v}^{(\ell)})$. Lastly, one has

$$\|\log(\mathrm{P}^{(\ell)}) - \log(\mathrm{P}^\star)\|_\infty \leqslant d_{\mathcal{H}}(\mathrm{u}^{(\ell)}, \mathrm{u}^\star) + d_{\mathcal{H}}(\mathrm{v}^{(\ell)}, \mathrm{v}^\star) \tag{45}$$

where $\mathrm{P}^\star$ is the unique solution of (28).

Proof. Notice that for any $(\mathrm{v},\mathrm{v}') \in (\mathbb{R}^m_{+,*})^2$, one has $d_{\mathcal{H}}(\mathrm{v},\mathrm{v}') = d_{\mathcal{H}}(\mathrm{v}/\mathrm{v}', \mathbb{1}_m) = d_{\mathcal{H}}(\mathbb{1}_m/\mathrm{v}, \mathbb{1}_m/\mathrm{v}')$, since indeed $d_{\mathcal{H}}(\mathrm{a}/\mathrm{v}, \mathrm{a}/\mathrm{v}') = d_{\mathcal{H}}(\mathrm{v},\mathrm{v}')$.
This shows that

$$d_{\mathcal{H}}(\mathrm{u}^{(\ell+1)}, \mathrm{u}^\star) = d_{\mathcal{H}}\left(\frac{\mathrm{a}}{\mathrm{Kv}^{(\ell)}}, \frac{\mathrm{a}}{\mathrm{Kv}^\star}\right) = d_{\mathcal{H}}(\mathrm{Kv}^{(\ell)}, \mathrm{Kv}^\star) \leqslant \lambda(\mathrm{K}) d_{\mathcal{H}}(\mathrm{v}^{(\ell)}, \mathrm{v}^\star).$$

where we used Theorem 4. This shows (43). One also has, using the triangular inequality

$$\begin{aligned} d_{\mathcal{H}}(\mathrm{u}^{(\ell)}, \mathrm{u}^\star) &\leqslant d_{\mathcal{H}}(\mathrm{u}^{(\ell+1)}, \mathrm{u}^{(\ell)}) + d_{\mathcal{H}}(\mathrm{u}^{(\ell+1)}, \mathrm{u}^\star) \leqslant d_{\mathcal{H}}\left(\frac{\mathrm{a}}{\mathrm{Kv}^{(\ell)}}, \mathrm{u}^{(\ell)}\right) + \lambda(\mathrm{K}) d_{\mathcal{H}}(\mathrm{u}^{(\ell)}, \mathrm{u}^\star) \\ &= d_{\mathcal{H}}\left(\mathrm{a}, \mathrm{u}^{(\ell)} \odot (\mathrm{Kv}^{(\ell)})\right) + \lambda(\mathrm{K}) d_{\mathcal{H}}(\mathrm{u}^{(\ell)}, \mathrm{u}^\star), \end{aligned}$$

which gives the first part of (44) since $\mathrm{u}^{(\ell)} \odot (\mathrm{Kv}^{(\ell)}) = \mathrm{P}^{(\ell)}\mathbb{1}_m$ (the second one being similar).
The proof of (45) follows from □

5 Dual Problem

5.1 Discrete dual

The Kantorovich problem (19) is a linear program so that one can equivalently compute its value by solving a dual linear program.
Proposition 31 (Discrete Kantorovich duality)**.** *One has*

$$\mathrm{L_C(a,b)} = \max_{(\mathrm{f,g})\in\mathrm{R(a,b)}} \langle \mathrm{f,\ a}\rangle + \langle \mathrm{g,\ b}\rangle \tag{46}$$

where the set of admissible potentials is

$$\mathrm{R(a,b)} \coloneqq \{(\mathrm{f,g}) \in \mathbb{R}^n \times \mathbb{R}^m \ ; \ \forall (i,j) \in [\![n]\!] \times [\![m]\!], \mathrm{f} \oplus \mathrm{g} \leqslant \mathrm{C}\} \tag{47}$$

Proof. For the sake of completeness, let us derive this dual problem using Lagrangian duality. The Lagrangian associated to (19) reads

$$\min_{\mathrm{P}\geqslant 0} \max_{(\mathrm{f,g})\in\mathbb{R}^n\times\mathbb{R}^m} \langle \mathrm{C,\ P}\rangle + \langle \mathrm{a} - \mathrm{P}\mathbb{1}_m,\ \mathrm{f}\rangle + \langle \mathrm{b} - \mathrm{P}^\top\mathbb{1}_n,\ \mathrm{g}\rangle. \tag{48}$$

For a linear program, if the primal constraint set is non-empty, one can always exchange the min and the max and get the same value. We thus consider $\max_{(\mathrm{f,g})\in\mathbb{R}^n\times\mathbb{R}^m} \langle \mathrm{a,\ f}\rangle + \langle \mathrm{b,\ g}\rangle + \min_{\mathrm{P}\geqslant 0} \langle \mathrm{C} - \mathrm{f}\mathbb{1}_m^\top - \mathbb{1}_n \mathrm{g}^\top,\ \mathrm{P}\rangle$. We conclude by remarking that

$$\min_{\mathrm{P}\geqslant 0} \langle \mathrm{Q,\ P}\rangle = \begin{cases} 0 & \text{if} \quad \mathrm{Q} \geqslant 0 \\ -\infty & \text{otherwise} \end{cases}$$

so that the constraint reads $\mathrm{C} - \mathrm{f}\mathbb{1}_m^\top - \mathbb{1}_n\mathrm{g}^\top = \mathrm{C} - \mathrm{f}\oplus\mathrm{g} \geqslant 0$. ☐

$$\mathrm{Supp(P)} \subset \{(i,j) \in [\![n]\!] \times [\![m]\!] \ ; \ \mathrm{f}_i + \mathrm{g}_j = \mathrm{C}_{i,j}\}\,. \tag{49}$$

5.2 General formulation

Proposition 32 (Kantorovich duality)**.** *Assume that $\mathcal{X}$ and $\mathcal{Y}$ are compact metric spaces and that $c \in \mathcal{C}(\mathcal{X}\times\mathcal{Y})$. Then*

$$\mathcal{L}_c(\alpha,\beta) = \max_{(f,g)\in\mathcal{R}(c)} \int_{\mathcal{X}} f(x)\mathrm{d}\alpha(x) + \int_{\mathcal{Y}} g(y)\mathrm{d}\beta(y), \tag{50}$$

where the set of admissible dual potentials is

$$\mathcal{R}(c) \coloneqq \{(f,g) \in \mathcal{C}(\mathcal{X}) \times \mathcal{C}(\mathcal{Y}) \ ; \ \forall (x,y), f(x) + g(y) \leqslant c(x,y)\}\,. \tag{51}$$

Here, (f,g) is a pair of continuous functions, and are often called "Kantorovich potentials". The same formula extends under the usual lower-semicontinuity and integrability assumptions, replacing maxima by suprema when dual optimizers need not exist.

Proof. Weak duality is immediate: if $f(x)+g(y) \leqslant c(x,y)$ and $\pi \in \mathcal{U}(\alpha,\beta)$, then $\int f\mathrm{d}\alpha + \int g\mathrm{d}\beta = \int (f(x)+g(y))\mathrm{d}\pi(x,y) \leqslant \int c\mathrm{d}\pi$. Taking the supremum over admissible potentials and the infimum over couplings gives "$\leqslant$".
For the reverse inequality, view the primal problem as a linear program over the locally convex space of Radon measures, paired with $\mathcal{C}(\mathcal{X}\times\mathcal{Y})$. The affine map $\pi \mapsto (\pi_1,\pi_2)$ is continuous for the weak topology, the feasible set is non-empty because it contains $\alpha\otimes\beta$, and the cost is continuous and bounded on compact sets. The separating-hyperplane theorem applied to the convex set of attainable triples $\left\{\left(\pi_1,\pi_2,\int c\mathrm{d}\pi + r\right) : \pi \geqslant 0,\ r \geqslant 0\right\}$ gives a continuous affine separator, hence functions (f,g) and a scalar multiplier which can be normalized so that $f\oplus g \leqslant c$. The separating inequality then states that the supremum over such potentials is at least the primal value. This proves equality. The same argument is the infinite-dimensional analogue of the finite linear-programming proof in Proposition 31. ☐

$$\mathrm{Supp}(\pi) \subset \{(x,y) \in \mathcal{X}\times\mathcal{Y} \ ; \ f(x) + g(y) = c(x,y)\}\,. \tag{52}$$

5.3 c-transforms

Best-response potentials and the c-transform. $\sup_{g\in\mathcal{C}(\mathcal{Y})} \left\{\int g\mathrm{d}\beta \ ; \ \forall (x,y), g(y) \leqslant c(x,y) - f(x)\right\}$. $\forall y \in \mathcal{Y}, \quad g(y) \leqslant f^c(y)$
Definition 9 (c-transform)**.** *For a function $f : \mathcal{X} \to \mathbb{R}\cup\{-\infty\}$, its c-transform is*

$$\forall y \in \mathcal{Y}, \quad f^c(y) \coloneqq \inf_{x\in\mathcal{X}} c(x,y) - f(x). \tag{53}$$

For a function $g : \mathcal{Y} \to \mathbb{R}\cup\{-\infty\}$, the $\bar{c}$-transform associated with $\bar{c}(y,x) = c(x,y)$ is $\forall x \in \mathcal{X}, \quad g^{\bar{c}}(x) \coloneqq \inf_{y\in\mathcal{Y}} c(x,y) - g(y)$.

Proposition 33 (c-transforms solve the semi-relaxed problems)**.** *For fixed f, the maximizers of the dual objective over all g such that $f\oplus g \leqslant c$ are exactly the functions satisfying $g = f^c$ β-almost everywhere. Equivalently, f^c gives the value of the one-marginal primal problem $\inf_{\pi:\,\pi_2=\beta} \int c(x,y)\mathrm{d}\pi(x,y) - \int f(x)\mathrm{d}\pi_1(x) = \int f^c(y)\mathrm{d}\beta(y)$. Symmetrically, for fixed g, the maximizers over f are the functions satisfying $f = g^{\bar{c}}$ α-almost everywhere.*

Proof. The constraint $f(x)+g(y) \leqslant c(x,y)$ for all x is equivalent, for each y, to $g(y) \leqslant \inf_x c(x,y) - f(x) = f^c(y)$. Since β is nonnegative, the largest possible value of $\int g \mathrm{d}\beta$ is obtained by saturating this pointwise upper bound on the support of β. The proof for $f = g^{\bar{c}}$ is identical after exchanging the two marginals.
For the primal formula, disintegrate any feasible π as $\pi(\mathrm{d}x, \mathrm{d}y) = \pi_y(\mathrm{d}x)\beta(\mathrm{d}y)$. Then

$$\int c\mathrm{d}\pi - \int f \mathrm{d}\pi_1 = \int \left(\int (c(x,y) - f(x))\mathrm{d}\pi_y(x) \right) \mathrm{d}\beta(y) \geqslant \int f^c(y)\mathrm{d}\beta(y).$$

If minimizers admit a measurable selection, equality is obtained by choosing π_y supported on minimizers of $x \mapsto c(x,y) - f(x)$. Otherwise one uses approximate measurable selections and lets the approximation error vanish. □

Note that these partial minimizations define maximizers on the support of α and β, respectively, while the definitions (53) define functions on the whole spaces $\mathcal{X}$ and $\mathcal{Y}$. This is thus a canonical way to extend solutions of (50) to the whole space.
Proposition 34 (Lipschitz stability of c-transforms)**.** *If c is L-Lipschitz with respect to the second variable, then f^c is L-Lipschitz.*

Proof. Indeed, using the fact that $|\inf(A) - \inf(B)| \leqslant \sup |A - B|$ for two functions A and B, one obtains $|F(y) - F(y')| = |\inf_x(F_x(y)) - \inf_x(F_x(y'))| \leqslant \sup_x |F_x(y) - F_x(y')| \leqslant \sup_x Ld(y,y') = Ld(y,y')$. □

Euclidean case.

$$\int \|x-y\|^2 \mathrm{d}\pi(x,y) = \mathrm{cst} - 2\int \langle x,\, y\rangle \mathrm{d}\pi(x,y) \quad \text{where} \quad \mathrm{cst} = \int \|x\|^2 \mathrm{d}\alpha(x) + \int \|y\|^2 \mathrm{d}\beta(y).$$

$h^*(y) \coloneqq \sup_x \langle x,\, y\rangle - h(x)$.
Remark 14 (Proof of Brenier's theorem). $\mathrm{supp}(\pi) \subset \{(x,y)\,;\, \varphi(x) + \varphi^*(y) = \langle x,\, y\rangle\}$ where $\varphi = -f^{cc}$ is convex and $-g = \varphi^*$. In the special case where α has a density, a convex function is differentiable Lebesgue-almost everywhere, hence α-almost everywhere. Thus $\partial\varphi(x)$ is a singleton for α-almost every x, and it is legitimate to use $T = \nabla\varphi$ as an optimal transport map.

The failure of alternate optimization.
Proposition 35 (Algebra of c-transforms)**.**

$$\text{(i) } f \leqslant f' \Rightarrow f^c \geqslant f'^c, \qquad \text{(ii) } f^{c\bar{c}} \geqslant f, \qquad \text{(iii) } g^{\bar{c}c} \geqslant g, \qquad \text{(iv) } f^{c\bar{c}c} = f^c.$$

Proof. The first inequality (i) follows from the definition of c-transforms (because of the $-$ sign). To prove (ii), expanding the definition of $f^{c\bar{c}}$ we have

$$\left(f^{c\bar{c}}\right)(x) = \min_y c(x,y) - f^c(y) = \min_y c(x,y) - \min_{x'}(c(x',y) - f(x')).$$

Now, since $-\min_{x'} c(x',y) - f(x') \geqslant -(c(x,y) - f(x))$, we recover $(f^{c\bar{c}})(x) \geqslant \min_y c(x,y) - c(x,y) + f(x) = f(x)$. The relation $g^{\bar{c}c} \geqslant g$ is obtained in the same way.
Now, to prove (iv), we first apply (ii) and then (i) with $f' = f^{c\bar{c}}$ to have $f^c \geqslant f^{c\bar{c}c}$. Then we apply (iii) to $g = f^c$ to obtain $f^c \leqslant f^{c\bar{c}c}$.

□

$$(f,g) \mapsto (f, f^c) \mapsto (f^{cc}, f^c) \mapsto (f^{cc}, f^{ccc}) = (f^{cc}, f^c)\ldots \tag{54}$$

6 Semi-discrete and $\mathcal{W}_1$

6.1 Semi-dual

$\sup_{f,g \in \mathcal{C}(\mathcal{X}) \times \mathcal{C}(\mathcal{Y})} \mathcal{E}(f,g)$ where $\mathcal{E}$ takes into account the constraints one can "marginalize" it with respect to g by minimizing over it to obtain the following "semi-dual" problem

$$\sup_{f \in \mathcal{C}(\mathcal{X})} \tilde{\mathcal{E}}(f) \coloneqq \mathcal{E}(f, f^c) = \sup_g \mathcal{E}(f,g) = \int_{\mathcal{X}} f\mathrm{d}\alpha + \int_{\mathcal{Y}} f^c \mathrm{d}\beta. \tag{55}$$

6.2 Semi-discrete

$$\forall\, \mathrm{g} \in \mathbb{R}^m,\ \forall\, x \in \mathcal{X}, \quad \mathrm{g}^{\bar{c}}(x) \coloneqq \min_{j \in [\![m]\!]} c(x, y_j) - \mathrm{g}_j. \tag{56}$$

$$\mathcal{L}_c(\alpha,\beta) = \max_{\mathrm{g} \in \mathbb{R}^m} \mathcal{E}(\mathrm{g}) \coloneqq \int_{\mathcal{X}} \mathrm{g}^{\bar{c}}(x)\mathrm{d}\alpha(x) + \sum_j \mathrm{g}_j \mathrm{b}_j. \tag{57}$$

$\mathbb{L}_j(\mathrm{g}) \coloneqq \{x \in \mathcal{X}\,;\, \forall\, j' \neq j, c(x,y_j) - \mathrm{g}_j \leqslant c(x, y_{j'}) - \mathrm{g}_{j'}\}$

$$\mathcal{E}(\mathrm{g}) = \sum_{j=1}^m \int_{\mathbb{L}_j(\mathrm{g})} (c(x,y_j) - \mathrm{g}_j)\,\mathrm{d}\alpha(x) + \langle \mathrm{g},\, \mathrm{b}\rangle. \tag{58}$$

Proposition 36 (Gradient of the semi-discrete dual)**.** *If α has a density with respect to Lebesgue measure and if c is smooth away from the diagonal, then $\mathcal{E}$ is differentiable and* $\forall\, j \in [\![m]\!], \quad \nabla\mathcal{E}(\mathrm{g})_j = \mathrm{b}_j - \int_{\mathbb{L}_j(\mathrm{g})} \mathrm{d}\alpha.$

Proof. One has

$$\mathcal{E}(\mathrm{g}+\varepsilon\delta_j) - \mathcal{E}(\mathrm{g}) - \varepsilon\left(\mathrm{b}_j - \int_{\mathbb{L}_j(\mathrm{g})} \mathrm{d}\alpha\right) = \sum_k \int_{\mathbb{L}_k(\mathrm{g}+\varepsilon\delta_j)} c(x,y_k)\mathrm{d}\alpha(x) - \int_{\mathbb{L}_k(\mathrm{g})} c(x,y_k)\mathrm{d}\alpha(x).$$

Most of the terms on the right-hand side vanish (because most of the Laguerre cells associated with $\mathrm{g}+\varepsilon\delta_j$ are equal to those of g) and the only terms remaining correspond to neighboring cells (j,k) such that $\mathbb{L}_j(\mathrm{g}) \cap \mathbb{L}_k(\mathrm{g}) \neq \emptyset$ (for the cost $\|x-y\|^2$ and $\mathrm{g}=0$ this forms the Delaunay triangulation).
On these pairs, the right integral differs on a volume of the order of ε (since α has a density), and the function being integrated only varies on the order of ε (since the cost is smooth). So the right-hand side is of the order of ε^2. ☐

Stochastic optimization.

$$\mathcal{E}(\mathrm{g}) = \int_{\mathcal{X}} E(\mathrm{g},x)\mathrm{d}\alpha(x) = \mathbb{E}_X(E(\mathrm{g},X)) \quad \text{where} \quad E(\mathrm{g},x) := \mathrm{g}^{\bar{c}}(x) - \langle \mathrm{g}, \mathrm{b}\rangle, \tag{59}$$

$\nabla_{\mathrm{g}} E(x,\mathrm{g}) = (\mathbb{1}_{\mathbb{L}_j(\mathrm{g})}(x) - \mathrm{b}_j)_{j=1}^m \in \mathbb{R}^m$ where $\mathbb{1}_{\mathbb{L}_j(\mathrm{g})}$ is the indicator function of the Laguerre cell.

$$\mathrm{g}^{(\ell+1)} := \mathrm{g}^{(\ell)} + \tau_\ell \nabla_{\mathrm{g}} E(\mathrm{g}^{(\ell)}, x_\ell). \tag{60}$$

$$\tau_\ell := \frac{\tau_0}{1+\ell/\ell_0}, \tag{61}$$

where ℓ_0 indicates roughly the number of iterations serving as a warmup phase.
$\mathcal{E}(\mathrm{g}^\star) - \mathbb{E}(\mathcal{E}(\mathrm{g}^{(\ell)})) = O\left(\frac{1}{\sqrt{\ell}}\right)$, where $\mathrm{g}^\star$ is a solution of (59) and where $\mathbb{E}$ indicates an expectation with respect to the i.i.d. sampling of $(x_\ell)_\ell$ performed at each iteration.

Optimal quantization. $\mathcal{Q}_m(\alpha) = \min_{Y=(y_j)_{j=1}^m,\ \mathrm{b}\in\Sigma_m} \mathcal{W}_p\left(\alpha, \sum_j \mathrm{b}_j\delta_{y_j}\right)$. The only setting where this problem is simple is the 1-D case, where monotone rearrangement gives the optimal sampling $y_j = \mathcal{C}_\alpha^{-1}(j/m)$.
Solving explicitly for the minimization over b in the formula (57) (exchanging the role of the min and the max) shows that necessarily, at optimality, one has $\mathrm{g}=0$, so that the optimal transport maps the Voronoi cells $\mathbb{L}_j(\mathrm{g}=0)$, which we denote $\mathbb{V}_j(Y)$ to highlight the dependence on the quantization points $Y=(y_j)_j$
$\mathbb{V}_j(Y) = \{x\,;\, \forall\, j', c(x,y_j) \leqslant c(x,y_{j'})\}$. $\mathcal{Q}_m(\alpha) = \min_Y \mathcal{F}(Y) := \int_{\mathcal{X}} \min_{1\leqslant j\leqslant m} c(x,y_j)\mathrm{d}\alpha(x)$. $y_j \in \operatorname{argmin}_y \int_{\mathbb{V}_j(Y)} c(x,y)\mathrm{d}\alpha(x)$. $y_j = \frac{\int_{\mathbb{V}_j(Y)} x\mathrm{d}\alpha(x)}{\int_{\mathbb{V}_j(Y)}\mathrm{d}\alpha}$.

6.3 Wasserstein-1 norm

c-transform for $\mathcal{W}_1$. Here we assume that d is a distance on $\mathcal{X}=\mathcal{Y}$, and we solve the OT problem with the ground cost $c(x,y)=d(x,y)$. In the following, we denote the Lipschitz constant of a function $f \in \mathcal{C}(\mathcal{X})$ as

$$\mathrm{Lip}(f) := \sup\left\{\frac{|f(x)-f(y)|}{d(x,y)}\ ;\ (x,y)\in\mathcal{X}^2, x\neq y\right\}. \tag{62}$$

Proposition 37 (c-transforms and 1-Lipschitz functions)**.** *Suppose $\mathcal{X}=\mathcal{Y}$ and $c(x,y)=d(x,y)$. Then, there exists g such that $f=g^c$ if and only* $\mathrm{Lip}(f)\leqslant 1$. *Furthermore, if* $\mathrm{Lip}(f)\leqslant 1$, *then $f^c=-f$.*

Proof. First, suppose $f=g^c$ for some g. Then, for $x,y\in\mathcal{X}$,

$$\begin{aligned}|f(x)-f(y)| &= \left|\inf_{z\in\mathcal{X}}[d(x,z)-g(z)] \quad - \quad \inf_{z\in\mathcal{X}}[d(y,z)-g(z)]\right| \\ &\leqslant \sup_{z\in\mathcal{X}}|d(x,z)-d(y,z)| \leqslant d(x,y).\end{aligned}$$

The first equality follows from the definition of g^c, the next inequality from the identity $|\inf A - \inf B| \leqslant \sup|A-B|$, and the last from the reversed triangle inequality. This shows that $\mathrm{Lip}(f)\leqslant 1$.
If f is 1-Lipschitz, for all $x,y\in\mathcal{X}$, $f(y)-d(x,y)\leqslant f(x)\leqslant f(y)+d(x,y)$, which shows that

$$\begin{aligned}f^c(y) &= \inf_{x\in\mathcal{X}}[d(x,y)-f(x)] \geqslant \inf_{x\in\mathcal{X}}[d(x,y)-f(y)-d(x,y)] = -f(y),\\ f^c(y) &= \inf_{x\in\mathcal{X}}[d(x,y)-f(x)] \leqslant \inf_{x\in\mathcal{X}}[d(x,y)-f(y)+d(x,y)] = -f(y),\end{aligned}$$

because $\inf_x d(x,y)=0$ (for $x=y$) and thus $f^c=-f$.
Applying this property to $-f$ which is also 1-Lipschitz shows that $(-f)^c=f$ so that f is c-concave (i.e. it is the c-transform of a function). ☐

Using the iterative c-transform scheme (54), one can replace the dual variable (f, g) by $(f^{cc}, f^c) = (-f^c, f^c)$, or equivalently by any pair $(f, -f)$ where f is 1-Lipschitz.

$$\mathcal{W}_1(\alpha, \beta) = \max_f \left\{ \int_{\mathcal{X}} f \mathrm{d}(\alpha - \beta) \; ; \; \mathrm{Lip}(f) \leqslant 1 \right\}. \tag{63}$$

For discrete measures of the form (2), writing $\alpha - \beta = \sum_k \mathrm{m}_k \delta_{z_k}$ with $z_k \in \mathcal{X}$ and $\sum_k \mathrm{m}_k = 0$, the optimization (63) can be rewritten as

$$\mathcal{W}_1(\alpha, \beta) = \max_{(\mathrm{f}_k)_k} \left\{ \sum_k \mathrm{f}_k \mathrm{m}_k \; ; \; \forall (k, \ell), |\mathrm{f}_k - \mathrm{f}_\ell| \leqslant d(z_k, z_\ell), \right\} \tag{64}$$

When using $d(x, y) = |x - y|$ with $\mathcal{X} = \mathbb{R}$, we can reduce the number of constraints by ordering the z_k's via $z_1 \leqslant z_2 \leqslant \ldots$.
$\mathcal{W}_1(\alpha, \beta) = \max_{(\mathrm{f}_k)_k} \left\{ \sum_k \mathrm{f}_k \mathrm{m}_k \; ; \; \forall k, |\mathrm{f}_{k+1} - \mathrm{f}_k| \leqslant z_{k+1} - z_k \right\},$

$\mathcal{W}_1$ on Euclidean spaces.

$$\mathcal{W}_1(\alpha, \beta) = \sup_f \left\{ \int_{\mathbb{R}^d} f(\mathrm{d}\alpha - \mathrm{d}\beta) \; ; \; \|\nabla f\|_\infty \leqslant 1 \right\}. \tag{65}$$

$\iota_{\|\cdot\|_{\mathbb{R}^d} \leqslant 1}(u) = \max_v \langle u, \, v \rangle - \|v\|_{\mathbb{R}^d}$

$$\begin{aligned} \mathcal{W}_1(\alpha, \beta) &= \sup_f \inf_{s(x) \in \mathbb{R}^d} \int_{\mathbb{R}^d} f \mathrm{d}\xi - \int \langle \nabla f(x), \, s(x) \rangle \mathrm{d}x + \int \|s(x)\|_{\mathbb{R}^d} \mathrm{d}x \\ &= \inf_{s(x) \in \mathbb{R}^d} \int \|s(x)\| \mathrm{d}x + \sup_f \int f(x)(\mathrm{d}\xi - \mathrm{div}(s)\mathrm{d}x) \end{aligned}$$

$$\mathcal{W}_1(\alpha, \beta) = \inf_s \left\{ \int_{\mathbb{R}^d} \|s(x)\|_{\mathbb{R}^d} \mathrm{d}x \; ; \; \mathrm{div}(s) = \alpha - \beta \right\}, \tag{66}$$

7 Divergences and Dual Norms

7.1 Dual norms (Integral Probability Metrics)

$$\|\alpha\|_B \coloneqq \sup_f \left\{ \int_{\mathcal{X}} f(x) \mathrm{d}\alpha(x) \; ; \; f \in B \right\}. \tag{67}$$

Example 2 (Total variation). $B = \{ f \in \mathcal{C}(\mathcal{X}) \; ; \; \|f\|_\infty \leqslant 1 \}$.
Example 3 ($\mathcal{W}_1$ norm). $\mathcal{W}_1$ as defined in (65), is a special case of dual norm (67), using
$B = \{ f \; ; \; \mathrm{Lip}(f) \leqslant 1 \}$
Example 4 (Flat norm and Dudley metric). If the set B is bounded and separates measures, then $\| \cdot \|_B$ is a norm on the whole space $\mathcal{M}(\mathcal{X})$ of finite measures.
This is not the case of $\mathcal{W}_1$, which is only defined for α such that $\int_{\mathcal{X}} \mathrm{d}\alpha = 0$ (otherwise $\|\alpha\|_B = +\infty$).

$$B = \{ f \; ; \; \mathrm{Lip}(f) \leqslant 1 \quad \text{and} \quad \|f\|_\infty \leqslant 1 \}. \tag{68}$$
$$B = \{ f \; ; \; \|\nabla f\|_\infty + \|f\|_\infty \leqslant 1 \}.$$

Proposition 38 (Metrization by dual norms)**.** *Assume $\mathcal{X}$ is compact, $B = -B$, and that all measures below are probability measures. (i) If $\mathcal{C}(\mathcal{X}) \subset \overline{\mathrm{Span}(B)}$ (i.e. if the span of B is dense in continuous functions for the sup-norm $\| \cdot \|_\infty$), then $\|\alpha_k - \alpha\|_B \to 0$ implies $\alpha_k \rightharpoonup \alpha$. (ii) If $B \subset \mathcal{C}(\mathcal{X})$ is compact for $\| \cdot \|_\infty$, then $\alpha_k \rightharpoonup \alpha$ implies $\|\alpha_k - \alpha\|_B \to 0$.*

Proof. (i) If $\|\alpha_k - \alpha\|_B \to 0$, then by duality and the symmetry of B, for any $f \in B$, $|\langle f, \, \alpha_k - \alpha \rangle| \leqslant \|\alpha_k - \alpha\|_B$, so $\langle f, \, \alpha_k \rangle \to \langle f, \, \alpha \rangle$. By linearity, this convergence extends to $\mathrm{Span}(B)$.
Let $u \in \mathcal{C}(\mathcal{X})$ and choose $h \in \mathrm{Span}(B)$ with $\|u - h\|_\infty \leqslant \eta$. Since α_k and α have mass one, $|\langle u, \, \alpha_k - \alpha \rangle| \leqslant |\langle h, \, \alpha_k - \alpha \rangle| + 2\eta$. Taking the limsup in k and then letting $\eta \to 0$ proves weak convergence.
(ii) We assume $\alpha_k \rightharpoonup \alpha$ and we consider a sub-sequence α_{n_k} such that $\|\alpha_{n_k} - \alpha\|_B \longrightarrow \limsup_k \|\alpha_k - \alpha\|_B$ Since B is compact, the maximum appearing in the definition of $\|\alpha_{n_k} - \alpha\|_B$ is reached, so there exists $f_{n_k} \in B$ such that $\langle \alpha_{n_k} - \alpha, \, f_{n_k} \rangle = \|\alpha_{n_k} - \alpha\|_B$. Once again, by compactness, we can extract from $(f_{n_k})_k$ a (not relabelled for simplicity) subsequence converging uniformly to some $f \in B$. One has $\|\alpha_{n_k} - \alpha\|_B = \langle \alpha_{n_k} - \alpha, \, f_{n_k} \rangle$, and this quantity converges to 0 because $\langle \alpha_{n_k} - \alpha, \, f_{n_k} \rangle = \langle \alpha_{n_k} - \alpha, \, f \rangle + \langle \alpha_{n_k}, \, f_{n_k} - f \rangle - \langle \alpha, \, f_{n_k} - f \rangle$ and these three terms go to zero because $\alpha_{n_k} - \alpha \rightharpoonup 0$ for the first one and $\|f_{n_k} - f\|_\infty \to 0$ for the last two. ☐

Corollary 3 (Wasserstein metrizes weak convergence)**.** *On a compact space, the Wasserstein-p distance metrizes the weak convergence.*

Proof. For $\mathcal{W}_1$, take $B = \{ f \; ; \; \mathrm{Lip}(f) \leqslant 1 \}$. The span of B contains all Lipschitz functions, and Lipschitz functions are dense in $\mathcal{C}(\mathcal{X})$ on a compact metric space.
For the converse implication, constants do not change the pairing with $\alpha_k - \alpha$ because both measures have mass one. We may therefore normalize every $f \in B$ by imposing $f(x_0) = 0$ for a fixed $x_0 \in \mathcal{X}$. The normalized class is uniformly bounded by $\mathrm{diam}(\mathcal{X})$ and equicontinuous, hence compact in $\| \cdot \|_\infty$ by Ascoli–Arzelà. Proposition 38 then applies.
Proposition 17 shows that $\mathcal{W}_p$ has the same topology as $\mathcal{W}_1$ on a compact space, so every $\mathcal{W}_p$ metrizes convergence in law. ☐

Dual RKHS Norms and Maximum Mean Discrepancies.

Definition 10 (Positive and conditionally positive kernels). *A symmetric function k defined on $\mathcal{X}\times\mathcal{X}$ is said to be positive definite if for any $n \geqslant 0$, for any family $x_1,\dots,x_n \in \mathcal{X}$ the matrix $(k(x_i,x_j))_{i,j}$ is positive semidefinite, i.e. for all $r \in \mathbb{R}^n$*

$$\sum_{i,j=1}^{n} r_i r_j k(x_i,x_j) \geqslant 0, \tag{69}$$

The kernel is said to be conditionally positive definite if positivity only holds in (69) for zero-mean vectors r, i.e. such that $\langle r,\, \mathbb{1}_n\rangle = 0$.

Example 5 (Riesz, energy and Matérn-type kernels). $-\iint \|x-y\| \mathrm{d}\xi(x)\mathrm{d}\xi(y)$

Definition 11 (Kernel norm and MMD). *If k is positive definite, or conditionally positive definite and restricted to signed measures of total mass zero, the associated kernel norm is*

$$\|\xi\|_k^2 := \int_{\mathcal{X}\times\mathcal{X}} k(x,y)\mathrm{d}\xi(x)\mathrm{d}\xi(y). \tag{70}$$

For two probability measures, $\|\alpha-\beta\|_k$ is the maximum mean discrepancy associated with k.

$\|\alpha\|_k^2 = \mathbb{E}_{X,X'}(k(X,X'))$.

Proposition 39 (Kernel norm as an RKHS dual norm). *Let $\mathcal{H}$ be the RKHS with reproducing kernel k, and assume the kernel mean embedding $m_\xi := \int k(x,\cdot)\,\mathrm{d}\xi(x)$ is well-defined for the signed measure ξ. Then $\|\xi\|_k = \sup_{\|h\|_{\mathcal{H}}\leqslant 1} \int h(x)\,\mathrm{d}\xi(x)$, so $\|\cdot\|_k$ is the dual norm in the sense of (67) associated with the RKHS unit ball.*

Proof. By the reproducing property,

$$\int h(x)\,\mathrm{d}\xi(x) = \left\langle h, \int k(x,\cdot)\,\mathrm{d}\xi(x)\right\rangle_{\mathcal{H}} = \langle h, m_\xi\rangle_{\mathcal{H}}.$$

Cauchy–Schwarz gives $\sup_{\|h\|_{\mathcal{H}}\leqslant 1}\int h\,\mathrm{d}\xi = \|m_\xi\|_{\mathcal{H}}$. Finally, $\|m_\xi\|_{\mathcal{H}}^2 = \iint k(x,y)\,\mathrm{d}\xi(x)\,\mathrm{d}\xi(y)$, which is exactly (70). □

Remark 15 (Universal kernels).

In the special case where α is a discrete measure, one thus has the simple expression

$$\|\alpha\|_k^2 = \sum_{i=1}^{n}\sum_{i'=1}^{n} \mathrm{a}_i \mathrm{a}_{i'} \mathrm{k}_{i,i'} = \langle \mathrm{k}\mathrm{a},\, \mathrm{a}\rangle \quad \text{where} \quad \mathrm{k}_{i,i'} := k(x_i,x_{i'}).$$

$$\|\alpha-\beta\|_k^2 = \sum_{i,i'} \mathrm{a}_i\mathrm{a}_{i'} k(x_i,x_{i'}) + \sum_{j,j'} \mathrm{b}_j\mathrm{b}_{j'} k(y_j,y_{j'}) - 2\sum_{i,j}\mathrm{a}_i\mathrm{b}_j k(x_i,y_j). \tag{71}$$

7.2 φ-divergences

Definition 12 (Entropy function). *A function $\varphi : \mathbb{R}\to\mathbb{R}\cup\{\infty\}$ is an entropy function if it is lower semicontinuous, convex,* $\operatorname{dom}\varphi \subset [0,\infty[$*, and satisfies the feasibility condition* $\operatorname{dom}\varphi \cap (0,+\infty) \neq \emptyset$*. The speed of growth of φ at ∞ is described by*

$$\varphi'_\infty = \lim_{x\to+\infty} \varphi(x)/x \in \mathbb{R}\cup\{\infty\}.$$

If $\varphi'_\infty = \infty$, then φ grows faster than any linear function and φ is said to be *superlinear*.

Definition 13 (φ-Divergences). *Let φ be an entropy function. For $\alpha,\beta\in\mathcal{M}(\mathcal{X})$, let $\frac{\mathrm{d}\alpha}{\mathrm{d}\beta}\beta + \alpha^\perp$ be the Lebesgue decomposition*[1] *of α with respect to β. The divergence $\mathcal{D}_\varphi$ is defined by*

$$\mathcal{D}_\varphi(\alpha|\beta) := \int_{\mathcal{X}} \varphi\left(\frac{\mathrm{d}\alpha}{\mathrm{d}\beta}\right)\mathrm{d}\beta + \varphi'_\infty \alpha^\perp(\mathcal{X}) \tag{72}$$

if α,β are nonnegative and ∞ otherwise.

$$\alpha = \sum_i \mathrm{a}_i\delta_{x_i} \quad \text{and} \quad \beta = \sum_i \mathrm{b}_i\delta_{x_i} \tag{73}$$

$$\mathrm{D}_\varphi(\mathrm{a}|\mathrm{b}) = \sum_{i\in\mathrm{Supp}(\mathrm{b})} \varphi\left(\frac{\mathrm{a}_i}{\mathrm{b}_i}\right)\mathrm{b}_i + \varphi'_\infty \sum_{i\notin\mathrm{Supp}(\mathrm{b})} \mathrm{a}_i, \tag{74}$$

where $\mathrm{Supp}(\mathrm{b}) := \{i \in [\![n]\!]\ ;\ \mathrm{b}_i \neq 0\}$.

Proposition 3 (Basic properties of φ-divergences). *If φ is an entropy function, then $\mathcal{D}_\varphi$ is jointly 1-homogeneous, convex and weak-* lower semicontinuous in (α,β).*

[1] The Lebesgue decomposition theorem asserts that, given β, α admits a unique decomposition as the sum of two measures $\alpha^{\mathrm{ac}} + \alpha^\perp$ such that α^{ac} is absolutely continuous with respect to β and $\alpha^\perp$ and β are singular.

Proof. One defines the associated perspective function

$$\forall\,(u,v)\in(\mathbb{R}_+)^2,\quad \psi(u,v)=\begin{cases}\varphi(u/v)v & \text{if}\quad v\neq 0\\ u\varphi'_\infty & \text{if}\quad v=0\end{cases}$$

The joint 1-homogeneity follows from the definition of this perspective. $\mathrm{D}_\varphi(\mathrm{a}|\mathrm{b})=\sum_i\psi(\mathrm{a}_i,\mathrm{b}_i)$, and it is enough to show that ψ is convex on $(\mathbb{R}_+)^2$. We first prove this on $\mathbb{R}_+\times\mathbb{R}_+^*$; the extension to $v=0$ follows by lower semicontinuity of the recession value $u\varphi'_\infty$.
Indeed, for any $\lambda\in[0,1]$, $\tau=1-\lambda$

$$\varphi\left(\frac{\tau u_1+\lambda u_2}{\tau v_1+\lambda v_2}\right)(\tau v_1+\lambda v_2)=\varphi\left(\frac{\tau v_1}{\tau v_1+\lambda v_2}\frac{u_1}{v_1}+\frac{\lambda v_2}{\tau v_1+\lambda v_2}\frac{u_2}{v_2}\right)(\tau v_1+\lambda v_2)\leqslant\tau\varphi\left(\frac{u_1}{v_1}\right)v_1+\lambda\varphi\left(\frac{u_2}{v_2}\right)v_2.$$

In the general measure case, weak-* lower semicontinuity is the standard lower-semicontinuity theorem for convex integral functionals with recession extension; in the discrete case it is immediate from the lower semicontinuity of ψ.

□

Proposition 4 (Non-negativity of φ-divergences)**.** *Assume that φ is normalized by $\varphi(1)=0$. For probability distributions $(\alpha,\beta)\in\mathcal{M}_+^1(\mathcal{X})$, one has $\mathcal{D}_\varphi(\alpha|\beta)\geqslant 0$. If φ is strictly convex, then one has $\mathcal{D}_\varphi(\alpha|\beta)=0$ if and only if $\alpha=\beta$.*

Proof. Let $m=\alpha+\beta$ and write $a=\frac{\mathrm{d}\alpha}{\mathrm{d}m}$ and $b=\frac{\mathrm{d}\beta}{\mathrm{d}m}$. Using the perspective function ψ from the previous proof, $\mathcal{D}_\varphi(\alpha|\beta)=\int\psi(a,b)\mathrm{d}m$. Since α and β are probabilities, $\int a\mathrm{d}m=\int b\mathrm{d}m=1$. Jensen's inequality and $\psi(1,1)=\varphi(1)=0$ give $\mathcal{D}_\varphi(\alpha|\beta)\geqslant\psi\left(\int a\mathrm{d}m,\int b\mathrm{d}m\right)=0$. If φ is strictly convex, equality in Jensen forces $a=b$ m-almost everywhere, hence $\alpha=\beta$. In the general non-probability case, if $\varphi\geqslant 0$ then the divergence is positive by construction. □

Example 6 (Kullback–Leibler divergence).

$$\varphi_{\mathrm{KL}}(s)=\begin{cases}s\log(s)-s+1 & \text{for } s>0,\\ 1 & \text{for } s=0,\\ +\infty & \text{otherwise.}\end{cases}\tag{75}$$

Example 7 (Total variation).

$$\varphi_{\mathrm{TV}}(s)=\begin{cases}|s-1| & \text{for } s\geqslant 0,\\ +\infty & \text{otherwise.}\end{cases}\tag{76}$$

It actually defines a norm on the full space of measures $\mathcal{M}(\mathcal{X})$ where

$$\mathrm{TV}(\alpha|\beta)=\|\alpha-\beta\|_{\mathrm{TV}},\quad\text{where}\quad\|\alpha\|_{\mathrm{TV}}=|\alpha|(\mathcal{X})=\int_{\mathcal{X}}\mathrm{d}|\alpha|(x).\tag{77}$$

If α has a density ρ_α on $\mathcal{X}=\mathbb{R}^d$, then the TV norm is the L^1 norm on functions, $\|\alpha\|_{\mathrm{TV}}=\int_{\mathcal{X}}|\rho_\alpha(x)|\mathrm{d}x=\|\rho_\alpha\|_{L^1}$.
If α is discrete as in (73), then the TV norm is the ℓ^1 norm of vectors in $\mathbb{R}^n$, $\|\alpha\|_{\mathrm{TV}}=\sum_i|\mathrm{a}_i|=\|\mathrm{a}\|_{\ell^1}$.
Remark 16 (Strong vs. weak topology). $\mathcal{W}_1(\alpha,\beta)\leqslant R\|\alpha-\beta\|_{\mathrm{TV}}$

Main families of φ-divergences. $\varphi_\gamma(s)=\frac{s^\gamma-\gamma s+\gamma-1}{\gamma(\gamma-1)}\qquad(\gamma\neq 0,1)$

$$\mathrm{JS}(\alpha,\beta)^2=\frac{1}{2}\mathrm{KL}\left(\alpha\middle|\frac{\alpha+\beta}{2}\right)+\frac{1}{2}\mathrm{KL}\left(\beta\middle|\frac{\alpha+\beta}{2}\right),$$

Remark 17 (φ-divergences versus Bregman divergences).
Proposition 5 (Dual expression). *A φ-divergence can be expressed using the Legendre transform $\varphi^{*,\geqslant 0}(s):=\sup_{t\in\mathbb{R}^+}st-\varphi(t)$ (notice that we restrict the function to the positive real) of φ as*

$$\mathcal{D}_\varphi(\alpha|\beta)=\sup_{f:\mathcal{X}\to\mathbb{R}}\int_{\mathcal{X}}f(x)\mathrm{d}\alpha(x)-\int_{\mathcal{X}}\varphi^{*,\geqslant 0}(f(x))\mathrm{d}\beta(x).\tag{78}$$

which equivalently reads that the Legendre transform of $\mathcal{D}_\varphi(\cdot|\beta)$ reads

$$\forall\,f\in\mathcal{C}(\mathcal{X}),\quad\mathcal{D}_\varphi^*(f|\beta)=\int_{\mathcal{X}}\varphi^{*,\geqslant 0}(f(x))\mathrm{d}\beta(x).\tag{79}$$

Proof. We first consider the superlinear case $\varphi'_\infty=+\infty$, so that $\mathcal{D}_\varphi(\alpha|\beta)=+\infty$ if α does not have a density $\rho\geqslant 0$ with respect to β, $\mathrm{d}\alpha=\rho\mathrm{d}\beta$. Thus the Legendre-Fenchel transform of $\mathcal{D}_\varphi(\cdot|\beta)$ reads

$$\begin{aligned}\mathcal{D}_\varphi^*(f|\beta)&=\sup_{\rho\geqslant 0}\int_{\mathcal{X}}f(x)\rho(x)\mathrm{d}\beta(x)-\int_{\mathcal{X}}\varphi(\rho(x))\mathrm{d}\beta(x)\\&=\int_{\mathcal{X}}\sup_{\rho(x)\geqslant 0}\left(f(x)\rho(x)-\varphi(\rho(x))\right)\mathrm{d}\beta(x)=\int_{\mathcal{X}}\varphi^{*,\geqslant 0}(f(x))\mathrm{d}\beta(x).\end{aligned}$$

Fenchel–Moreau then gives the displayed dual expression. For a general entropy, the same argument is applied to the perspective with its recession term; the singular part is exactly encoded by the effective domain of $\varphi^{*,\geqslant 0}$. □

7.3 GANs via Duality

$$\min_\theta \|\alpha_\theta - \beta\|_B = \min_\theta \sup_{f\in B} \int_{\mathcal{X}} f(x)\mathrm{d}(\alpha_\theta-\beta)(x) = \min_\theta \sup_{f\in B} \int_{\mathcal{Z}} f(g_\theta(z))\mathrm{d}\zeta - \frac{1}{m}\sum_j f(y_j).$$

$$\min_\theta \mathcal{D}_\varphi(\alpha_\theta|\beta) = \min_\theta \sup_f \int_{\mathcal{X}} f(x)\mathrm{d}\alpha_\theta(x) - \mathcal{D}_\varphi^*(f|\beta) = \min_\theta \sup_f \int_{\mathcal{X}} f(g_\theta(z))\mathrm{d}\zeta(z) - \frac{1}{m}\sum_j \varphi^*(f(y_j)).$$

8 Advanced Topics on Entropic Regularization

8.1 Dual of Sinkhorn

Discrete dual.
Proposition 40 (Dual of entropic OT)**.** *One has*

$$\mathrm{L}_{\mathrm{C}}^{\varepsilon}(\mathrm{a},\mathrm{b}) = \max_{\mathrm{f}\in\mathbb{R}^n,\mathrm{g}\in\mathbb{R}^m} \langle \mathrm{f},\,\mathrm{a}\rangle + \langle \mathrm{g},\,\mathrm{b}\rangle - \varepsilon\sum_{i,j}\exp\left(\frac{\mathrm{f}_i+\mathrm{g}_j-\mathrm{C}_{i,j}}{\varepsilon}\right)\mathrm{a}_i\mathrm{b}_j + \varepsilon. \tag{80}$$

The optimal (f,g) *are linked to scalings* (u,v) *appearing in* (29) *through*

$$\mathrm{u}_i = \mathrm{a}_i e^{\mathrm{f}_i/\varepsilon} \quad \textit{and} \quad \mathrm{v}_j = \mathrm{b}_j e^{\mathrm{g}_j/\varepsilon}. \tag{81}$$

Proof. We introduce Lagrange multipliers and consider $\min_{\mathrm{P}\geqslant 0}\max_{\mathrm{f},\mathrm{g}} \langle \mathrm{C},\,\mathrm{P}\rangle + \varepsilon\mathrm{KL}(\mathrm{P}|\mathrm{a}\otimes\mathrm{b}) + \langle \mathrm{a}-\mathrm{P}\mathbb{1},\,\mathrm{f}\rangle + \langle \mathrm{b}-\mathrm{P}^\top\mathbb{1},\,\mathrm{g}\rangle$. Finite-dimensional convex duality allows us to exchange the minimum over P with the maximum over (f,g), giving

$$\max_{\mathrm{f},\mathrm{g}} \langle \mathrm{f},\,\mathrm{a}\rangle + \langle \mathrm{g},\,\mathrm{b}\rangle + \varepsilon\min_{\mathrm{P}\geqslant 0}\left(\mathrm{KL}(\mathrm{P}|\mathrm{a}\otimes\mathrm{b}) - \langle\frac{\mathrm{f}\oplus\mathrm{g}-\mathrm{C}}{\varepsilon},\,\mathrm{P}\rangle\right) = \langle \mathrm{f},\,\mathrm{a}\rangle + \langle \mathrm{g},\,\mathrm{b}\rangle - \varepsilon\mathrm{KL}^*\left(\frac{\mathrm{f}\oplus\mathrm{g}-\mathrm{C}}{\varepsilon}|\mathrm{a}\otimes\mathrm{b}\right).$$

One concludes by using (79) for $\varphi(r) = r\log(r) - r + 1$ $\mathrm{KL}^*(H|\mathrm{a}\otimes\mathrm{b}) = \sum_{i,j}\varphi^*(H_{i,j})\mathrm{a}_i\mathrm{b}_j$. Indeed, the scalar maximization $\varphi^*(s) = \sup_{r\geqslant 0}\{rs - r\log r + r - 1\}$ has first-order condition $s - \log r = 0$, hence $r = e^s$ and $\varphi^*(s) = e^s - 1$. ☐

Discrete soft c-transforms.

$$\mathrm{a}_i - e^{\frac{\mathrm{f}_i}{\varepsilon}}\mathrm{a}_i\sum_j \exp\left(\frac{\mathrm{g}_j-\mathrm{C}_{i,j}}{\varepsilon}\right)\mathrm{b}_j = 0$$

$$\mathrm{f}_i = -\varepsilon\log\sum_j\exp\left(\frac{\mathrm{g}_j-\mathrm{C}_{i,j}}{\varepsilon}\right)\mathrm{b}_j.$$

$\min_{\mathrm{b}}^{\varepsilon}(h) := -\varepsilon\log\sum_j e^{-h_j/\varepsilon}\mathrm{b}_j$

$$\mathrm{f}_i = \min_{\mathrm{b}}^{\varepsilon}(\mathrm{C}_{i,\cdot}-\mathrm{g}). \tag{82}$$

$$\mathrm{g}_j = \min_{\mathrm{a}}^{\varepsilon}(\mathrm{C}_{\cdot,j}-\mathrm{f}). \tag{83}$$

$\min_{\mathrm{b}}^{\varepsilon}(h-\mathrm{cst}) = \min_{\mathrm{b}}^{\varepsilon}(h) - \mathrm{cst}$

Continuous dual and soft-transforms. For generic (not necessarily discrete) input measures (α,β), the dual problem (80) reads

$$\sup_{f,g\in\mathcal{C}(\mathcal{X})\times\mathcal{C}(\mathcal{Y})} \int_{\mathcal{X}} f(x)\mathrm{d}\alpha(x) + \int_{\mathcal{Y}} g(y)\mathrm{d}\beta(y) - \varepsilon\int_{\mathcal{X}\times\mathcal{Y}}\left(e^{\frac{f\oplus g-c}{\varepsilon}} - 1\right)\mathrm{d}\alpha\otimes\mathrm{d}\beta \tag{84}$$

The corresponding soft c-transform, which minimizes this dual problem with respect to either f or g, reads

$$\forall y\in\mathcal{Y},\quad f^{c,\varepsilon}(y) := -\varepsilon\log\left(\int_{\mathcal{X}} e^{\frac{-c(x,y)+f(x)}{\varepsilon}}\mathrm{d}\alpha(x)\right),$$

$$\forall x\in\mathcal{X},\quad g^{\bar{c},\varepsilon}(x) := -\varepsilon\log\left(\int_{\mathcal{Y}} e^{\frac{-c(x,y)+g(y)}{\varepsilon}}\mathrm{d}\beta(y)\right).$$

The soft c-transform, besides being useful to define Sinkhorn's algorithm, is also important to show the existence of dual solutions.
Proposition 41 (Existence and uniqueness of entropic dual potentials)**.** *Assume that $\mathcal{X}$ and $\mathcal{Y}$ are compact and that c is continuous. The dual problem* (84) *has solutions, and the set of solutions is of the form* $(f^\star+\lambda, g^\star-\lambda)$ *for* $\lambda\in\mathbb{R}$.

Proof. Normalize potentials by imposing $\int f \mathrm{d}\alpha = 0$. Replacing any pair (f, g) by the corresponding soft c-transforms does not decrease the dual objective, because each soft transform is the exact maximizer in one block variable. For transformed potentials, the oscillations are bounded by the oscillation of c: $\|f\|_V + \|g\|_V \leqslant 2(\sup c - \inf c)$. Moreover the modulus of continuity of the soft transforms is controlled by the modulus of continuity of c; for instance $|g^{\bar{c},\varepsilon}(x) - g^{\bar{c},\varepsilon}(x')| \leqslant \sup_y |c(x, y) - c(x', y)|$. After normalization, maximizing sequences are therefore uniformly bounded and equicontinuous. Arzela–Ascoli gives a uniformly converging subsequence, and continuity of the dual objective gives a maximizer.
Uniqueness follows from strict convexity of $H \mapsto \int e^{H/\varepsilon} \mathrm{d}\alpha \otimes \beta$ on the image of the map $(f, g) \mapsto H = f \oplus g - c$, modulo constants. If two optimal pairs exist, their midpoint is also optimal; strict convexity forces the two functions $f \oplus g$ to agree $\alpha \otimes \beta$-almost everywhere. Since the potentials are continuous on compact supports, this implies $f - f'$ is constant and $g - g'$ is the opposite constant. □

Remark 18 (Convexity properties). Similarly to the unregularized case studied in Remark 14, in the particular case $c(x, y) = -\langle x, y\rangle$, the function $f^{c,\varepsilon}$ is always concave. Hence, when $c(x, y) = \|x - y\|^2/2$, the optimal potentials are of the form $f^\star(x) = \|x\|^2 - \varphi^\star(x)$ where $\varphi^\star$ is a convex function.
Remark 19 (Gaussian marginals).

8.2 Convergence Analysis

Sublinear dual-gap convergence by Bregman projections. Sinkhorn is cyclic coordinate ascent on the smooth dual problem and, equivalently, alternating KL projection on the two marginal constraint sets. We denote the dual objective in (84) by

$$\mathcal{D}_\varepsilon(f, g) := \int f \mathrm{d}\alpha + \int g \mathrm{d}\beta - \varepsilon \int \left(e^{(f \oplus g - c)/\varepsilon} - 1\right) \mathrm{d}\alpha \otimes \mathrm{d}\beta.$$

Proposition 42 (Pinsker inequality)**.** *If $p, q \in \Sigma_n$, then $\|p - q\|_1^2 \leqslant 2\,\mathrm{KL}(p|q)$.*

Proof. It is enough to prove the scalar inequality $u \log u - u + 1 \geqslant \frac{(u-1)^2}{2(u+1)}$ $\quad u \geqslant 0$. Indeed, after bringing the right-hand side to the left, the resulting function h satisfies $h(1) = h'(1) = 0$ and $h''(u) = \frac{1}{u} - \frac{4}{(u+1)^3} = \frac{(u+1)^3 - 4u}{u(u+1)^3} \geqslant 0$, so h is convex and minimized at $u = 1$. Multiplying by q_i and summing with $u = p_i/q_i$ gives $\mathrm{KL}(p|q) \geqslant \frac{1}{2}\sum_i \frac{(p_i - q_i)^2}{p_i + q_i}$. Cauchy–Schwarz yields

$$\|p - q\|_1^2 = \left(\sum_i \frac{|p_i - q_i|}{\sqrt{p_i + q_i}} \sqrt{p_i + q_i}\right)^2 \leqslant \left(\sum_i \frac{(p_i - q_i)^2}{p_i + q_i}\right)\left(\sum_i (p_i + q_i)\right),$$

and $\sum_i (p_i + q_i) = 2$. □

Proposition 43 (A compact $O(1/k)$ dual rate)**.** *Assume that $\mathcal{X}$ and $\mathcal{Y}$ are compact, that c is bounded, and write $R = \sup c - \inf c$. Let (f_k, g_k) be Sinkhorn dual iterates normalized by $\int f_k \mathrm{d}\alpha = 0$, and let $\Delta_k := \mathcal{D}_\varepsilon(f^\star, g^\star) - \mathcal{D}_\varepsilon(f_k, g_k)$ be the dual suboptimality gap for the entropic dual objective (84). Then there exists a numerical constant C such that $\Delta_k \leqslant \frac{CR^2}{\varepsilon(k+1)}$.*

Proof. First, the soft c-transform bounds the oscillation of every normalized iterate and every normalized optimum: $\|f_k\|_V + \|g_k\|_V + \|f^\star\|_V + \|g^\star\|_V \leqslant CR$. Here $\|h\|_V := \sup h - \inf h$ is the variation seminorm, the same projective norm used in Hilbert's metric in Section 4.5. It is natural because dual potentials can be shifted by constants without changing the coupling.
Second, each Sinkhorn half-step is an exact KL projection. The Pythagorean identity for KL projections gives the ascent identity $\mathcal{D}_\varepsilon(f_{k+1}, g_{k+1}) - \mathcal{D}_\varepsilon(f_k, g_k) = \varepsilon\big[\mathrm{KL}(\pi^\star|\pi_k) - \mathrm{KL}(\pi^\star|\pi_{k+1})\big]$, where $\pi_k = e^{(f_k \oplus g_k - c)/\varepsilon}\alpha \otimes \beta$ and $\pi^\star$ is the optimal entropic coupling. The KL drop controls the marginal residuals through Pinsker's inequality, Proposition 42. Third, convexity of the exponential dual objective gives a one-step estimate of the form $\Delta_k^2 \leqslant \frac{CR^2}{\varepsilon}\big(\mathcal{D}_\varepsilon(f_{k+1}, g_{k+1}) - \mathcal{D}_\varepsilon(f_k, g_k)\big)$. This is the usual Bregman-projection estimate: the dual gap is controlled by the product of a bounded dual radius and the marginal residual corrected by the next projection, and the residual squared is controlled by the KL drop. Summing the reciprocal inequality obtained from the last display yields $\frac{1}{\Delta_{k+1}} - \frac{1}{\Delta_k} \geqslant \frac{\varepsilon}{CR^2}$, and therefore $\Delta_k \leqslant CR^2/(\varepsilon(k+1))$. □

Corollary 4 (Iteration complexity for entropic accuracy)**.** *Under the assumptions of Proposition 43, reaching dual objective accuracy $\Delta_k \leqslant \delta$ is guaranteed after $k + 1 \geqslant \frac{CR^2}{\varepsilon\delta}$ Sinkhorn cycles.*

Proof. The proposition gives $\Delta_k \leqslant CR^2/(\varepsilon(k+1))$. Solving this inequality for k gives the displayed sufficient condition. □

Hilbert-metric linear convergence. $\|f_k - f^\star\|_V = O(\lambda^k)$ and $\|g_k - g^\star\|_V = O(\lambda^k)$

$$\|\log \frac{\mathrm{d}\pi_k}{\mathrm{d}\pi^\star}\|_\infty = \|(f_k - f^\star) \oplus (g_k - g^\star)\|_\infty \leqslant \|f_k - f^\star\|_V + \|g_k - g^\star\|_V$$

where the contraction ratio only depends on the radius $R := \sup_{x,y} c(x, y)$ of the cost, $\lambda = \frac{\sqrt{\eta}-1}{\sqrt{\eta}+1} \leqslant \tanh(R/(2\varepsilon)) < 1$. One also has the following bounds
$\|f_k - f^\star\|_V \leqslant \frac{\|\log \frac{\mathrm{d}\pi_{k,1}}{\mathrm{d}\alpha}\|_\infty}{1-\lambda}$

Order-monotone convergence.

Proposition 44 (Monotone fixed-point route to Sinkhorn convergence)**.** *Let c be bounded and continuous on compact spaces, and define the double Sinkhorn map, normalized by subtracting its α-mean, $\mathcal{A}(f) \coloneqq (f^{c,\varepsilon})^{\bar{c},\varepsilon} - \int (f^{c,\varepsilon})^{\bar{c},\varepsilon}\mathrm{d}\alpha$. The map $\mathcal{A}$ is order preserving modulo additive constants. If an initialization is chosen so that $f_0 \leqslant \mathcal{A}(f_0)$ after normalization, then the iterates $f_{k+1} = \mathcal{A}(f_k)$ increase pointwise up to constants, remain uniformly bounded in oscillation, and converge to a fixed point. A convenient practical choice is $f_0 = 0$, followed if necessary by subtracting a sufficiently large constant so that the subsolution inequality holds. This fixed point is the entropic potential, hence the associated Sinkhorn scalings converge.*

Proof. The soft c-transform is order reversing: if $g \leqslant g'$, then $-\varepsilon \log \int e^{(g-c)/\varepsilon}\mathrm{d}\beta \geqslant -\varepsilon \log \int e^{(g'-c)/\varepsilon}\mathrm{d}\beta$. The composition of two order-reversing transforms is therefore order preserving. The transform also commutes with additive constants in the projective sense, which is why one fixes a normalization. Starting from a subsolution gives $f_0 \leqslant f_1$, and order preservation gives $f_k \leqslant f_{k+1}$ for all k. Soft-transform oscillation bounds, controlled by $\sup c - \inf c$, prevent escape to infinity after normalization. Monotone pointwise convergence, compactness of equicontinuous soft transforms, and continuity of $\mathcal{A}$ then give a fixed point. Uniqueness of entropic potentials up to constants, Proposition 22 and the dual uniqueness statement above identify this fixed point with the Sinkhorn solution. □

8.3 Sinkhorn Divergences

Entropic bias. $\alpha_\varepsilon = \underset{\beta}{\mathrm{argmin}}\, \mathcal{L}_c^\varepsilon(\alpha, \beta)$

Proposition 45 (Large-temperature entropic bias)**.** *Assume that c is bounded and continuous. Then $\mathcal{L}_c^\varepsilon(\alpha, \beta) \to \iint c(x, y)\mathrm{d}\alpha(x)\mathrm{d}\beta(y)$ as $\varepsilon \to +\infty$.*

Proof. Let $(f_\varepsilon, g_\varepsilon)$ be optimal dual potentials, normalized by $\int g_\varepsilon \mathrm{d}\beta = 0$. The soft c-transform equation gives

$$f_\varepsilon(x) = -\varepsilon \log \int \exp\Big(\frac{g_\varepsilon(y) - c(x, y)}{\varepsilon}\Big)\, \mathrm{d}\beta(y).$$

For bounded c, the oscillations of normalized entropic potentials are bounded uniformly in ε by the oscillation of c. Hence the log-sum-exp expansion is uniform: $f_\varepsilon(x) = -\int (g_\varepsilon(y) - c(x, y))\mathrm{d}\beta(y) + O(\varepsilon^{-1}) = \int c(x, y)\mathrm{d}\beta(y) + O(\varepsilon^{-1})$. At optimality the exponential penalty in the dual integrates to zero, so $\mathcal{L}_c^\varepsilon(\alpha, \beta) = \int f_\varepsilon \mathrm{d}\alpha + \int g_\varepsilon \mathrm{d}\beta = \iint c(x, y)\mathrm{d}\alpha(x)\mathrm{d}\beta(y) + O(\varepsilon^{-1})$. This proves the limit. □

$$\alpha_\varepsilon \to \min_\beta \langle \int c(x, \cdot)\mathrm{d}\alpha(x),\, \beta\rangle = \delta_{y^\star(\alpha)} \quad \text{where} \quad y^\star(\alpha) = \underset{y}{\mathrm{argmin}} \int c(x, y)\mathrm{d}\alpha(x).$$

Sinkhorn divergences.

$$\bar{\mathcal{L}}_c^\varepsilon(\alpha, \beta) \coloneqq \mathcal{L}_c^\varepsilon(\alpha, \beta) - \frac{1}{2}\mathcal{L}_c^\varepsilon(\alpha, \alpha) - \frac{1}{2}\mathcal{L}_c^\varepsilon(\beta, \beta). \tag{85}$$

Lemma 3 (Entropic dual cost at optimum)**.** *Let $(f_{\alpha,\beta}, g_{\alpha,\beta})$ be optimal dual potentials, normalized arbitrarily. One has*

$$\mathcal{L}_c^\varepsilon(\alpha, \beta) = \langle f_{\alpha,\beta},\, \alpha\rangle + \langle g_{\alpha,\beta},\, \beta\rangle. \tag{86}$$

Proof. We first notice that at optimality, the relation $f_{\alpha,\beta} = -\varepsilon \log \int_{\mathcal{Y}} e^{\frac{g_{\alpha,\beta}(y) - c(x,y)}{\varepsilon}}\mathrm{d}\beta(y)$ after taking the exponential, equivalently reads

$$1 = \int_{\mathcal{Y}} e^{\frac{f_{\alpha,\beta}(x) + g_{\alpha,\beta}(y) - c(x,y)}{\varepsilon}}\mathrm{d}\beta(y) \quad \Longrightarrow \quad \int_{\mathcal{X}\times\mathcal{Y}} \Big(e^{\frac{f_{\alpha,\beta} \oplus g_{\alpha,\beta} - c}{\varepsilon}} - 1\Big)\, \mathrm{d}(\alpha \otimes \beta) = 0.$$

Plugging this in formula (84), one obtains the result. □

Proposition 46 (Asymptotics of Sinkhorn divergences)**.** *Assume that the two measures are supported on the same space and that c is bounded, continuous and satisfies $c(x, x) = 0$. One has $\bar{\mathcal{L}}_c^\varepsilon(\alpha, \beta) \to \mathcal{L}_c(\alpha, \beta)$ when $\varepsilon \to 0$ and*

$$\bar{\mathcal{L}}_c^\varepsilon(\alpha, \beta) \to \frac{1}{2}\int -c\mathrm{d}(\alpha - \beta) \otimes \mathrm{d}(\alpha - \beta) \quad \textit{when} \quad \varepsilon \to +\infty.$$

Proof. For discrete measures, the convergence is already proved in (36); we now give a general treatment. **Case $\varepsilon \to 0$.** The first limit follows from the standard Γ-convergence argument for entropic optimal transport: the entropy term is lower semicontinuous along weakly converging couplings, while any finite-cost coupling can be approximated by couplings with finite entropy. Since $c(x, x) = 0$, the two self-costs in the debiased expression converge to zero, and the cross term converges to $\mathcal{L}_c(\alpha, \beta)$.

Case $\varepsilon \to +\infty$. We denote by $(f_\varepsilon, g_\varepsilon)$ optimal dual potentials. After normalizing them and using boundedness of c, their oscillations stay uniformly bounded, so the following expansion is uniform. The optimality condition on f_ε (equivalently the Sinkhorn fixed point on f_ε) reads

$$\begin{aligned} f_\varepsilon &= -\varepsilon \log \int \exp\left(\frac{g_\varepsilon(y) - c(\cdot, y)}{\varepsilon}\right)\mathrm{d}\beta(y) = -\varepsilon \log \int \left(1 + \frac{g_\varepsilon(y) - c(\cdot, y)}{\varepsilon} + o(1/\varepsilon)\right)\mathrm{d}\beta(y) \\ &= -\varepsilon \log\left(1 + \frac{1}{\varepsilon}\int (g_\varepsilon(y) - c(\cdot, y))\mathrm{d}\beta(y) + o(1/\varepsilon)\right) = -\int g_\varepsilon \mathrm{d}\beta + \int c(\cdot, y)\mathrm{d}\beta(y) + o(1). \end{aligned}$$

Plugging this relation in the dual expression (86) $\mathcal{L}_c^\varepsilon(\alpha,\beta) = \int f_\varepsilon \mathrm{d}\alpha + \int g_\varepsilon \mathrm{d}\beta = \iint c(x,y)\mathrm{d}\alpha(x)\mathrm{d}\beta(y) + o(1)$. Applying this expansion to (α,β), (α,α) and (β,β) gives

$$\bar{\mathcal{L}}_c^\varepsilon(\alpha,\beta) \to \int c\,\mathrm{d}\alpha\otimes\mathrm{d}\beta - \frac{1}{2}\int c\,\mathrm{d}\alpha\otimes\mathrm{d}\alpha - \frac{1}{2}\int c\,\mathrm{d}\beta\otimes\mathrm{d}\beta = -\frac{1}{2}\int c\,\mathrm{d}(\alpha-\beta)\otimes\mathrm{d}(\alpha-\beta).$$

□

In the case where $-c$ defines a conditionally positive definite kernel, $\bar{\mathcal{L}}_c^\varepsilon(\alpha,\beta)$ converges to the square of a Hilbertian kernel norm. A typical example is $c(x,y) = \|x-y\|^p$ for $0<p<2$, which corresponds to the energy-distance kernel.

Proposition 47 (Non-negativity of Sinkhorn divergences)**.** *If $k(x,y) = e^{-c(x,y)/\varepsilon}$ is positive definite, then $\bar{\mathcal{L}}_c^\varepsilon(\alpha,\beta) \geqslant 0$.*

Proof. In the following, we denote by $(f_{\alpha,\beta}, g_{\alpha,\beta})$ optimal dual potentials for the dual Schrödinger problem between α and β. We denote by $f_{\alpha,\alpha} = g_{\alpha,\alpha}$ (one can assume they are equal by symmetry) the solution for the problem between α and itself. Using the suboptimal function $(f_{\alpha,\alpha}, g_{\beta,\beta})$ in the dual maximization problem, and using relation (86) for the simplified expression of the dual cost, one obtains

$$\mathcal{L}_c^\varepsilon(\alpha,\beta) \geqslant \langle f_{\alpha,\alpha},\, \alpha\rangle + \langle g_{\beta,\beta},\, \beta\rangle - \varepsilon\langle e^{\frac{f_{\alpha,\alpha}\oplus g_{\beta,\beta}-c}{\varepsilon}} - 1,\, \alpha\otimes\beta\rangle$$

But one has $\langle f_{\alpha,\alpha},\, \alpha\rangle = \frac{1}{2}\mathcal{L}_c^\varepsilon(\alpha,\alpha)$, and similarly for β, so that the previous inequality equivalently reads

$$\frac{1}{\varepsilon}\bar{\mathcal{L}}_c^\varepsilon(\alpha,\beta) \geqslant 1 - \langle e^{\frac{f_{\alpha,\alpha}\oplus g_{\beta,\beta}-c}{\varepsilon}},\, \alpha\otimes\beta\rangle = 1 - \langle\tilde{\alpha},\, \tilde{\beta}\rangle_k$$

where $\tilde{\alpha} = e^{f_{\alpha,\alpha}/\varepsilon}\alpha$, $\tilde{\beta} = e^{f_{\beta,\beta}/\varepsilon}\beta$ and we introduced the inner product (which is valid because k is positive definite) $\langle\tilde{\alpha},\, \tilde{\beta}\rangle_k \coloneqq \int k(x,y)\mathrm{d}\tilde{\alpha}(x)\mathrm{d}\tilde{\beta}(y)$. One notes that the Sinkhorn fixed point equation, once exponentiated, reads $e^{f_{\alpha,\alpha}/\varepsilon} \odot [k(\tilde{\alpha})] = 1$ and hence $\|\tilde{\alpha}\|_k^2 = \langle k(\tilde{\alpha}),\, \tilde{\alpha}\rangle = \langle e^{f_{\alpha,\alpha}/\varepsilon}\odot k(\tilde{\alpha}),\, \alpha\rangle = \langle 1,\, \alpha\rangle = 1$ and similarly $\|\tilde{\beta}\|_k^2 = 1$. Therefore, by Cauchy–Schwarz, one has $1 - \langle\tilde{\alpha},\, \tilde{\beta}\rangle_k \geqslant 0$. □

Remark 20 (Strict positivity).

8.4 Sample Complexity

Proposition 48 (Empirical OT has dimension-dependent rates)**.** *Let α be a probability distribution with a density bounded above and below on $[0,1]^d$, and let $\hat{\alpha}_n = n^{-1}\sum_{i=1}^n \delta_{X_i}$ be its empirical measure. For $d > 2p$, the typical scale of empirical OT is $\mathbb{E}\,\mathcal{W}_p(\hat{\alpha}_n,\alpha) \asymp n^{-1/d}$. The exponent changes in low dimension, but the important message is that exact OT deteriorates with the ambient dimension.*

Proof. For the upper bound, partition $[0,1]^d$ into dyadic cubes. At scale 2^{-j}, the empirical mass fluctuation over the cells is of order $n^{-1/2}2^{jd/2}$, while moving this excess mass inside cells costs 2^{-j}. Summing the multiscale contributions up to the scale where the expected number of samples per cell is of order one gives 2^{-J} with $2^{Jd} \simeq n$, hence $n^{-1/d}$. For the lower bound, a packing argument shows that a positive fraction of cells at the critical scale are empty or overfull with non-negligible probability, forcing mass to travel at least a distance comparable to $n^{-1/d}$. The matching upper and lower bounds give the displayed rate. □

Proposition 49 (MMD has a parametric empirical rate)**.** *Let k be a bounded positive definite kernel with RKHS $\mathcal{H}_k$, and define $\mathrm{MMD}_k(\alpha,\beta) = \|\int k(x,\cdot)\mathrm{d}(\alpha-\beta)(x)\|_{\mathcal{H}_k}$. If $\hat{\alpha}_n$ is the empirical measure of n i.i.d. samples from α, then $\mathbb{E}\,\mathrm{MMD}_k(\hat{\alpha}_n,\alpha)^2 = \frac{1}{n}\Big(\mathbb{E}k(X,X) - \mathbb{E}k(X,X')\Big)$, where X, X' are independent with law α. In particular, if $k(x,x) \leqslant \kappa^2$, then $\mathbb{E}\,\mathrm{MMD}_k(\hat{\alpha}_n,\alpha) \leqslant \kappa/\sqrt{n}$.*

Proof. Let $\Phi(x) = k(x,\cdot)$ be the feature map and $m_\alpha = \mathbb{E}\Phi(X)$. Then $\mathrm{MMD}_k(\hat{\alpha}_n,\alpha) = \|\frac{1}{n}\sum_{i=1}^n(\Phi(X_i) - m_\alpha)\|_{\mathcal{H}_k}$. Independence cancels the cross terms after taking the squared norm and expectation, giving $\mathbb{E}\,\mathrm{MMD}_k(\hat{\alpha}_n,\alpha)^2 = \frac{1}{n}\mathbb{E}\|\Phi(X) - m_\alpha\|_{\mathcal{H}_k}^2 = \frac{1}{n}\Big(\mathbb{E}k(X,X) - \mathbb{E}k(X,X')\Big)$. The last bound follows from Jensen's inequality and $k(x,x) \leqslant \kappa^2$. □

Proposition 50 (Sinkhorn divergences interpolate the rates)**.** *Assume that α and β are supported in a compact subset of $\mathbb{R}^d$ and that the cost is smooth. For fixed $\varepsilon > 0$, the debiased Sinkhorn divergence satisfies a parametric empirical bound of the form*

$$\mathbb{E}\left|\bar{\mathcal{L}}_c^\varepsilon(\hat{\alpha}_n,\hat{\beta}_m) - \bar{\mathcal{L}}_c^\varepsilon(\alpha,\beta)\right| \leqslant C_{c,d}\,\varepsilon^{-d/2}\left(\frac{1}{\sqrt{n}} + \frac{1}{\sqrt{m}}\right),$$

up to constants depending on smoothness and support diameter. Thus regularization removes the $n^{-1/d}$ curse for fixed ε, while the prefactor deteriorates as $\varepsilon \to 0$.

Proof. The proof follows the empirical-process argument. By the envelope theorem, the fluctuation of $\mathcal{L}_c^\varepsilon$ with respect to its first marginal is controlled by the class of entropic dual potentials. The soft c-transform smooths these potentials at spatial scale $\sqrt{\varepsilon}$ for a quadratic-type cost. Covering a bounded d-dimensional domain at this scale gives an effective complexity of order $\varepsilon^{-d/2}$. Standard Rademacher or Dudley entropy bounds then give an empirical-process fluctuation of order $\varepsilon^{-d/2}/\sqrt{n}$ for each marginal. Applying the same estimate to the three terms defining the debiased divergence gives the stated bound. □

Remark 21 (No free lunch when approximating exact OT).

$$\left|\bar{\mathcal{L}}_c^\varepsilon(\alpha,\beta) - \mathcal{L}_c(\alpha,\beta)\right| \leqslant C\varepsilon, \qquad \mathbb{E}\left|\bar{\mathcal{L}}_c^\varepsilon(\hat{\alpha}_n,\hat{\beta}_n) - \bar{\mathcal{L}}_c^\varepsilon(\alpha,\beta)\right| \leqslant C_{c,d}\varepsilon^{-d/2}n^{-1/2}.$$

Balancing the two terms gives $\varepsilon \simeq n^{-1/(d+2)}$ and total error of order $n^{-1/(d+2)}$. Equivalently, target accuracy η requires choosing $\varepsilon \simeq \eta$ and $n \simeq \eta^{-(d+2)}$ samples under this bound.

9 Extensions

9.1 Unbalanced OT

Relaxed formulation.

$$\mathrm{UW}_c(\alpha,\beta) = \inf_{\pi\in\mathcal{M}_+(\mathcal{X}\times\mathcal{Y})} \int_{\mathcal{X}\times\mathcal{Y}} c(x,y)\mathrm{d}\pi(x,y) + \mathcal{D}_{\psi_1}(\pi_1|\alpha) + \mathcal{D}_{\psi_2}(\pi_2|\beta) \tag{87}$$

$\mathrm{UW}_{c,\tau}(\alpha,\beta) = \inf_{\pi\geqslant 0}\int c\mathrm{d}\pi + \tau\mathcal{D}_{\bar\psi_1}(\pi_1|\alpha) + \tau\mathcal{D}_{\bar\psi_2}(\pi_2|\beta)$.

Proposition 51 (Small-transport-scale limit for KL penalties)**.** *Assume $c\geqslant 0$ and $c(x,x)=0$ on a common space, and take $\bar\psi_1=\bar\psi_2=\mathrm{KL}$. Then*

$$\lim_{\tau\downarrow 0}\frac{1}{\tau}\mathrm{UW}_{c,\tau}(\alpha,\beta) = \inf_{\rho\in\mathcal{M}_+(\mathcal{X})}\mathrm{KL}(\rho|\alpha)+\mathrm{KL}(\rho|\beta) = \int(\sqrt{\rho_\alpha}-\sqrt{\rho_\beta})^2,$$

when α,β have densities with respect to a common reference measure. Thus the KL marginal relaxation contains the squared Hellinger distance as its local mass-variation limit.

Proof. For the upper bound, restrict to diagonal plans $\pi=(\mathrm{Id},\mathrm{Id})_\sharp\rho$, whose transport cost is zero and whose two marginals are both ρ. Conversely, after division by τ, any transport cost is nonnegative and can be discarded in the lower bound, so only the best common marginal survives at the scale τ. For KL, minimizing pointwise over the density r of ρ gives $r\log(r/a)-r+a+r\log(r/b)-r+b$. The optimality condition is $\log(r/a)+\log(r/b)=0$, hence $r=\sqrt{ab}$, and the minimum is $a+b-2\sqrt{ab}=(\sqrt{a}-\sqrt{b})^2$. □

Proposition 52 (Dual of unbalanced optimal transport)**.** *One has equality between* (87) *and* $\mathrm{UW}_c(\alpha,\beta) = \sup_{f\oplus g\leqslant c} -\mathcal{D}^*_{\psi_1}(-f|\alpha)-\mathcal{D}^*_{\psi_2}(-g|\beta)$.

Proof. One gets a dual by using the formula (79) for the dual of a divergence

$$\inf_{\pi\in\mathcal{M}_+(\mathcal{X}\times\mathcal{Y})}\ \sup_{(f,g)\in\mathcal{C}(\mathcal{X})\times\mathcal{C}(\mathcal{Y})} \langle c,\,\pi\rangle + \langle -f,\,\pi_1\rangle + \langle -g,\,\pi_2\rangle - \mathcal{D}^*_{\psi_1}(-f|\alpha) - \mathcal{D}^*_{\psi_2}(-g|\beta).$$

Exchanging the min and the max gives

$$\sup_{(f,g)} -\mathcal{D}^*_{\psi_1}(-f|\alpha)-\mathcal{D}^*_{\psi_2}(-g|\beta) + \inf_{\pi\geqslant 0}\langle c-(f\oplus g),\,\pi\rangle = \sup_{f\oplus g\leqslant c} -\mathcal{D}^*_{\psi_1}(-f|\alpha)-\mathcal{D}^*_{\psi_2}(-g|\beta).$$

□

Reverse formulation.

$$\begin{aligned}&\int_{\mathcal{X}\times\mathcal{Y}} c(x,y)\mathrm{d}\pi(x,y) + \mathcal{D}_{\psi_1}(\pi_1|\alpha) + \mathcal{D}_{\psi_2}(\pi_2|\beta)\\ &= \int_{\mathcal{X}\times\mathcal{Y}}\left(c(x,y)+\psi_1\left(\frac{\mathrm{d}\pi_1}{\mathrm{d}\alpha}(x)\right)\frac{\mathrm{d}\alpha}{\mathrm{d}\pi_1}(x)+\psi_2\left(\frac{\mathrm{d}\pi_2}{\mathrm{d}\beta}(y)\right)\frac{\mathrm{d}\beta}{\mathrm{d}\pi_2}(y)\right)\mathrm{d}\pi(x,y).\end{aligned}$$

$$\forall\,(c,r,s)\in\mathbb{R}^3,\quad L_c(r,s) \coloneqq c + r\psi_1(1/r) + s\psi_2(1/s), \tag{88}$$

where we implicitly assume that $r\psi_1(1/r) = (\psi_1')^\infty$ when $r=0$, one has
$\mathrm{UW}_c(\alpha,\beta) = \inf_{\pi\in\mathcal{M}_+(\mathcal{X}\times\mathcal{Y})}\int L_{c(x,y)}(F(x),G(y))\mathrm{d}\pi(x,y) + \psi_1(0)\alpha^\perp(\mathcal{X}) + \psi_2(0)\beta^\perp(\mathcal{Y})$ $\alpha = F\pi_1+\alpha^\perp$ and $\beta = G\pi_2+\beta^\perp$. Thus $F=\mathrm{d}\alpha/\mathrm{d}\pi_1$ and $G=\mathrm{d}\beta/\mathrm{d}\pi_2$ on the absolutely continuous parts.

Homogeneous formulation.

$$H_c(r,s) \coloneqq \inf_{\theta>0}\theta L_c(r/\theta,s/\theta). \tag{89}$$

$$\mathrm{HW}_c(\alpha,\beta) = \inf_{\pi\in\mathcal{M}_+(\mathcal{X}\times\mathcal{Y})}\int H_{c(x,y)}(F(x),G(y))\mathrm{d}\pi(x,y) + \psi_1(0)\alpha^\perp(\mathcal{X}) + \psi_2(0)\beta^\perp(\mathcal{Y}). \tag{90}$$

Proposition 53 (Homogenization does not change the unbalanced cost)**.** *One has* $\mathrm{HW}_c(\alpha,\beta)=\mathrm{UW}_c(\alpha,\beta)$.

Proof. The inequality $\mathrm{HW}\leqslant\mathrm{UW}$ follows from $H_c\leqslant L_c$. For the reverse inequality, take a feasible measure π in the homogeneous formulation. By definition of the perspective transform, for every (x,y) and every $\eta>0$ there exists a scale $\theta(x,y)>0$ such that $H_{c(x,y)}(F(x),G(y))+\eta\geqslant\theta(x,y)L_{c(x,y)}\big(F(x)/\theta(x,y),G(y)/\theta(x,y)\big)$. Replacing π by the rescaled measure $\tilde\pi=\theta\pi$ and the densities by F/θ and G/θ gives an admissible competitor for the reverse formulation with cost no larger than the homogeneous cost plus $\eta\pi(\mathcal{X}\times\mathcal{Y})$. Letting $\eta\to 0$ yields $\mathrm{UW}\leqslant\mathrm{HW}$. The singular terms are unchanged because the same rescaling is performed before taking the Lebesgue decomposition of the marginals. □

Conic lifting. The last formulation is obtained by lifting the above optimization problem on the cone space $\mathfrak{C}[\mathcal{X}] := (\mathcal{X} \times \mathbb{R}_+)/\sim$ associated to $\mathcal{X}$ (and similarly for $\mathcal{Y}$). Here, the equivalence relation $\sim$ "glues" together all the points $(x, 0)$, which are often called the "apex" of the cone.
$\Delta((x,r),(y,s)) := H_{c(x,y)}(r^p, s^p)^{1/p}$.

- $\mathcal{D}_\psi = \mathrm{KL}$, $p = 2$ and $c(x,y) = -\log\cos^2(d(x,y) \wedge \frac{\pi}{2})$ in which case Δ is the so-called cone metric associated to d and reads $\Delta((x,r),(y,s))^2 = r^2 + s^2 - 2rs\cos(d(x,y) \wedge \frac{\pi}{2})$.
- $\mathcal{D}_\psi = \mathrm{KL}$, $p = 2$ and $c(x,y) = d(x,y)^2$, so that $\Delta((x,r),(y,s))^2 = r^2 + s^2 - 2rse^{-d(x,y)^2/2}$.
- $\mathcal{D}_\psi = \mathrm{TV}$, $p = 1$ and $c(x,y) = d(x,y)$, so that $\Delta((x,r),(y,s)) = r + s - (r \wedge s)(2 - d(x,y))_+$.

$$\mathrm{CW}(\alpha,\beta) = \inf_{\gamma \in \mathcal{M}_+(\mathfrak{C}[\mathcal{X}]^2)} \left\{ \int \Delta((x,r),(y,s))^p \mathrm{d}\gamma((x,r),(y,s))\ ;\ \int r^p \mathrm{d}\gamma_1(\cdot, r) = \alpha,\ \int s^p \mathrm{d}\gamma_2(\cdot, s) = \beta \right\}.$$

Theorem 6 (Cone formulation of unbalanced OT)**.** *One has* $\mathrm{UW} = \mathrm{HW} = \mathrm{CW}$*. If* Δ *is a distance, so is* $\mathrm{CW}^{1/p}$*.*

Proof. The equality $\mathrm{UW} = \mathrm{HW}$ is the previous proposition. To prove $\mathrm{HW} = \mathrm{CW}$, disintegrate an admissible cone coupling γ with respect to its spatial variables (x,y) and radii (r,s). The cone marginal constraints say precisely that the spatial marginals are recovered after weighting by r^p and s^p. Because $\Delta((x,r),(y,s))^p = H_{c(x,y)}(r^p, s^p)$, integrating the cone cost gives the homogeneous objective. Conversely, any homogeneous competitor can be lifted to the cone by placing, over each (x,y), radii whose p-th powers are the two density factors appearing in H_c.
If Δ is a distance on the cone, then $\mathrm{CW}^{1/p}$ is the usual p-Wasserstein distance between the lifted measures under the linear cone-marginal constraints. Symmetry and the triangle inequality follow from the corresponding Wasserstein properties and the gluing lemma on the cone. If the distance is zero, an optimal cone coupling is concentrated on the diagonal of the cone, so the weighted projections agree and therefore $\alpha = \beta$. □

Entropic regularization.

$$\inf_{\pi \in \mathcal{M}_+(\mathcal{X}\times\mathcal{Y})} \int_{\mathcal{X}\times\mathcal{Y}} c(x,y)\mathrm{d}\pi(x,y) + \mathcal{D}_{\psi_1}(\pi_1|\alpha) + \mathcal{D}_{\psi_2}(\pi_2|\beta) + \varepsilon\mathcal{D}_\varphi(\pi|\alpha\otimes\beta)$$

$\sup_{f,g}\ -\mathcal{D}^*_{\psi_1}(-f|\alpha) - \mathcal{D}^*_{\psi_2}(-g|\beta) - \varepsilon\mathcal{D}^*_\varphi\left(\frac{f\oplus g - c}{\varepsilon}\right)$. When using $\mathcal{D}_\varphi = \mathrm{KL}$, the primal-dual relation reads $\pi = e^{\frac{f\oplus g - c}{\varepsilon}}$ and when also considering $\mathcal{D}_{\psi_1} = \mathcal{D}_{\psi_2} = \tau\,\mathrm{KL}$, the dual reads

$$\sup_{f,g}\ -\tau\left(\langle e^{\frac{-f}{\tau}}, \alpha\rangle + \langle e^{\frac{-g}{\tau}}, \beta\rangle - 2\right) - \varepsilon\iint_{\mathcal{X}\times\mathcal{Y}} e^{\frac{f\oplus g - c}{\varepsilon}}\mathrm{d}\alpha\otimes\mathrm{d}\beta + \varepsilon.$$

$$f = -\frac{\tau\varepsilon}{\tau+\varepsilon}\log\int_{\mathcal{Y}}\exp\left(\frac{g(y) - c(\cdot,y)}{\varepsilon}\right)\mathrm{d}\beta(y),$$
$$g = -\frac{\tau\varepsilon}{\tau+\varepsilon}\log\int_{\mathcal{X}}\exp\left(\frac{f(x) - c(x,\cdot)}{\varepsilon}\right)\mathrm{d}\alpha(x).$$

9.2 OT Barycenters

Fréchet means. For discrete input histograms $\{\mathrm{b}_s\}_{s=1}^S$, with $\mathrm{b}_s \in \Sigma_{n_s}$, and weights $\lambda \in \Sigma_S$, a Wasserstein barycenter can be computed by minimizing

$$\min_{\mathrm{a}\in\Sigma_n} \sum_{s=1}^S \lambda_s \mathrm{L}_{\mathrm{C}_s}(\mathrm{a}, \mathrm{b}_s), \tag{91}$$

where the cost matrices $\mathrm{C}_s \in \mathbb{R}^{n\times n_s}$ are prescribed.
Given a set of input measures $(\beta_s)_s$ defined on some space $\mathcal{X}$, the barycenter problem becomes

$$\min_{\alpha\in\mathcal{M}_+^1(\mathcal{X})} \sum_{s=1}^S \lambda_s \mathcal{L}_c(\alpha, \beta_s). \tag{92}$$

Remark 22 (Two measures give a Wasserstein geodesic).
Example 8 (Dirac inputs). Problem (92) generalizes the computation of barycenters of points $(x_s)_{s=1}^S \in \mathcal{X}^S$ to arbitrary measures. Indeed, if $\beta_s = \delta_{x_s}$ is a single Dirac mass, then a solution to (92) is $\delta_{x^\star}$ where $x^\star$ is a Fréchet mean of the points $(x_s)_s$.
Remark 23 (Mean of a quadratic barycenter). For $c(x,y) = \|x-y\|^2$, the mean of the barycenter $\alpha^\star$ is necessarily the barycenter of the means,
$\int_{\mathcal{X}} x\mathrm{d}\alpha^\star(x) = \sum_s \lambda_s \int_{\mathcal{X}} x\mathrm{d}\beta_s(x)$.
Proposition 6 (Convexity of the OT cost)**.** $(\alpha,\beta) \mapsto \mathcal{L}_c(\alpha,\beta)$ *is convex.*

Proof. Let (α_0, β_0) and (α_1, β_1) be two pairs of probability measures and let $t \in [0,1]$. For $\varepsilon > 0$, choose couplings $\pi_i \in \mathcal{U}(\alpha_i, \beta_i)$ such that $\int c\mathrm{d}\pi_i \leqslant \mathcal{L}_c(\alpha_i,\beta_i) + \varepsilon \quad \text{for} \quad i = 0, 1$. Then $\pi_t := (1-t)\pi_0 + t\pi_1$ is a coupling between $(1-t)\alpha_0 + t\alpha_1$ and $(1-t)\beta_0 + t\beta_1$. Hence $\mathcal{L}_c((1-t)\alpha_0 + t\alpha_1, (1-t)\beta_0 + t\beta_1) \leqslant (1-t)\mathcal{L}_c(\alpha_0,\beta_0) + t\mathcal{L}_c(\alpha_1,\beta_1) + \varepsilon$. Letting $\varepsilon \to 0$ gives the claim. □

1-D Case.

Proposition 54 (Quantile barycenters on the line)**.** *For $\mathcal{X} = \mathbb{R}$ and $c(x,y) = |x-y|^2$, the quantile function of a Wasserstein barycenter is the weighted average of the input quantile functions: $\mathcal{C}_{\alpha^\star}^{-1}(r) = \sum_{s=1}^S \lambda_s \mathcal{C}_{\beta_s}^{-1}(r) \quad$ for $\quad r \in [0,1]$.*

Proof. The one-dimensional formula (11) gives $\sum_s \lambda_s \mathcal{W}_2^2(\alpha, \beta_s) = \int_0^1 \sum_s \lambda_s \left|\mathcal{C}_\alpha^{-1}(r) - \mathcal{C}_{\beta_s}^{-1}(r)\right|^2 \mathrm{d}r$. The minimization decouples pointwise in r. For each fixed r, the minimizer of $z \mapsto \sum_s \lambda_s |z - \mathcal{C}_{\beta_s}^{-1}(r)|^2$ is the weighted average $\sum_s \lambda_s \mathcal{C}_{\beta_s}^{-1}(r)$. This function is nondecreasing because it is a positive weighted sum of nondecreasing quantile functions, hence it is itself a valid quantile function. □

Gaussian Case.

Remark 24 (Gaussian barycenters). $\boldsymbol{\Sigma} \mapsto \sum_s \lambda_s \mathcal{B}(\boldsymbol{\Sigma}, \boldsymbol{\Sigma}_s)^2$, $\boldsymbol{\Sigma} = \sum_s \lambda_s \left(\boldsymbol{\Sigma}^{1/2} \boldsymbol{\Sigma}_s \boldsymbol{\Sigma}^{1/2}\right)^{1/2}$.

Sinkhorn for barycenters.

$$\min_{\mathrm{a} \in \Sigma_n} \sum_{s=1}^S \lambda_s \mathrm{L}^\varepsilon_{\mathrm{C}_s}(\mathrm{a}, \mathrm{b}_s) \tag{93}$$

$$\min_{(\mathrm{P}_s)_s} \left\{ \sum_s \lambda_s \mathrm{KL}(\mathrm{P}_s | \mathrm{K}_s) \;;\; \forall s, \mathrm{P}_s{}^\top \mathbb{1}_n = \mathrm{b}_s,\ \mathrm{P}_1 \mathbb{1}_{n_1} = \ldots = \mathrm{P}_S \mathbb{1}_{n_S} \right\} \tag{94}$$

where we denoted $\mathrm{K}_s \coloneqq e^{-\mathrm{C}_s/\varepsilon}$. Here, the barycenter a is implicitly encoded in the row marginals of all the couplings $\mathrm{P}_s \in \mathbb{R}^{n \times n_s}$ as $\mathrm{a} = \mathrm{P}_1 \mathbb{1}_{n_1} = \ldots = \mathrm{P}_S \mathbb{1}_{n_S}$.

$$\mathrm{P}_s = \mathrm{diag}(\mathrm{u}_s) \mathrm{K}_s \,\mathrm{diag}(\mathrm{v}_s), \tag{95}$$

$$\forall\, s \in [\![1, S]\!], \quad \mathrm{v}_s^{(\ell+1)} \coloneqq \frac{\mathrm{b}_s}{\mathrm{K}_s{}^\top \mathrm{u}_s^{(\ell)}}, \tag{96}$$

$$\forall\, s \in [\![1, S]\!], \quad \mathrm{u}_s^{(\ell+1)} \coloneqq \frac{\mathrm{a}^{(\ell+1)}}{\mathrm{K}_s \mathrm{v}_s^{(\ell+1)}}, \tag{97}$$

$$\text{where} \quad \mathrm{a}^{(\ell+1)} \coloneqq \prod_s (\mathrm{K}_s \mathrm{v}_s^{(\ell+1)})^{\lambda_s}. \tag{98}$$

Proposition 55 (Dual of entropic barycenters)**.** *The optimal $(\mathrm{u}_s, \mathrm{v}_s)$ appearing in (95) can be written as $(\mathrm{u}_s, \mathrm{v}_s) = (e^{\mathrm{f}_s/\varepsilon}, e^{\mathrm{g}_s/\varepsilon})$ where $(\mathrm{f}_s, \mathrm{g}_s)_s$ are the solutions of the following program (whose value matches the one of (93))*

$$\max_{(\mathrm{f}_s, \mathrm{g}_s)_s} \left\{ \sum_s \lambda_s \left(\langle \mathrm{g}_s,\, \mathrm{b}_s \rangle - \varepsilon \langle \mathrm{K}_s e^{\mathrm{g}_s/\varepsilon},\, e^{\mathrm{f}_s/\varepsilon} \rangle \right) \;;\; \sum_s \lambda_s \mathrm{f}_s = 0 \right\}. \tag{99}$$

Proof. Introducing Lagrange multipliers in (94) leads to

$$\min_{(\mathrm{P}_s)_s, \mathrm{a}} \max_{(\mathrm{f}_s, \mathrm{g}_s)_s} \sum_s \lambda_s \Big(\varepsilon \mathrm{KL}(\mathrm{P}_s | \mathrm{K}_s) + \langle \mathrm{a} - \mathrm{P}_s \mathbb{1}_{n_s},\, \mathrm{f}_s \rangle + \langle \mathrm{b}_s - \mathrm{P}_s{}^\top \mathbb{1}_n,\, \mathrm{g}_s \rangle \Big).$$

Strong duality holds, so one can exchange the minimum and the maximum and obtain

$$\max_{(\mathrm{f}_s, \mathrm{g}_s)_s} \sum_s \lambda_s \left(\langle \mathrm{g}_s,\, \mathrm{b}_s \rangle + \min_{\mathrm{P}_s} \varepsilon \mathrm{KL}(\mathrm{P}_s | \mathrm{K}_s) - \langle \mathrm{P}_s,\, \mathrm{f}_s \oplus \mathrm{g}_s \rangle \right) + \min_{\mathrm{a}} \langle \sum_s \lambda_s \mathrm{f}_s,\, \mathrm{a} \rangle.$$

The explicit minimization on a gives the constraint $\sum_s \lambda_s \mathrm{f}_s = 0$ together with

$$\max_{(\mathrm{f}_s, \mathrm{g}_s)_s} \sum_s \lambda_s \langle \mathrm{g}_s,\, \mathrm{b}_s \rangle - \varepsilon \mathrm{KL}^* \left(\frac{\mathrm{f}_s \oplus \mathrm{g}_s}{\varepsilon} | \mathrm{K}_s \right)$$

where $\mathrm{KL}^*(\cdot | \mathrm{K}_s)$ is the Legendre transform (79) of the function $\mathrm{KL}(\cdot | \mathrm{K}_s)$.
This Legendre transform reads

$$\mathrm{KL}^*(\mathrm{U} | \mathrm{K}) = \sum_{i,j} \mathrm{K}_{i,j} (e^{\mathrm{U}_{i,j}} - 1), \tag{100}$$

which shows the desired formula.
To show (100), since this function is separable, one needs to compute $\forall\,(u,k) \in \mathbb{R} \times \mathbb{R}_+^*$, $\ \mathrm{KL}^*(u|k) \coloneqq \max_{r \geqslant 0} ur - (r \log(r/k) - r + k)$ whose optimality condition reads $u = \log(r/k)$, i.e. $r = ke^u$, hence the result. The case $k = 0$ follows by lower semicontinuity and gives zero. □

9.3 Multimarginal OT

Definition, example and properties. The multi-marginal formulation replaces a coupling between two measures by a joint distribution with several prescribed marginals. Given measures $(\alpha_s)_{s=1}^S$ on spaces $(\mathcal{X}_s)_{s=1}^S$ and a cost $c : \mathcal{X}_1 \times \cdots \times \mathcal{X}_S \to \mathbb{R}$, the problem reads $\min_{\pi \in \mathcal{U}(\alpha_1,\ldots,\alpha_S)} \int_{\mathcal{X}_1\times\cdots\times\mathcal{X}_S} c(x_1,\ldots,x_S)\mathrm{d}\pi(x_1,\ldots,x_S)$, where $\mathcal{U}(\alpha_1,\ldots,\alpha_S)$ is the set of probability measures whose s-th marginal is α_s. This is still a linear program in the discrete setting, but its ambient dimension grows exponentially with the number of marginals.

Multi-marginal formulation of barycenters. $c(x_1,\ldots,x_S) = \min_{x\in\mathbb{R}^d} \sum_{s=1}^S \lambda_s \|x - x_s\|^2$.

Proposition 56 (Multi-marginal formula for quadratic barycenters)**.** *Let $\beta_1,\ldots,\beta_S \in \mathcal{P}_2(\mathbb{R}^d)$ and $\lambda \in \Sigma_S$. Define $B(x_1,\ldots,x_S) = \sum_{s=1}^S \lambda_s x_s$, $c_{\mathrm{bar}}(x_1,\ldots,x_S) = \min_x \sum_s \lambda_s\|x - x_s\|^2$. If $\pi^\star$ solves the multi-marginal OT problem with marginals $(\beta_s)_s$ and cost c_{bar}, then $\alpha^\star = B_\sharp\pi^\star$ is a Wasserstein barycenter. Conversely, every barycenter is obtained this way from an optimal multi-marginal plan.*

Proof. For any candidate barycenter α and couplings $\pi_s \in \mathcal{U}(\alpha,\beta_s)$, glue the couplings along their common α marginal to obtain a joint law of $(X, Y_1,\ldots,Y_S)$. Conditioning on $(Y_s)_s$ and minimizing over X gives $\sum_s \lambda_s\mathbb{E}\|X - Y_s\|^2 \geqslant \mathbb{E}\min_x \sum_s \lambda_s\|x - Y_s\|^2 = \mathbb{E}c_{\mathrm{bar}}(Y_1,\ldots,Y_S)$. Taking the infimum over the couplings gives that the barycenter value is at least the multi-marginal value. Conversely, from an optimal multi-marginal plan $\pi^\star$, set $X = B(Y_1,\ldots,Y_S)$. The couplings between X and each Y_s are feasible for the barycenter problem and attain exactly the multi-marginal cost, proving equality and the formula. □

Corollary 5 (Gaussian and discrete barycenters)**.** *Quadratic Wasserstein barycenters of Gaussian measures are Gaussian. If the input measures are discrete, then there exists a barycenter supported on the set of weighted averages $\sum_s \lambda_s x_{s,i_s}$ of one support point from each input; in particular, if the s-th input has n_s atoms, a barycenter exists with at most $\prod_s n_s$ atoms.*

Proof. Let the input Gaussians have means $\mathbf{m}_s$ and covariances $\mathbf{\Sigma}_s$. For any candidate barycenter α with mean $\mathbf{m}$ and covariance $\mathbf{\Sigma}$, Gelbrich's inequality gives $\mathcal{W}_2^2(\alpha,\beta_s) \geqslant \|\mathbf{m} - \mathbf{m}_s\|^2 + \mathcal{B}(\mathbf{\Sigma},\mathbf{\Sigma}_s)^2$, with equality when α is Gaussian. Therefore the barycenter objective is bounded below by a function depending only on $(\mathbf{m},\mathbf{\Sigma})$, and this lower bound is attained by the Gaussian measure with the minimizing mean and covariance. Hence at least one barycenter is Gaussian, and uniqueness in the usual nondegenerate setting gives the Gaussian barycenter mentioned above. For discrete inputs, any multi-marginal optimizer is supported on the finite product of the input supports, and B maps this product to at most $\prod_s n_s$ points. □

Entropic regularization of multi-marginal. $\inf_{\pi\in\mathcal{U}(\alpha_1,\ldots,\alpha_S)} \int c\mathrm{d}\pi + \varepsilon\,\mathrm{KL}(\pi|\alpha_1\otimes\cdots\otimes\alpha_S)$.

$$\mathrm{d}\pi^\star(x_1,\ldots,x_S) = \exp\left(\frac{\sum_s f_s(x_s) - c(x_1,\ldots,x_S)}{\varepsilon}\right)\prod_s \mathrm{d}\alpha_s(x_s),$$

9.4 Sliced Wasserstein

For measures on $\mathbb{R}^d$, sliced Wasserstein distances replace one high-dimensional transport problem by many one-dimensional ones. For $\theta \in \mathbb{S}^{d-1}$, let $P_\theta(x) = \langle x,\,\theta\rangle$ be the projection on direction θ. The sliced distance is

$$\mathrm{SW}_p^p(\alpha,\beta) := \int_{\mathbb{S}^{d-1}} \mathcal{W}_p^p\left((P_\theta)_\sharp\alpha,(P_\theta)_\sharp\beta\right)\mathrm{d}\theta. \tag{101}$$

Proposition 57 (Metric properties of sliced Wasserstein)**.** *For $p \geqslant 1$, SW_p is a distance on $\mathcal{P}_p(\mathbb{R}^d)$. Moreover, SW_p metrizes weak convergence together with convergence of the p-th moment. Finally, $\mathrm{SW}_p(\alpha,\beta) \leqslant \mathcal{W}_p(\alpha,\beta)$, and, for $p = 2$ with the uniform probability measure on the sphere, $\mathrm{SW}_2^2(\alpha,\beta) \leqslant \frac{1}{d}\mathcal{W}_2^2(\alpha,\beta)$.*

Proof. Non-negativity and symmetry follow from the one-dimensional Wasserstein distance. For the triangle inequality, apply the triangle inequality of $\mathcal{W}_p$ for each direction θ and then Minkowski's inequality in $L^p(\mathbb{S}^{d-1})$.
If $\mathrm{SW}_p(\alpha,\beta) = 0$, then $(P_\theta)_\sharp\alpha = (P_\theta)_\sharp\beta$ for almost every direction, hence for all directions by continuity of characteristic functions. The Cramer–Wold theorem then implies $\alpha = \beta$. This proves that SW_p is a distance.
The bound $\mathrm{SW}_p \leqslant \mathcal{W}_p$ follows because P_θ is 1-Lipschitz, so $\mathcal{W}_p((P_\theta)_\sharp\alpha,(P_\theta)_\sharp\beta) \leqslant \mathcal{W}_p(\alpha,\beta)$ for every θ. For $p = 2$, using any coupling π between α and β, $\int_{\mathbb{S}^{d-1}}\int|\langle x-y,\,\theta\rangle|^2\mathrm{d}\pi(x,y)\mathrm{d}\theta = \frac{1}{d}\int\|x-y\|^2\mathrm{d}\pi(x,y)$. Optimizing over π gives the sharper displayed inequality.
The weak-convergence statement follows from the same Cramer–Wold mechanism plus the moment condition: convergence in SW_p gives convergence of almost all one-dimensional projections and tightness of the p-moments; conversely, weak convergence with p-moment convergence implies convergence of projected $\mathcal{W}_p$ distances and dominated convergence on the sphere. Detailed proofs and statistical refinements can be found in. □

Remark 25 (Hilbert embedding for SW_2). In one dimension, $\mathcal{W}_2$ is the $L^2(0,1)$ distance between quantile functions. Hence $\mathrm{SW}_2^2(\alpha,\beta) = \int_{\mathbb{S}^{d-1}}\int_0^1 \left|F_{\theta,\alpha}^{-1}(u) - F_{\theta,\beta}^{-1}(u)\right|^2 \mathrm{d}u\,\mathrm{d}\theta$, where $F_{\theta,\alpha}^{-1}$ is the quantile of $(P_\theta)_\sharp\alpha$. Thus SW_2 is a Hilbertian distance after embedding each measure into its field of projected quantiles.

Definition 14 (Max-, min- and subspace-sliced variants)**.** *The max-sliced distance replaces the average over directions by the supremum $\mathrm{MaxSW}_p(\alpha,\beta) = \sup_{\theta\in\mathbb{S}^{d-1}}\mathcal{W}_p((P_\theta)_\sharp\alpha,(P_\theta)_\sharp\beta)$. Min-sliced transport-plan variants such as min-SWGG instead choose a direction whose lifted 1-D matching has minimal full-dimensional cost; they provide a cheap transport plan and an upper bound on $\mathcal{W}_2^2$ rather than a lower-dimensional distance. Subspace-sliced variants replace lines by k-dimensional orthogonal projectors $U^\top x$, with $U^\top U = I_k$. Max-sliced variants are useful in generative modeling.*

Proposition 58 (Basic bounds for sliced variants)**.** *With normalized surface measure,* $\mathrm{SW}_p(\alpha,\beta) \leqslant \mathrm{MaxSW}_p(\alpha,\beta) \leqslant \mathcal{W}_p(\alpha,\beta)$*. For min-SWGG-type lifted plans, the induced cost is an upper bound on* $\mathcal{W}_2^2(\alpha,\beta)$*. For k-dimensional subspace projections, the corresponding projected Wasserstein distance is bounded above by* $\mathcal{W}_p$ *for every fixed subspace.*

Proof. The first inequality follows because an average is bounded by a supremum. The second and the subspace bound follow because orthogonal projections are 1-Lipschitz, so pushing any admissible coupling forward gives a feasible coupling for the projected problem with no larger cost. A lifted one-dimensional matching is an admissible coupling between the original measures; evaluating the full quadratic cost on any admissible coupling can only be larger than the optimal $\mathcal{W}_2^2$ value. □

Definition 15 (Subspace robust Wasserstein)**.** *For a coupling* π *between measures on* $\mathbb{R}^d$*, define its displacement covariance* $M_\pi = \int (x-y)(x-y)^\top \mathrm{d}\pi(x,y)$*. The subspace robust Wasserstein cost of rank k is* $\mathrm{SRW}_{2,k}^2(\alpha,\beta) := \inf_{\pi\in\mathcal{U}(\alpha,\beta)} \sum_{\ell=1}^k \lambda_\ell(M_\pi)$*, where* $\lambda_1 \geqslant \cdots \geqslant \lambda_d$ *are eigenvalues. Taking the trace, i.e. summing all eigenvalues, recovers* $\mathcal{W}_2^2$*.*

Proposition 59 (Metric behavior of robust subspace costs)**.** *The relaxed covariance-gauge formulation above is a convex minimization over couplings; replacing the trace gauge by the spectral partial-sum gauge emphasizes the largest displacement directions and satisfies* $\mathrm{SRW}_{2,k} \leqslant \mathcal{W}_2$ *when the eigenvalue sum is normalized consistently.*

Proof. Metricity of the projected max formulation follows from applying the Wasserstein metric after each orthogonal projection and taking a supremum over projections that separates points. The bound by $\mathcal{W}_2$ again follows from 1-Lipschitzness of projections. For the relaxed formulation, M_π depends linearly on π, and the sum of the largest k eigenvalues is a convex spectral gauge; minimizing a convex function over the convex coupling set gives a convex problem. Since the trace is the sum of all eigenvalues and dominates the normalized partial sum, the value is bounded by the usual quadratic OT cost. This is the finite-dimensional robust-subspace viewpoint; related spectral Wasserstein gauges give analogous constructions. □

9.5 Linear Optimal Transport

Fix a reference absolutely continuous measure ρ and let T_α be the Brenier map pushing ρ to α. The linear OT embedding is

$$\alpha \mapsto T_\alpha - \mathrm{Id} \in L^2(\rho;\mathbb{R}^d), \qquad \mathrm{LOT}_\rho(\alpha,\beta) = \|T_\alpha - T_\beta\|_{L^2(\rho)}. \tag{102}$$

Proposition 60 (Local stability of linear OT)**.** *Assume that the measures are supported on a fixed convex compact set, with densities bounded above and below, and that the Brenier maps from* ρ *are regular. Then locally around* ρ*,* $\mathcal{W}_2(\alpha,\beta) \leqslant \mathrm{LOT}_\rho(\alpha,\beta)$ *and* $\mathrm{LOT}_\rho(\alpha,\beta) \leqslant C\,\mathcal{W}_2(\alpha,\beta)^\eta$ *for constants* $C>0$ *and* $\eta\in(0,1]$ *depending on regularity.*

Proof. The first inequality is immediate: $(T_\alpha, T_\beta)_\sharp\rho$ is a feasible coupling between α and β. The reverse local estimate is a standard stability statement for the Monge–Ampère equation under the stated regularity assumptions: changes in the target measure control changes in the Brenier potential in Hölder norms, hence control $T_\alpha - T_\beta$ in $L^2(\rho)$. In simple one-dimensional settings, quantile functions make this exact with $\eta = 1$. □

9.6 Weak Optimal Transport

Given $\pi\in\mathcal{U}(\alpha,\beta)$, disintegrate it as $\pi(\mathrm{d}x,\mathrm{d}y) = \pi_x(\mathrm{d}y)\alpha(\mathrm{d}x)$. A weak OT problem has the form $\inf_{\pi\in\mathcal{U}(\alpha,\beta)} \int C(x,\pi_x)\mathrm{d}\alpha(x)$, where $C(x,\cdot)$ is convex on probability measures. The classical Kantorovich problem is recovered when $C(x,\nu) = \int c(x,y)\mathrm{d}\nu(y)$.

9.7 Gromov–Wasserstein

Proposition 61 (Euclidean GW is controlled by Wasserstein)**.** *Let* α,β *be probability measures on* $\mathbb{R}^d$*, equipped with the Euclidean distance, and take* $\Delta(u,v) = |u-v|$ *in* (104)*. Then* $\mathcal{GW}((\mathbb{R}^d,\|\cdot\|,\alpha),(\mathbb{R}^d,\|\cdot\|,\beta)) \leqslant 2\,\mathcal{W}_p(\alpha,\beta)$*.*

Proof. Let π be any coupling between α and β. For two independent pairs $(X,Y),(X',Y')\sim\pi$, the reverse triangle inequality gives $\big|\|X-X'\|-\|Y-Y'\|\big| \leqslant \|X-Y\|+\|X'-Y'\|$. Taking the L^p norm and using Minkowski gives a bound by $2(\int\|x-y\|^p\mathrm{d}\pi)^{1/p}$. Optimizing over π proves the claim. □

Optimal transport needs a ground cost C to compare histograms (a, b), and thus cannot be used directly if the histograms are not defined on the same underlying space, or if one cannot pre-register these spaces to define a ground cost.
To address this issue, one can instead use a weaker requirement: two matrices $\mathrm{D}\in\mathbb{R}^{n\times n}$ and $\mathrm{D}'\in\mathbb{R}^{m\times m}$ are available and represent relationships between the points on which the histograms are defined. A typical scenario is when these matrices are powers of distance matrices.

$$\mathrm{GW}((\mathrm{a},\mathrm{D}),(\mathrm{b},\mathrm{D}'))^p := \min_{\mathrm{P}\in\mathrm{U}(\mathrm{a},\mathrm{b})} \mathcal{E}_{\mathrm{D},\mathrm{D}'}(\mathrm{P}) := \sum_{i,j,i',j'} \Delta(\mathrm{D}_{i,i'},\mathrm{D}'_{j,j'})^p \mathrm{P}_{i,j}\mathrm{P}_{i',j'}, \tag{103}$$

where $p\geqslant 1$ and Δ is a distance on $\mathbb{R}$, typically $\Delta(u,v) = |u-v|$.

General setting. The general setting corresponds to computing couplings between metric measure spaces $\mathbb{X} = (\mathcal{X}, d_\mathcal{X}, \alpha)$ and $\mathbb{Y} = (\mathcal{Y}, d_\mathcal{Y}, \beta)$ where $(d_\mathcal{X}, d_\mathcal{Y})$ are distances and (α,β) are measures on their respective spaces.

$$\mathcal{GW}(\mathbb{X},\mathbb{Y})^p := \min_{\pi\in\mathcal{U}(\alpha,\beta)} \int_{\mathcal{X}^2\times\mathcal{Y}^2} \Delta(d_\mathcal{X}(x,x'), d_\mathcal{Y}(y,y'))^p \mathrm{d}\pi(x,y)\mathrm{d}\pi(x',y'). \tag{104}$$

Theorem 7 (Gromov–Wasserstein metric modulo isometries)**.** *$\mathcal{GW}$ defines a distance between metric measure spaces up to isometries.*

Proof. If $\mathcal{GW}(\mathbb{X},\mathbb{Y})=0$ and π is an optimal plan, then $d_{\mathcal{X}}(x,x')=d_{\mathcal{Y}}(y,y')$ holds $\pi\otimes\pi$-almost everywhere. By continuity, this equality holds on $\mathrm{supp}(\pi)^2$.
$d_\pi((x,y),(x',y')) \coloneqq \frac{1}{2}d_{\mathcal{X}}(x,x')+\frac{1}{2}d_{\mathcal{Y}}(y,y')$.
One has, for $((x,y),(x',y'))\in\mathrm{supp}(\pi)^2$, $d_{\mathcal{X}}(x,x')=d_{\mathcal{Y}}(y,y')$ and thus $d_{\mathcal{X}}(\psi(x,y),\psi(x',y'))=d_{\mathcal{X}}(x,x')=d_{\mathcal{Y}}(y,y')=d_\pi((x,y),(x',y'))$, thus ψ is an isometry, hence injective. Indeed, since $\psi_\sharp\pi=\alpha$, there exists a sequence $(x_k,y_k)\in\mathrm{supp}(\pi)$ such that $x_k\to x$.
Since $d_{\mathcal{X}}(x_k,x_{k'})=d_{\mathcal{Y}}(y_k,y_{k'})$ the sequence $(y_k)_k$ is Cauchy, and since $\mathrm{supp}(\pi)$ is closed, it is complete and thus (x_k,y_k) is converging to some $(x,y)\in\mathrm{supp}(\pi)$. The same argument for the second projection shows that this support space is also isometric to $\mathbb{Y}$.
Given an optimal π and ξ between three spaces $(\mathcal{X},\mathcal{Y},\mathcal{Z})$ with measures (α,β,γ), we use the gluing Lemma 2 to define σ with marginals $(P_{\mathcal{X},\mathcal{Y}})_\sharp\sigma=\pi$ and $(P_{\mathcal{Y},\mathcal{Z}})_\sharp\sigma=\xi$. We denote $\rho\coloneqq(P_{\mathcal{X},\mathcal{Z}})_\sharp\sigma$ the composition of the two couplings, so that

$$\begin{aligned}
\mathcal{GW}(\mathbb{X},\mathbb{Z}) &\leqslant \Big(\int_{\mathcal{X}\times\mathcal{Z}}\Delta(d_{\mathcal{X}}(x,x'),d_{\mathcal{Z}}(z,z'))^p\mathrm{d}\rho(x,z)\mathrm{d}\rho(x',z')\Big)^{1/p}\\
&\leqslant \Big(\int_{\mathcal{X}\times\mathcal{Y}\times\mathcal{Z}}(\Delta(d_{\mathcal{X}}(x,x'),d_{\mathcal{Y}}(y,y'))+\Delta(d_{\mathcal{Y}}(y,y'),d_{\mathcal{Z}}(z,z')))^p\mathrm{d}\sigma(x,y,z)\mathrm{d}\sigma(x',y',z')\Big)^{1/p}\\
&\leqslant \Big(\int_{\mathcal{X}\times\mathcal{Y}\times\mathcal{Z}}\Delta(d_{\mathcal{X}}(x,x'),d_{\mathcal{Y}}(y,y'))^p\mathrm{d}\sigma(x,y,z)\mathrm{d}\sigma(x',y',z')\Big)^{1/p}\\
&\quad+\Big(\int_{\mathcal{X}\times\mathcal{Y}\times\mathcal{Z}}\Delta(d_{\mathcal{Y}}(y,y'),d_{\mathcal{Z}}(z,z'))^p\mathrm{d}\sigma(x,y,z)\mathrm{d}\sigma(x',y',z')\Big)^{1/p}\\
&\leqslant \mathcal{GW}(\mathbb{X},\mathbb{Y})+\mathcal{GW}(\mathbb{Y},\mathbb{Z}),
\end{aligned}$$

where the first inequality follows from the feasibility of ρ, the second is the triangular inequality of Δ, and the third is the Minkowski inequality. ☐

Remark 26 (Gromov–Wasserstein geodesics). $d_t((x_0,x_1),(x_0',x_1'))\coloneqq(1-t)d_{\mathcal{X}_0}(x_0,x_0')+td_{\mathcal{X}_1}(x_1,x_1')$.

Entropic regularization and iterative solver.

$$\min_{\mathrm{P}\in\mathrm{U}(\mathrm{a},\mathrm{b})}\mathcal{E}_{\mathrm{D},\mathrm{D}'}(\mathrm{P})-\varepsilon\mathrm{H}(\mathrm{P}). \tag{105}$$

$$\mathrm{P}^{(\ell+1)}\coloneqq\min_{\mathrm{P}\in\mathrm{U}(\mathrm{a},\mathrm{b})}\langle\mathrm{P},\,\mathrm{C}^{(\ell)}\rangle-\varepsilon H(\mathrm{P})\quad\text{where} \tag{106}$$

$\mathrm{C}^{(\ell)}\coloneqq\mathrm{D}^{\odot 2}\mathrm{a}\,\mathbb{1}_m^\top+\mathbb{1}_n(\mathrm{D}'^{\odot 2}\mathrm{b})^\top-2\mathrm{D}\,\mathrm{P}^{(\ell)}\,\mathrm{D}'^\top$,

9.8 Metric learning and inverse OT

Metric learning and derivatives of OT.
Proposition 62 (Derivative with respect to the cost)**.** *In the discrete setting, assume that the optimal coupling for $\mathrm{L}_{\mathrm{C}}(\mathrm{a},\mathrm{b})$ is unique and denote it by $\mathrm{P}^\star(\mathrm{C})$. Then $\mathrm{C}\mapsto\mathrm{L}_{\mathrm{C}}(\mathrm{a},\mathrm{b})$ is differentiable at C and $\nabla_{\mathrm{C}}\mathrm{L}_{\mathrm{C}}(\mathrm{a},\mathrm{b})=\mathrm{P}^\star(\mathrm{C})$.*

Proof. The value is the minimum of affine functions of C, $\mathrm{L}_{\mathrm{C}}(\mathrm{a},\mathrm{b})=\min_{\mathrm{P}\in\mathrm{U}(\mathrm{a},\mathrm{b})}\langle\mathrm{C},\,\mathrm{P}\rangle$. The envelope theorem, or equivalently Danskin's theorem, states that the subdifferential with respect to C is the convex hull of the optimal couplings. If the optimizer is unique, this subdifferential is the singleton $\{\mathrm{P}^\star(\mathrm{C})\}$, hence the value is differentiable with the displayed gradient. ☐

Thus, if the cost is parameterized as C_θ, gradients of losses involving OT values are obtained by backpropagating through the inner product $\langle\mathrm{P}^\star(\mathrm{C}_\theta),\,\partial_\theta\mathrm{C}_\theta\rangle$. The difficulty is not differentiating a solved OT problem, but learning a cost for which the resulting matching has the desired semantic behavior; this is a bilevel and usually non-convex optimization problem.

Inverse Optimal Transport. $\inf_{\pi\in\mathcal{U}(\alpha,\beta)}\int c(x,y)\mathrm{d}\pi(x,y)$. A useful statistical methodology is to measure the suboptimality of the observed plan through a Fenchel–Young loss. Write the score as $s=-c$ and define the convex regularized prediction value
$G_\varepsilon(s)=\sup_{\pi\in\mathcal{U}(\alpha,\beta)}\int s\mathrm{d}\pi-\varepsilon\,\mathrm{KL}(\pi|\alpha\otimes\beta)$. $\mathcal{L}_\varepsilon(c;\widehat{\pi})=G_\varepsilon(-c)+G_\varepsilon^*(\widehat{\pi})+\int c\mathrm{d}\widehat{\pi}$ is nonnegative by Fenchel's inequality and vanishes exactly when $\widehat{\pi}\in\partial G_\varepsilon(-c)$, i.e. when $\widehat{\pi}$ satisfies the regularized optimality conditions for c. Entropic regularization is important here because it makes the forward map smoother and provides gradients with respect to c, at the price of a bias that must be controlled statistically.
$C_\theta=\sum_{r=1}^R\theta_rC^{(r)}$, $\quad\theta\in\Theta$, where Θ is convex and the matrices $C^{(r)}$ encode features, graph distances or a Mahalanobis parameterization.
Proposition 63 (Convex dual-gap formulation of inverse OT)**.** *Let $\widehat{P}\in\mathrm{U}(a,b)$ be an observed coupling and let C_θ depend affinely on $\theta\in\Theta$, where Θ is convex. The condition that $\widehat{P}$ is optimal for the cost C_θ is equivalent to the existence of dual potentials (f,g) such that $f_i+g_j\leqslant(C_\theta)_{i,j}$ and $\sum_{i,j}\widehat{P}_{i,j}\big((C_\theta)_{i,j}-f_i-g_j\big)=0$.*

$$\min_{\theta\in\Theta,f,g}R(\theta)+\lambda\sum_{i,j}\widehat{P}_{i,j}\big((C_\theta)_{i,j}-f_i-g_j\big)\quad\textit{subject to}\quad f_i+g_j\leqslant(C_\theta)_{i,j}\quad\forall i,j. \tag{107}$$

Proof. For a fixed cost C_θ, Kantorovich duality gives $\min_{P\in \mathrm{U}(a,b)}\langle C_\theta,\, P\rangle = \max_{f_i+g_j\leqslant (C_\theta)_{i,j}}\langle f,\, a\rangle+\langle g,\, b\rangle$. Since $\widehat{P}$ has marginals (a,b), every dual feasible pair satisfies $\langle C_\theta,\, \widehat{P}\rangle - \langle f,\, a\rangle - \langle g,\, b\rangle = \sum_{i,j}\widehat{P}_{i,j}\big((C_\theta)_{i,j}-f_i-g_j\big)\geqslant 0$. It vanishes if and only if $\widehat{P}$ reaches the dual value and is therefore optimal. If C_θ is affine and Θ and R are convex, the constraints and objective in (107) are convex, proving the relaxation claim. $\square$

$\widehat{P}_{i,j}\approx a_ib_j\exp\left(\frac{f_i+g_j-(C_\theta)_{i,j}}{\varepsilon}\right)$,

10 Wasserstein (gradient) Flows

10.1 Evolutions over the Space of Measures

$$\frac{\mathrm{d}x(t)}{\mathrm{d}t} = v_t(x(t)), \tag{108}$$

$$\frac{\partial \alpha_t}{\partial t} + \mathrm{div}(v_t\alpha_t) = 0. \tag{109}$$

This PDE is only valid if α_t has a smooth density, and for general measures, this PDE (109) should be understood in the weak sense, allowing it to be defined even for discrete measures with particles evolving according to (108). Specifically, for any smooth test function $(t,x)\to\varphi(t,x)$ supported in time (in particular it should vanish at $t=0,1$),

$$\int_0^1\int_{\mathbb{R}^d}\left(\partial_t\varphi(t,x)+\langle v_t(x),\nabla_x\varphi(t,x)\rangle\right)\mathrm{d}\alpha_t(x)\mathrm{d}t = 0. \tag{110}$$

Proposition 64 (Lagrangian flows solve the continuity equation)**.** *Consider a smooth flow $T_t:\mathbb{R}^d\to\mathbb{R}^d$ and denote $\alpha_t := (T_t)_\sharp\alpha_0$. Define the Eulerian velocity field by $v_t(T_t(y)) := \partial_t T_t(y)$. Then (α_t, v_t) solves the continuity equation in the weak sense* (110)*.*

Proof. Let $\varphi(t,x)$ be a smooth test function vanishing at $t=0$ and $t=1$. Using $\alpha_t=(T_t)_\sharp\alpha_0$ and the chain rule,

$$\frac{\mathrm{d}}{\mathrm{d}t}\int\varphi(t,x)\mathrm{d}\alpha_t(x) = \frac{\mathrm{d}}{\mathrm{d}t}\int\varphi(t,T_t(y))\mathrm{d}\alpha_0(y) = \int\left(\partial_t\varphi(t,T_t(y))+\langle\nabla_x\varphi(t,T_t(y)),\,\partial_tT_t(y)\rangle\right)\mathrm{d}\alpha_0(y).$$

By the definition of v_t, this is

$$\int\left(\partial_t\varphi(t,x)+\langle\nabla_x\varphi(t,x),\,v_t(x)\rangle\right)\mathrm{d}\alpha_t(x).$$

Integrating in time and using the vanishing boundary values of φ gives (110). $\square$

From measure evolutions to vector fields.

$$\partial_t\alpha_t + \mathrm{div}(v_t\alpha_t) = 0. \tag{111}$$

$$\mathcal{H}_\alpha := \{v : \mathrm{div}(\alpha v) = 0\}.$$

Dacorogna and Moser inversion.

$$v_t = -\frac{1}{\alpha_t}\nabla\Delta^{-1}(\partial_t\alpha_t). \tag{112}$$

which was initially proposed by Dacorogna and Moser. A difficulty with this choice is that it is not well defined when α_t vanishes, and also it is not a gradient field, which might be desirable in some cases.

Least square inversion and gradient structure.

$$\min_v \frac{1}{2}\int_0^1\int_{\mathbb{R}^d}\|v_t(x)\|^2\,\mathrm{d}\alpha_t(x)\,\mathrm{d}t \quad \text{subject to} \quad \mathrm{div}(\alpha v)+\frac{\partial\alpha}{\partial t} = 0. \tag{113}$$

$$\min_v\max_\psi \frac{1}{2}\int_0^1\left[\int_{\mathbb{R}^d}\|v_t(x)\|^2\,\mathrm{d}\alpha_t(x) + \int_{\mathbb{R}^d}\psi_t(x)\left(\mathrm{div}(\alpha_tv_t)(x)+\partial_t\alpha_t(x)\right)\,\mathrm{d}x\right]\mathrm{d}t. \tag{114}$$

$$\min_v\max_\psi \int_0^1\left[\int_{\mathbb{R}^d}\left(\frac{1}{2}\|v_t(x)\|^2-\nabla\psi_t(x)\cdot v_t(x)\right)\,\mathrm{d}\alpha_t(x) + \int_{\mathbb{R}^d}\psi_t(x)\,\partial_t\alpha_t(x)\,\mathrm{d}x\right]\mathrm{d}t. \tag{115}$$

$v_t(x) = \nabla\psi_t(x)$.

$$v_t = -\nabla\Delta_\alpha^{-1}(\partial_t\alpha_t), \quad \text{with} \quad \Delta_\alpha(\varphi) := \mathrm{div}(\alpha_t\nabla\varphi). \tag{116}$$

10.2 Generative Models via Flow Matching

Stochastic interpolant. We assume that α_t is defined via a "projection" (in a loose sense) of a latent distribution $\pi \in \mathcal{P}(\mathbb{R}^{d'})$, using an operator $P_t : \mathbb{R}^{d'} \to \mathbb{R}^d$ where $d' \gg d$, i.e.

$$\forall t \in [0,1], \quad \alpha_t := (P_t)_\sharp \pi. \tag{117}$$

A common case is $d' = 2d$. We write $(x,y) \in \mathbb{R}^{d'} = \mathbb{R}^d \times \mathbb{R}^d$ and assume $P_0(x,y) = x$ and $P_1(x,y) = y$, so that π is a probabilistic coupling between α_0 and α_1, i.e. π has marginals (α_0, α_1).
If $\pi = \alpha \otimes \beta$ and $\alpha = \frac{1}{n}\sum_i \delta_{x_i}$, $\beta = \frac{1}{m}\sum_j \delta_{y_j}$, then α_t consists of $n \times m$ Dirac masses
$\alpha_t = \frac{1}{nm}\sum_{i,j} \delta_{P_t(x_i,y_j)}$. If $\pi = (\mathrm{Id}, T)_\sharp \alpha$ is a Brenier-type coupling, then $\alpha_t = ((1-t)\mathrm{Id} + tT)_\sharp \alpha$ is the so-called McCann OT interpolation.

Flow matching formula.
Proposition 65 (Flow matching vector field)**.** *For each fixed t, the solution of the following flow-matching problem*

$$\min_{v_t} \int_{\mathbb{R}^{d'}} \|v_t(P_t(u)) - [\partial_t P_t](u)\|^2 \,\mathrm{d}\pi(u). \tag{118}$$

or equivalently the conditional expectation

$$v_t(z) = \mathbb{E}_{u\sim\pi}\big([\partial_t P_t](u) \,\big|\, z = P_t(u)\big). \tag{119}$$

satisfies the continuity equation (111).

Proof. We now prove the flow matching formula (119), i.e. that it defines a valid flow between α_0 and α_1.
First, let us recall the definition of the interpolated measure α_t and the velocity field v_t.
According to (117), the measure α_t can be represented formally as: $\alpha_t(z) = \int \delta(z - P_t(u))\,\mathrm{d}\pi(u)$ and rigorously for any test function $\varphi(z)$ as:

$$\int \varphi(z)\,\mathrm{d}\alpha_t(z) = \int \varphi(P_t(u))\,\mathrm{d}\pi(u). \tag{120}$$

Following (119), the velocity field v_t can be written formally as: $v_t(z) = \frac{1}{\alpha_t(z)}\int \delta(z - P_t(u))[\partial_t P_t](u)\,\mathrm{d}\pi(u)$, and rigorously for any vector field $m(z)$ as:

$$\int \langle m(z), v_t(z)\rangle \,\mathrm{d}\alpha_t(z) = \int \langle m(P_t(u)), [\partial_t P_t](u)\rangle \,\mathrm{d}\pi(u). \tag{121}$$

We aim to prove that the measure α_t satisfies the continuity equation: $\frac{\partial \alpha_t}{\partial t} + \mathrm{div}(\alpha_t v_t) = 0$. In a rigorous sense, this means showing that for any smooth test function $\varphi(z)$,

$$\frac{\mathrm{d}}{\mathrm{d}t}\int \varphi(z)\,\mathrm{d}\alpha_t(z) - \int \langle v_t(z), \nabla\varphi(z)\rangle \,\mathrm{d}\alpha_t(z) = 0. \tag{122}$$

To prove (122), we compute both terms separately and show that they cancel out. First, consider the time derivative of $\int \varphi(z)\,\mathrm{d}\alpha_t(z)$. Using (120), we differentiate under the integral sign: $\frac{\mathrm{d}}{\mathrm{d}t}\int \varphi(z)\,\mathrm{d}\alpha_t(z) = \int \frac{\mathrm{d}}{\mathrm{d}t}\varphi(P_t(u))\,\mathrm{d}\pi(u)$. Applying the chain rule to $t \to \varphi \circ P_t$:

$$\frac{\mathrm{d}}{\mathrm{d}t}\varphi(P_t(u)) = \langle \nabla\varphi(P_t(u)), [\partial_t P_t](u)\rangle\,.$$

Thus:

$$\frac{\mathrm{d}}{\mathrm{d}t}\int \varphi(z)\,\mathrm{d}\alpha_t(z) = \int \langle \nabla\varphi(P_t(u)), [\partial_t P_t](u)\rangle \,\mathrm{d}\pi(u). \tag{123}$$

Next, for the term involving $\mathrm{div}(\alpha_t v_t)$, we use the definition of v_t in (121) with $m(z) = \nabla\varphi(z)$. Substituting this into the expression for v_t, we get: $\int \langle v_t(z), \nabla\varphi(z)\rangle \,\mathrm{d}\alpha_t(z) = \int \langle \nabla\varphi(P_t(u)), [\partial_t P_t](u)\rangle \,\mathrm{d}\pi(u)$. Comparing this result with (123), we see that: $\frac{\mathrm{d}}{\mathrm{d}t}\int \varphi(z)\,\mathrm{d}\alpha_t(z) - \int \langle v_t(z), \nabla\varphi(z)\rangle \,\mathrm{d}\alpha_t(z) = 0$. ☐

Connection with diffusion models. In the special case where $P_t(x,y) = (1-t)x + ty$ is a linear interpolation and $\pi = \alpha \otimes \beta$, then α_t is a convolution of rescaled versions of α_0 and α_1. Then the flow matching (118) boils down to

$$\min_{(v_t)_t} \int_{\mathbb{R}^d \times \mathbb{R}^d} \|v_t((1-t)x + ty) - (y - x)\|^2 \,\mathrm{d}\alpha_0(x)\mathrm{d}\alpha_1(y).$$

If furthermore, α_0 is an isotropic Gaussian, this is exactly (up to an exponential change of variable in the time variable) the diffusion model method.
Proposition 7 (Tweedie identity)**.** *Let W be a random vector in $\mathbb{R}^d$ with density β. For $\sigma > 0$, observe $Z = W + \sigma\varepsilon$, where $\varepsilon \sim \mathcal{N}(0, I_d)$ is independent of W. Denote by $\beta_\sigma = \beta * \mathcal{N}(0, \sigma^2 I_d)$ the density of Z. Then $\mathbb{E}[W \mid Z = z] = z + \sigma^2 \nabla \log \beta_\sigma(z)$ for all $z \in \mathbb{R}^d$.*

Proof. Bayes' rule gives the conditional density $p_{W|Z}(w \mid z) = \dfrac{\beta(w)\,\varphi_\sigma(z-w)}{\beta_\sigma(z)}$ with φ_σ the $\mathcal{N}(0, \sigma^2 I_d)$ density. Hence $\mathbb{E}[W \mid Z = z] = \frac{1}{\beta_\sigma(z)}\int_{\mathbb{R}^d} w\,\beta(w)\,\varphi_\sigma(z-w)\,\mathrm{d}w$. Integrating by parts in w and using $\nabla_z\varphi_\sigma(z-w) = -\sigma^{-2}(z-w)\,\varphi_\sigma(z-w)$ yields $\nabla_z\beta_\sigma(z) = \int \beta(w)\,\nabla_z\varphi_\sigma(z-w)\,\mathrm{d}w = -\sigma^{-2}\Big(z - \mathbb{E}[W \mid Z = z]\Big)\,\beta_\sigma(z)$. Rearranging finishes the proof. ☐

Proposition 8 (Optimal vector field)**.** *Let $X \sim \alpha$ and $Y \sim \mathcal{N}(0, I_d)$ be independent. For $t \in (0,1)$ set $Z_t = (1-t)\,X + t\,Y$, $\quad \alpha_t = Law(Z_t)$. The minimizer $v^\star$ of $\min_v \int_0^1 \iint_{\mathbb{R}^d\times\mathbb{R}^d} \big|y - x - v\big((1-t)x + ty, t\big)\big|^2 \,\mathrm{d}\alpha(x)\,\mathrm{d}\mathcal{N}(y)\,\mathrm{d}t$ is $v^\star(x,t) = -\frac{1}{1-t}\,x \;-\; \frac{t}{1-t}\,\nabla\log\alpha_t(x) \qquad (x \in \mathbb{R}^d,\ t \in (0,1))$.*

Proof. Fix $t \in (0,1)$ and write $W = (1-t)X$, $\sigma = t$, so that $Z_t = W + \sigma\,Y$ matches the setting of Proposition 7. Conditional expectations satisfy $v^\star(z,t) = \mathbb{E}[Y - X \mid Z_t = z] = \frac{1}{t}\,\mathbb{E}[Z_t - W \mid Z_t = z] - \frac{1}{1-t}\,\mathbb{E}[W \mid Z_t = z]$. Applying Proposition 7 to $\mathbb{E}[W \mid Z_t = z]$ and noting $\mathbb{E}[Y \mid Z_t = z] = -\,t\,\nabla\log\alpha_t(z)$ gives the claimed formula. □

When is the induced map optimal? Integrating the learned velocity gives a deterministic map from α_0 to α_1, but this map is not automatically the Brenier optimal map. It is optimal only in special cases where the accumulated flow remains the gradient of a convex potential.

10.3 Benamou–Brenier dynamic formulation of OT

Theorem 8 (Benamou–Brenier)**.** *For probability measures α_0, α_1 with finite second moments,*

$$\mathcal{W}_2^2(\alpha_0,\alpha_1) = \inf_{(\alpha_t,v_t)} \int_0^1 \int_{\mathbb{R}^d} |v_t(x)|^2\,\mathrm{d}\alpha_t(x)\,\mathrm{d}t, \tag{124}$$

where the infimum is over (α_t, v_t) solving $\partial_t\alpha_t + \nabla\cdot(\alpha_t v_t) = 0$ with $\alpha_{t=0} = \alpha_0$ and $\alpha_{t=1} = \alpha_1$. If α_0 has a density and T is the optimal Monge map $T_\sharp\alpha_0 = \alpha_1$, the minimizer is

$$\alpha_t = ((1-t)\mathrm{Id} + tT)_\sharp\alpha_0, \qquad v_t\big((1-t)x + tT(x)\big) = T(x) - x, \quad t \in [0,1]. \tag{125}$$

Proof. Dynamic $\le$ static. Let T solve the Monge problem and define (α_t, v_t) by (125). As $\dot{T}_t(x) = T(x) - x$ is t-independent, $\int_0^1\int |v_t|^2\,\mathrm{d}\alpha_t\,\mathrm{d}t = \int |T(x) - x|^2\,\mathrm{d}\alpha_0(x)$, so the dynamic cost is no larger.

Static $\le$ dynamic. Conversely, for a smooth deterministic path take the flow T_t defined by $\dot{T}_t = v_t \circ T_t$, $T_0 = \mathrm{Id}$. Then $\alpha_t = (T_t)_\sharp\alpha_0$ and $(T_1)_\sharp\alpha_0 = \alpha_1$. Jensen's inequality gives $|T_1(x) - x|^2 \le \int_0^1 |v_t(T_t(x))|^2\,\mathrm{d}t$. Integrating yields $\int |T_1(x) - x|^2\,\mathrm{d}\alpha_0 \le \int_0^1\int |v_t|^2\,\mathrm{d}\alpha_t\,\mathrm{d}t$. Thus the Monge cost is no larger than the dynamic one. For general finite-energy solutions of the continuity equation, the same argument is obtained by applying the superposition principle to lift the curve to a probability measure on absolutely continuous paths; this is the path-space formulation stated below. □

Remark 27 (Path-space formulation). Let $\mathcal{S} = C([0,1];\mathbb{R}^d)$ be the space of continuous paths endowed with the uniform topology. For $t \in [0,1]$ define the evaluation map
$P_t : \mathcal{S} \to \mathbb{R}^d, \qquad P_t(\gamma) = \gamma(t).$

$$\mathcal{W}_2^2(\alpha_0,\alpha_1) = \inf_{M\in\mathcal{P}(\mathcal{S})} \Big\{ \int_{\mathcal{S}}\int_0^1 |\dot\gamma(t)|^2\,\mathrm{d}t\,\mathrm{d}M(\gamma) : (P_0)_\sharp M = \alpha_0,\ (P_1)_\sharp M = \alpha_1 \Big\}.$$

If α_0 has a density, the minimizer M^* is unique. Its time marginals reproduce the optimal curve: $\alpha_t = (P_t)_\sharp M^*$ for all t. Furthermore, denoting $Q_t : \mathcal{S} \to \mathbb{R}^d, Q_t(\gamma) = \dot\gamma(t)$, the conditional law of the velocity is deterministic:
$(P_t, Q_t)_\sharp M^* = \alpha_t \otimes \delta_{v_t^*(x)}, \qquad t \in [0,1]$, where v_t^* is the optimal velocity field in the Benamou–Brenier formulation. Hence M^* concentrates on straight-line geodesics, assigning exactly one direction at each spatial point.

10.4 Wasserstein Gradient Flows

$$\alpha_{t+\tau} := \arg\min_\alpha \frac{1}{2\tau}\,\mathcal{W}_2(\alpha_t,\alpha)^2 + f(\alpha). \tag{126}$$

Euclidean gradient flows. If we restrict (126) to finite dimensions and assume $\alpha_t = \delta_{x(t)}$ and $\alpha = \delta_x$ (single Dirac measures), this matches the implicit Euler scheme:
$x(t+\tau) := \arg\min_x \frac{1}{2\tau}\|x - x(t)\|^2 + h(x)$, where $h(x) = f(\delta_x)$. Its solution is formally given by the implicit Euler formula:
$x(t+\tau) = (\mathrm{Id} + \tau\nabla h)^{-1}(x(t))$. $x(t+\tau) = (\mathrm{Id} - \tau\nabla h)(x(t)) = x(t) - \tau\nabla h(x(t))$.

$$\dot{x}(t) = -\nabla h(x(t)). \tag{127}$$

Wasserstein gradient formula. As $\tau \to 0$, under certain conditions on f, (126) defines a continuous evolution $t \mapsto \alpha_t$. As discussed earlier, this evolution can be described as a Lagrangian evolution (108).

$$\frac{\partial\alpha_t}{\partial t} + \mathrm{div}(-\nabla_{\mathcal{W}} f(\alpha_t)\alpha_t) = 0. \tag{128}$$

$\nabla_{\mathcal{W}} f(\alpha) = \nabla_x\varphi, \quad$ where $\varphi := \delta f(\alpha)$. $f((1-\tau)\alpha + \tau\beta) = f(\alpha + \tau\rho) = f(\alpha) + \tau\int[\delta f(\alpha)](x)\mathrm{d}\rho(x) + o(\tau)$, where $\rho = \beta - \alpha$ is a zero-mean measure.

Proposition 9 (Formal Wasserstein gradient)**.** *Assume that f admits a smooth first variation $\delta f(\alpha)$ and that α has a smooth positive density. For infinitesimal perturbations generated by a velocity field v through $(\mathrm{Id} + \tau v)_\sharp\alpha$, the differential of f is $\frac{\mathrm{d}}{\mathrm{d}\tau}_{|\tau=0} f\big((\mathrm{Id} + \tau v)_\sharp\alpha\big) = \int\langle\nabla\delta f(\alpha)(x),\, v(x)\rangle\mathrm{d}\alpha(x)$. Hence, for the Riemannian metric $\|v\|^2_{L^2(\alpha)} = \int\|v\|^2\mathrm{d}\alpha$, the Wasserstein gradient is the vector field $\nabla_{\mathcal{W}} f(\alpha) = \nabla\delta f(\alpha)$.*

Proof. The push-forward expansion gives, in the sense of distributions, $(\mathrm{Id}+\tau v)_\sharp\alpha = \alpha - \tau\,\mathrm{div}(\alpha v) + o(\tau)$. Using the definition of the first variation, $f((\mathrm{Id}+\tau v)_\sharp\alpha) = f(\alpha) - \tau\int \delta f(\alpha)\,\mathrm{div}(\alpha v)\mathrm{d}x + o(\tau)$. An integration by parts, with either compact support or vanishing boundary flux, gives $-\int \delta f(\alpha)\,\mathrm{div}(\alpha v)\mathrm{d}x = \int\langle\nabla\delta f(\alpha),\, v\rangle\mathrm{d}\alpha$. By definition of the Riesz representative for the $L^2(\alpha)$ metric, this representative is $\nabla\delta f(\alpha)$. □

From the JKO step to the velocity field.

$$\min_v \frac{1}{2\tau}\tau^2\|v\|^2_{L^2(\alpha_t)} + f((\mathrm{Id}+\tau v)_\sharp\alpha_t).$$

$$(\mathrm{Id}+\tau v)_\sharp\alpha_t = \alpha_t - \tau\,\mathrm{div}(v\alpha_t) + o(\tau)$$

$$f((\mathrm{Id}+\tau v)_\sharp\alpha_t) = f(\alpha_t) - \tau\int \delta f(\alpha_t)\,\mathrm{div}(v\alpha_t)\mathrm{d}x + o(\tau)$$

$$= f(\alpha_t) + \tau\int\langle\nabla_x\delta f(\alpha_t)(x), v(x)\rangle\mathrm{d}\alpha_t(x) + o(\tau)$$

$$\min_v f(\alpha_t) + \tau\int\left[\frac{1}{2}\|v(x)\|^2 + \langle\nabla_{\mathcal{W}} f(\alpha_t)(x), v(x)\rangle\right]\mathrm{d}\alpha_t(x) + o(\tau).$$

Discrete evolutions. If $f(\alpha)$ can be evaluated on discrete distributions and $\nabla_{\mathcal{W}}$ is continuous in this case, the flow (128) maintains the number of Dirac masses, $\alpha_t = \frac{1}{n}\sum_i \delta_{x_i(t)}$. The particles $X(t) := (x_i(t))_i$ evolve according to a system of coupled ODEs:

$$\dot{x}_i(t) = -n\nabla_{x_i}F(X(t)), \tag{129}$$

where $F(X) := f\left(\frac{1}{n}\sum_i\delta_{x_i}\right)$ and the factor n comes from the empirical Wasserstein metric $\frac{1}{n}\sum_i\|\dot{x}_i\|^2$.

Linear Functionals.

$$f(\alpha) = \int h(x)\mathrm{d}\alpha(x). \tag{130}$$

$\frac{\partial\alpha_t}{\partial t} + \mathrm{div}(-\nabla h\alpha_t) = 0.$

Shannon Neg-Entropy.

$$f(\alpha) = \int\log\left(\frac{\mathrm{d}\alpha}{\mathrm{d}x}(x)\right)\mathrm{d}\alpha(x). \tag{131}$$

$\partial_t\alpha_t = \Delta\alpha_t.$

$$f(\alpha) = \int g\left(\frac{\mathrm{d}\alpha}{\mathrm{d}x}\right)\mathrm{d}x, \tag{132}$$

$\frac{\partial\alpha_t}{\partial t} = \Delta(P(\alpha_t))$, where the pressure P satisfies $P'(s) = sg''(s)$.

Interaction Energies.

$$f(\alpha) := \iint k(x,y)\mathrm{d}\alpha(x)\mathrm{d}\alpha(y). \tag{133}$$

For a symmetric kernel k:
$\delta f(\alpha)(x) = 2\int k(x,y)\mathrm{d}\alpha(y), \quad \nabla_{\mathcal{W}} f(\alpha)(x) = 2\int\nabla_x k(x,y)\mathrm{d}\alpha(y)$. For $\alpha_0 = \frac{1}{n}\sum_i\delta_{x_i}$, the flow (128) implies particles $(x_i(t))_i$ obey:
$\dot{x}_i(t) = -\frac{2}{n}\sum_j\nabla k(x_i(t), x_j(t))$. If k is positive definite and one minimizes the squared MMD distance to a teacher distribution β, then

$$\|\alpha-\beta\|_k^2 = \iint k\mathrm{d}\alpha\mathrm{d}\alpha - 2\int\left(\int k(x,y)\mathrm{d}\beta(y)\right)\mathrm{d}\alpha(x) + \text{constant}.$$

Thus MMD training energies are exactly an interaction energy plus a linear potential; the teacher distribution appears through the potential $x\mapsto -2\int k(x,y)\mathrm{d}\beta(y)$.

Geodesics and geodesic convexity. $\alpha_t = ((1-t)P_0 + tP_1)_\sharp\pi^\star, \qquad t\in[0,1]$, where $P_0(x,y) = x$ and $P_1(x,y) = y$. If the optimal plan is induced by a Brenier map T, this reduces to $((1-t)\mathrm{Id} + tT)_\sharp\alpha_0$.

Definition 16 (Geodesic convexity)**.** *A functional f on $\mathcal{P}_2(\mathbb{R}^d)$ is geodesically convex if for every $\mathcal{W}_2$ geodesic $(\alpha_t)_t$, $f(\alpha_t)\leqslant(1-t)f(\alpha_0) + tf(\alpha_1)$. It is λ-geodesically convex if the right-hand side is improved by $-\frac{\lambda}{2}t(1-t)\,\mathcal{W}_2^2(\alpha_0,\alpha_1)$.*

Proposition 66 (Basic geodesically convex energies)**.**

1. *If h is convex, then $\alpha\mapsto\int h\mathrm{d}\alpha$ is geodesically convex; if h is λ-strongly convex, it is λ-geodesically convex.*
2. *If $W(x-y)$ is convex as a function of the displacement, then $\alpha\mapsto\frac{1}{2}\iint W(x-y)\mathrm{d}\alpha(x)\mathrm{d}\alpha(y)$ is geodesically convex.*
3. *Shannon entropy $\alpha\mapsto\int\rho\log\rho\,\mathrm{d}x$ is geodesically convex.*

4. The relative entropy $\mathrm{KL}(\alpha|\gamma)$ *with* $\mathrm{d}\gamma = e^{-V}\mathrm{d}x/Z$ *is* λ*-geodesically convex when* V *is* λ*-strongly convex.*

Proof. Along a Monge geodesic $X_t = (1-t)X_0 + tX_1$, convexity of h gives $h(X_t) \leqslant (1-t)h(X_0) + th(X_1)$, and strong convexity gives the additional quadratic term; integrating proves the first claim. The interaction claim follows similarly by applying convexity of W to pairwise differences $X_t - X_t' = (1-t)(X_0 - X_0') + t(X_1 - X_1')$ and integrating over two independent copies. The entropy claim is McCann's displacement convexity theorem; at the density level it follows from the concavity of the Jacobian determinant under the interpolation of optimal maps. Finally, $\mathrm{KL}(\alpha|\gamma) = \int \rho \log \rho \,\mathrm{d}x + \int V \mathrm{d}\alpha + \text{constant}$, so it is the sum of displacement-convex entropy and a λ-geodesically convex linear potential. □

Convergence of the flow.

Proposition 67 (Energy decay for convex Wasserstein flows)**.** *Assume formally that* f *is geodesically convex, admits a smooth first variation, and has a minimizer* $\alpha^\star$*. Let* $(\alpha_t)_t$ *be a smooth solution of the Wasserstein gradient flow* $\partial_t \alpha_t + \mathrm{div}(\alpha_t v_t) = 0, \quad v_t = -\nabla_{\mathcal{W}} f(\alpha_t)$*. Then* $\frac{\mathrm{d}}{\mathrm{d}t} f(\alpha_t) = -\int \|\nabla_{\mathcal{W}} f(\alpha_t)(x)\|^2 \,\mathrm{d}\alpha_t(x) \leqslant 0$*. If* T_t *is the optimal map from* α_t *to* $\alpha^\star$*, then* $f(\alpha_t) - f(\alpha^\star) \leqslant -\frac{\mathrm{d}}{\mathrm{d}t}\frac{1}{2}\mathcal{W}_2^2(\alpha_t, \alpha^\star)$*, and consequently* $f(\alpha_t) - f(\alpha^\star) \leqslant \frac{\mathcal{W}_2^2(\alpha_0,\alpha^\star)}{2t}$*. If* f *is* λ*-geodesically convex with* $\lambda > 0$*, then* $f(\alpha_t) - f(\alpha^\star) \leqslant e^{-2\lambda t}\big(f(\alpha_0) - f(\alpha^\star)\big)$*.*

Proof. The chain rule and Proposition 9 give $\frac{\mathrm{d}}{\mathrm{d}t} f(\alpha_t) = \int \langle \nabla_{\mathcal{W}} f(\alpha_t)(x),\, v_t(x)\rangle \,\mathrm{d}\alpha_t(x) = -\int \|\nabla_{\mathcal{W}} f(\alpha_t)(x)\|^2 \,\mathrm{d}\alpha_t(x)$. Geodesic convexity along the geodesic $((1-s)\mathrm{Id} + sT_t)_\sharp \alpha_t$ gives $f(\alpha^\star) - f(\alpha_t) \geqslant \int \langle \nabla_{\mathcal{W}} f(\alpha_t)(x),\, T_t(x) - x\rangle \,\mathrm{d}\alpha_t(x)$. Since $v_t = -\nabla_{\mathcal{W}} f(\alpha_t)$, this reads $f(\alpha_t) - f(\alpha^\star) \leqslant \int \langle v_t(x),\, T_t(x) - x\rangle \,\mathrm{d}\alpha_t(x)$. $\frac{\mathrm{d}}{\mathrm{d}t}\frac{1}{2}\mathcal{W}_2^2(\alpha_t, \alpha^\star) = \int \langle x - T_t(x),\, v_t(x)\rangle \,\mathrm{d}\alpha_t(x)$, which proves the differential inequality. Integrating it from 0 to t and using the monotonicity of $s \mapsto f(\alpha_s)$ gives

$$t\big(f(\alpha_t) - f(\alpha^\star)\big) \leqslant \int_0^t \big(f(\alpha_s) - f(\alpha^\star)\big)\,\mathrm{d}s \leqslant \frac{1}{2}\mathcal{W}_2^2(\alpha_0, \alpha^\star).$$

If f is λ-geodesically convex, the Wasserstein analogue of strong convexity gives the slope inequality $\int \|\nabla_{\mathcal{W}} f(\alpha_t)\|^2 \mathrm{d}\alpha_t \geqslant 2\lambda\big(f(\alpha_t) - f(\alpha^\star)\big)$. $\frac{\mathrm{d}}{\mathrm{d}t}\big(f(\alpha_t) - f(\alpha^\star)\big) \leqslant -2\lambda\big(f(\alpha_t) - f(\alpha^\star)\big)$, and Gronwall's lemma gives the exponential rate. □

Proposition 68 (Convex examples covered by the theory)**.** *The energy decay proposition applies to linear potentials* (130) *when* h *is convex, to interaction energies* (133) *when the interaction potential is convex in displacements, and to entropy-type functionals such as Shannon entropy* (131)*.*

Proof. linear potentials inherit convexity from the pointwise potential, interaction energies inherit it from pairwise displacement convexity, and Shannon entropy is displacement convex by McCann's theorem. □

10.5 Application: Training Two-Layer MLPs as Wasserstein Flows

$g_\theta(u) := \frac{1}{n}\sum_i \psi(\theta_i, u), \quad$ where $\psi(\theta_i, u) := a_i\langle u, w_i\rangle$, and $\theta_i := (w_i, a_i) \in \mathbb{R}^{d+1}$ represents the parameters of a neuron. Importantly, these functions are invariant under permutations of the neurons.
$G_\alpha(u) := \int_{\mathbb{R}^{d+1}} \psi(\theta, u)\,\mathrm{d}\alpha(\theta)$. $\min_\alpha f(\alpha) := \frac{1}{N}\sum_{k=1}^N \ell(G_\alpha(u_k), y_k)$.

$$\dot{\theta}_i = -n\nabla_{\theta_i} F(\theta), \quad \text{where } F(\theta) := f\left(\frac{1}{n}\sum_i \delta_{\theta_i}\right).$$

This is equivalent to the PDE (128) on the neuron density, where the Wasserstein gradient is used.

$$\delta f(\alpha)(\theta) = \frac{1}{N}\sum_{k=1}^N \left(\int \psi(\theta', u_k)\,\mathrm{d}\alpha(\theta') - y_k\right)\psi(\theta, u_k),$$

$\delta f(\alpha)(\theta) = \int k(\theta, \theta')\,\mathrm{d}\alpha(\theta') + g(\theta)$,

$$k(\theta, \theta') := \frac{1}{N}\sum_k \psi(\theta, u_k)\psi(\theta', u_k), \tag{134}$$

$$g(\theta) := -\frac{1}{N}\sum_k y_k\psi(\theta, u_k). \tag{135}$$

$\nabla_{\mathcal{W}} f(\alpha)(\theta) = \int \nabla_\theta k(\theta, \theta')\,\mathrm{d}\alpha(\theta') + \nabla_\theta g(\theta)$.

10.6 Application: One-Step Generative Models

Let ζ be a simple latent distribution and let $\alpha_\theta = (G_\theta)_\sharp\zeta$ be the model distribution. Assume that the target data distribution is β. A Wasserstein-flow construction chooses a discrepancy
$\mathcal{E}_\beta(\alpha)$,

$$\partial_t \mu_t + \mathrm{div}(\mu_t w_t) = 0, \qquad w_t(x) = -\nabla\delta_\alpha \mathcal{E}_\beta(\mu_t)(x). \tag{136}$$

$$\min_\eta \int_0^1 \int \|U_\eta(t, x) - w_t(x)\|^2 \,\mathrm{d}\mu_t(x)\,\mathrm{d}t. \tag{137}$$

$$\alpha_\theta^+ = (\mathrm{Id} + \tau U_\eta)_\sharp \alpha_\theta, \qquad \text{or equivalently} \qquad G_\theta^+(z) = G_\theta(z) + \tau U_\eta(G_\theta(z)).$$

Normalized drifting fields.

$$B_\varepsilon[\nu](x) := \frac{\int (y-x) K_\varepsilon(x,y)\, \mathrm{d}\nu(y)}{\int K_\varepsilon(x,y)\, \mathrm{d}\nu(y)}. \tag{138}$$

For the Gaussian kernel $K_\varepsilon(x,y) = \exp(-\|x-y\|^2/(2\varepsilon))$, this normalized field is a score of a smoothed density:

$$B_\varepsilon[\nu](x) = \varepsilon \nabla \log\Big(\int K_\varepsilon(x,y)\, \mathrm{d}\nu(y)\Big). \tag{139}$$

$$u_t(x) = B_\varepsilon[\beta](x) - B_\varepsilon[\mu_t](x) = \varepsilon \nabla \log \frac{\int K_\varepsilon(x,y)\, \mathrm{d}\beta(y)}{\int K_\varepsilon(x,y)\, \mathrm{d}\mu_t(y)}. \tag{140}$$

Proposition 69 (Drifting as a time-dependent Wasserstein gradient)**.** *Let μ_t be a smooth curve of positive densities and let $u_t = \nabla\varphi_t$ be a smooth time-dependent gradient field. Define the semi-relaxed functional*

$$\mathcal{R}_t(\alpha|\mu_t) := -\int \varphi_t(x)\, \mathrm{d}\alpha(x) + \int \varphi_t(x)\, \mathrm{d}\mu_t(x). \tag{141}$$

Here μ_t and φ_t are frozen when taking the first variation with respect to the first argument α. Then the continuity equation $\partial_t\mu_t + \mathrm{div}(\mu_t u_t) = 0$ is the formal Wasserstein gradient descent of the time-dependent functional $\alpha \mapsto \mathcal{R}_t(\alpha|\mu_t)$.

Proof. Since μ_t and φ_t are fixed in the variation with respect to α, the first variation is $\delta_\alpha \mathcal{R}_t(\alpha|\mu_t)(x) = -\varphi_t(x)$. By Proposition 9, $\nabla_{\mathcal{W}}\mathcal{R}_t(\alpha|\mu_t) = \nabla\delta_\alpha\mathcal{R}_t(\alpha|\mu_t) = -\nabla\varphi_t = -u_t$. The Wasserstein gradient-descent velocity is the negative of this gradient, namely u_t. Substituting this velocity in the continuity equation gives the claimed flow. □

Example 9 (Kernel drifting as a semi-relaxed divergence). For the Gaussian-kernel drift (140), set $\varphi_t(x) = \varepsilon \log \frac{\int K_\varepsilon(x,y)\,\mathrm{d}\beta(y)}{\int K_\varepsilon(x,y)\,\mathrm{d}\mu_t(y)}$. Then $u_t = \nabla\varphi_t$, so Proposition 69 shows that kernel drifting is the Wasserstein gradient descent of

$$\mathcal{R}_t^{\mathrm{drift}}(\alpha|\mu_t) = \varepsilon \int \log \frac{\int K_\varepsilon(x,y)\, \mathrm{d}\mu_t(y)}{\int K_\varepsilon(x,y)\, \mathrm{d}\beta(y)}\, \mathrm{d}\alpha(x) + \text{constant}.$$

Remark 28 (General fields and projection onto gradients). $\nabla\varphi_t = \underset{\nabla\varphi}{\operatorname{argmin}} \int \|\nabla\varphi(x) - b_t(x)\|^2\, \mathrm{d}\mu_t(x)$.

10.7 Application: Evolution in Depth of Transformers

Attention as a context-dependent velocity. After tokenization, embedding, and positional encoding, each input (from a set of tokens) is represented as a point cloud $(x_i)_{i=1}^n$ of n points in the space of vectorized tokens. An attention layer with skip connection and rescaling by $1/T$ (where T is the depth) defines a transformation of the tokens: $x_i \mapsto x_i + \frac{1}{T}\sum_j \frac{e^{\langle Qx_i, Kx_j\rangle} V x_j}{\sum_\ell e^{\langle Qx_i, Kx_\ell\rangle}}$, where $\theta = (K, Q, V)$ are the parameters of the attention layer, represented by three matrices.

Token measure evolution. To handle an arbitrary number of tokens, we define $\alpha = \frac{1}{n}\sum_i \delta_{x_i}$ as the empirical measure of tokens and rewrite the transformer mapping as: $x_i \mapsto x_i + \frac{1}{T}\Gamma_\theta[\alpha](x_i)$, $\Gamma_\theta[\alpha](x) := \int \frac{e^{\langle Qx, Ky\rangle} V y\, \mathrm{d}\alpha(y)}{\int e^{\langle Qx, Kz\rangle}\, \mathrm{d}\alpha(z)}$. $\alpha_{t+\tau} = (\mathrm{Id} + \tau\Gamma_{\theta_t}[\alpha_t])_\sharp \alpha_t$. $\partial_t\alpha_t + \mathrm{div}(\alpha_t\Gamma[\alpha_t]) = 0$.

Gradient structure and limitations. When $V = Q^\top K$, the field $\Gamma[\alpha]$ is a gradient vector field, since it is the gradient of the log-partition function associated with the attention weights. It is not, however, the gradient of a first variation, so this PDE is not a Wasserstein gradient flow.

10.8 Gaussian Closures as Finite-Dimensional Test Cases

Proposition 70 (Affine velocities preserve Gaussianity)**.** *Let $\alpha_t = \mathcal{N}(\mathbf{m}_t, \mathbf{\Sigma}_t)$, with $\mathbf{\Sigma}_t$ positive definite, solve the continuity equation with an affine velocity $v_t(x) = b_t + A_t(x - \mathbf{m}_t)$. Then α_t remains Gaussian and its moments solve $\dot{\mathbf{m}}_t = b_t$, $\dot{\mathbf{\Sigma}}_t = A_t\mathbf{\Sigma}_t + \mathbf{\Sigma}_t A_t^\top$. Conversely, any smooth Gaussian curve with positive definite covariance can be generated by such an affine velocity. If one wants the velocity to be a Wasserstein tangent gradient, one chooses the unique symmetric solution of the Lyapunov equation $A_t\mathbf{\Sigma}_t + \mathbf{\Sigma}_t A_t = \dot{\mathbf{\Sigma}}_t$.*

Proof. Let X_t follow the characteristic ODE $\dot{X}_t = b_t + A_t(X_t - \mathbf{m}_t)$. This linear ODE maps Gaussian random variables to Gaussian random variables. Taking expectation gives $\dot{\mathbf{m}}_t = b_t$. Writing $\tilde{X}_t = X_t - \mathbf{m}_t$, one has $\dot{\tilde{X}}_t = A_t\tilde{X}_t$, hence $\dot{\mathbf{\Sigma}}_t = \frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}(\tilde{X}_t\tilde{X}_t^\top) = A_t\mathbf{\Sigma}_t + \mathbf{\Sigma}_t A_t^\top$. For the converse, set $b_t = \dot{\mathbf{m}}_t$ and choose any matrix A_t satisfying the covariance equation. Since $\mathbf{\Sigma}_t$ is positive definite, the Lyapunov map $A \mapsto A\mathbf{\Sigma}_t + \mathbf{\Sigma}_t A$ is invertible on symmetric matrices, which gives the unique symmetric choice when a gradient velocity is required. In that case v_t is the gradient of the quadratic potential $x \mapsto \langle b_t, x\rangle + \langle A_t(x - \mathbf{m}_t), x - \mathbf{m}_t\rangle/2$. □

Example 10 (Flow matching and diffusion paths between Gaussians). $v_t(x) = \dot{\mathbf{m}}_t + A_t(x - \mathbf{m}_t)$, $\quad A_t\mathbf{\Sigma}_t + \mathbf{\Sigma}_t A_t = \dot{\mathbf{\Sigma}}_t$. $v_t(x) = \dot{\mathbf{m}}_t + \frac{\dot{s}_t}{s_t}(x - \mathbf{m}_t)$. $X_t = a_tX_0 + \sigma_t Z$, $\quad Z \sim \mathcal{N}(0, \mathrm{Id})$, has $\mathbf{m}_t = a_t\mathbf{m}_0$ and $\mathbf{\Sigma}_t = a_t^2\mathbf{\Sigma}_0 + \sigma_t^2\mathrm{Id}$. Thus, in the Gaussian case, diffusion paths and flow-matching paths reduce to the same mean-covariance bookkeeping, although the corresponding training objectives are different.

Remark 29 (Gaussian Wasserstein-gradient closures). Let $F(\mathbf{m}, \mathbf{\Sigma})$ be the restriction of a smooth functional f to Gaussian measures. The Wasserstein gradient flow constrained to the Gaussian family is formally

$$\dot{\mathbf{m}}_t = -\nabla_{\mathbf{m}} F(\mathbf{m}_t, \mathbf{\Sigma}_t), \qquad \dot{\mathbf{\Sigma}}_t = -2\big(\mathbf{\Sigma}_t \nabla_{\mathbf{\Sigma}} F(\mathbf{m}_t, \mathbf{\Sigma}_t) + \nabla_{\mathbf{\Sigma}} F(\mathbf{m}_t, \mathbf{\Sigma}_t)\mathbf{\Sigma}_t\big), \tag{142}$$

where $\nabla_{\mathbf{\Sigma}} F$ is the symmetric matrix derivative. For the KL flow toward $\beta = \mathcal{N}(\bar{\mathbf{m}}, \bar{\mathbf{\Sigma}})$, this gives the Ornstein–Uhlenbeck equations
$\dot{\mathbf{m}}_t = -\bar{\mathbf{\Sigma}}^{-1}(\mathbf{m}_t - \bar{\mathbf{m}}), \qquad \dot{\mathbf{\Sigma}}_t = 2\mathrm{Id} - \bar{\mathbf{\Sigma}}^{-1}\mathbf{\Sigma}_t - \mathbf{\Sigma}_t\bar{\mathbf{\Sigma}}^{-1}$. For the Sinkhorn divergence between Gaussian measures, one can similarly restrict the closed Gaussian formula to the variables $(\mathbf{m}, \mathbf{\Sigma})$.

Remark 30 (Other Gaussian test energies).

Example 11 (Linear mean-field networks). $S = \frac{1}{N}\sum_{k=1}^N u_k u_k^\top, \qquad r = \frac{1}{N}\sum_{k=1}^N y_k u_k, \qquad q_t = \int aw \, \mathrm{d}\alpha_t(a, w)$. For the square loss, set $h_t = Sq_t - r$. The first variation is
$\delta f(\alpha_t)(a, w) = a\langle w, \, h_t\rangle$.

$$-\nabla_{(a,w)}\delta f(\alpha_t)(a, w) = -\begin{pmatrix} 0 & h_t^\top \\ h_t & 0 \end{pmatrix}\begin{pmatrix} a \\ w \end{pmatrix}.$$

Example 12 (Gaussian kernel drifting). Let the target be $\beta = \mathcal{N}(\bar{\mathbf{m}}, \bar{\mathbf{\Sigma}})$ and assume $\mu_t = \mathcal{N}(\mathbf{m}_t, \mathbf{\Sigma}_t)$. For the Gaussian kernel $K_\varepsilon(x, y) = \exp(-\|x - y\|^2/(2\varepsilon))$, the normalized field (138) satisfies
$B_\varepsilon[\mu_t](x) = -\varepsilon(\mathbf{\Sigma}_t + \varepsilon\mathrm{Id})^{-1}(x - \mathbf{m}_t)$. Thus the drifting velocity (140) is affine and preserves Gaussianity. With
$A_t = (\mathbf{\Sigma}_t + \varepsilon\mathrm{Id})^{-1}, \qquad \bar{A} = (\bar{\mathbf{\Sigma}} + \varepsilon\mathrm{Id})^{-1}$, $\dot{\mathbf{m}}_t = \varepsilon\bar{A}(\bar{\mathbf{m}} - \mathbf{m}_t), \qquad \dot{\mathbf{\Sigma}}_t = \varepsilon\big((A_t - \bar{A})\mathbf{\Sigma}_t + \mathbf{\Sigma}_t(A_t - \bar{A})\big)$.

Example 13 (Gaussian closure of attention dynamics). For the transformer PDE, assume $\alpha = \mathcal{N}(\mathbf{m}, \mathbf{\Sigma})$. Since exponential tilting preserves Gaussianity,
$\frac{\int e^{\langle Qx,\, Ky\rangle}\, y \, \mathrm{d}\alpha(y)}{\int e^{\langle Qx,\, Kz\rangle}\, \mathrm{d}\alpha(z)} = \mathbf{m} + \mathbf{\Sigma}K^\top Qx$. $\Gamma_\theta[\alpha](x) = V\mathbf{m} + V\mathbf{\Sigma}K^\top Qx$ $\dot{\mathbf{m}}_t = (V_t + V_t\mathbf{\Sigma}_t K_t^\top Q_t)\mathbf{m}_t, \qquad \dot{\mathbf{\Sigma}}_t = B_t\mathbf{\Sigma}_t + \mathbf{\Sigma}_t B_t^\top, \qquad B_t = V_t\mathbf{\Sigma}_t K_t^\top Q_t$. When $V_t = Q_t^\top K_t$, the matrix $B_t = Q_t^\top K_t\mathbf{\Sigma}_t K_t^\top Q_t$ is symmetric positive semidefinite, matching the gradient-field case mentioned above. It is instead a tractable model of how attention can shear, amplify or contract a cloud of tokens through its covariance.